

SECTION PLATE t=0 | Year 2000 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

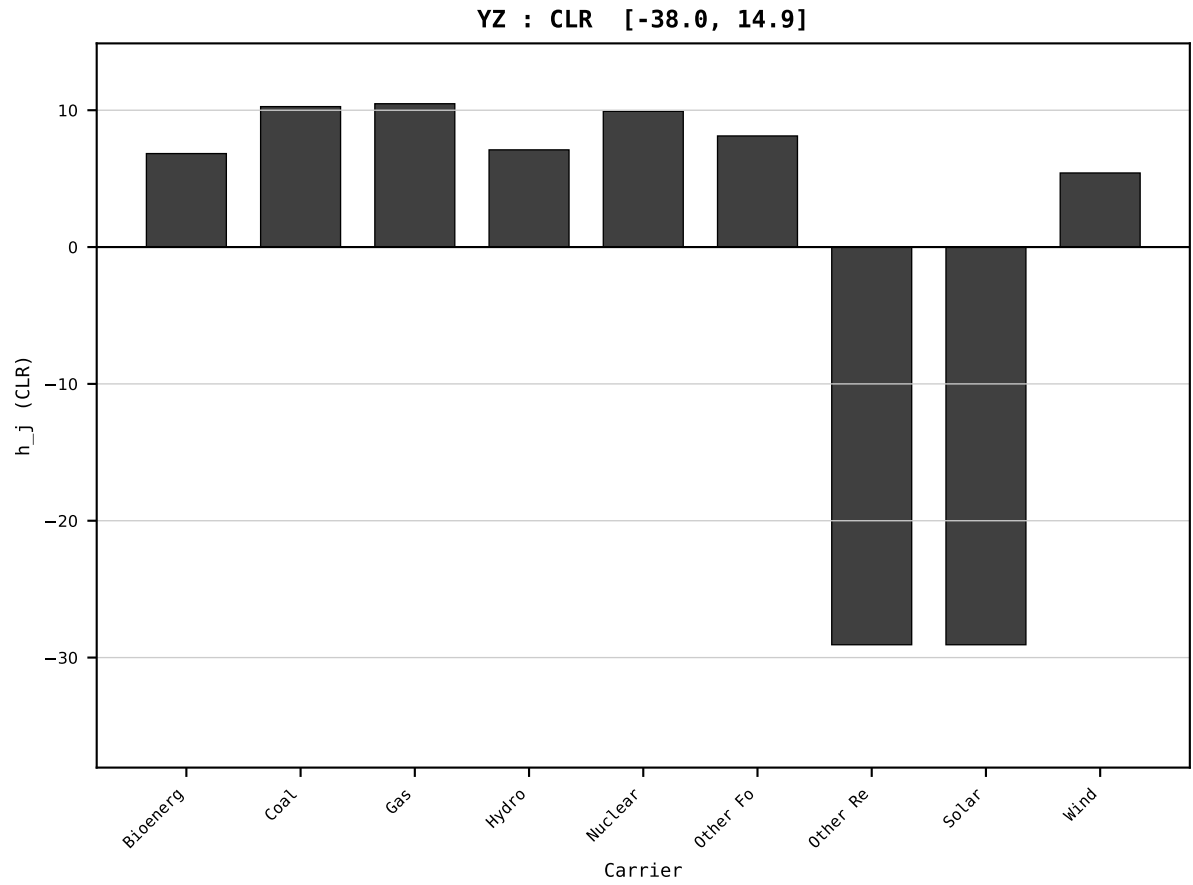
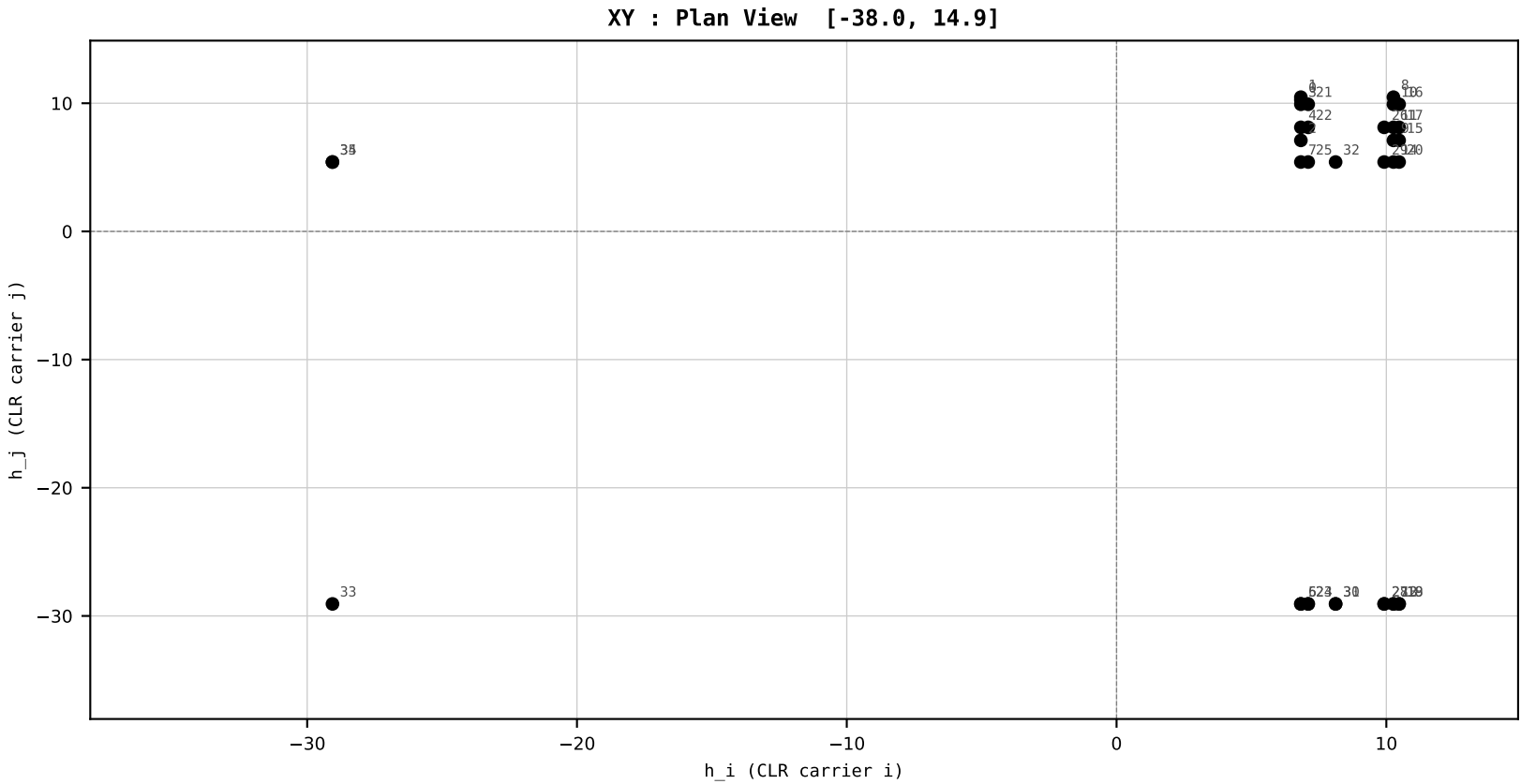
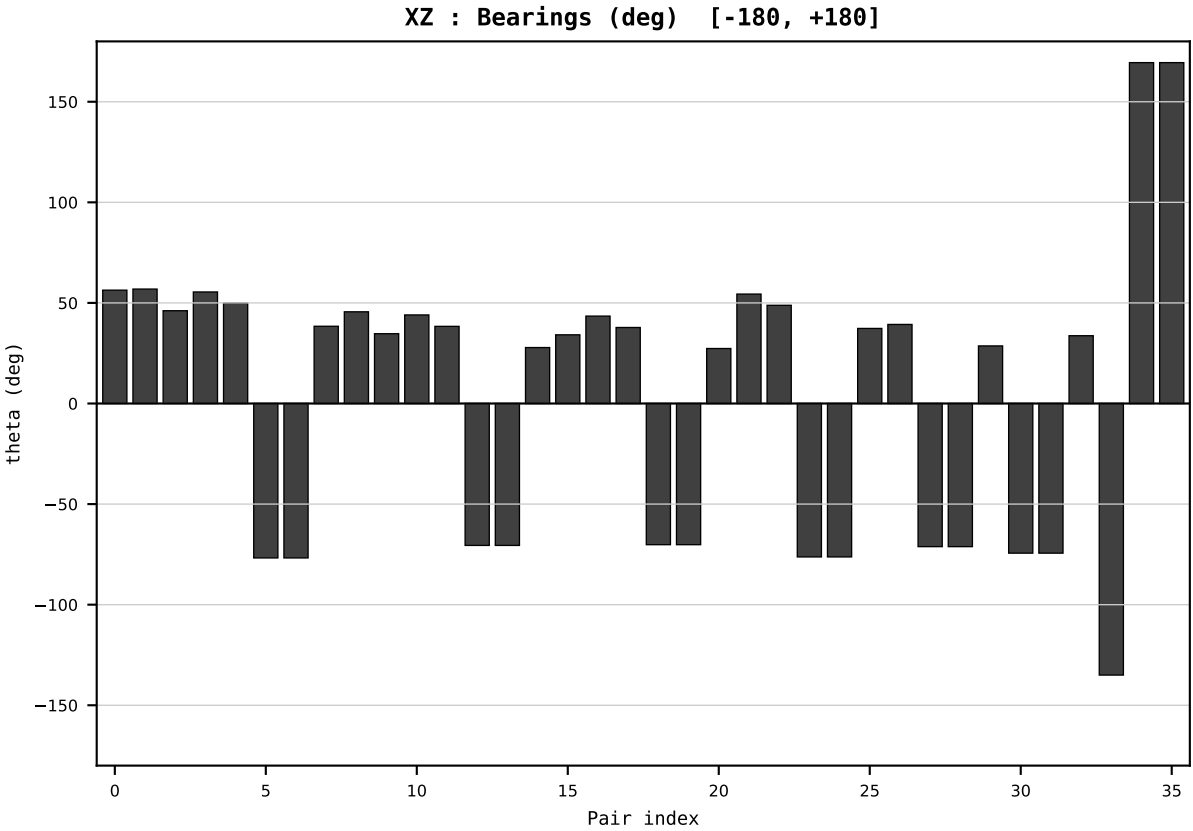
Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2000
t : 0 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.403740
Ring = Hs-3
E_metric= 1560.1795
kappa_HS= 148080000000000032.00
omega = 0.0000 deg
d_A = 0.000000
Helm = ---
Helm d = ---
DR = 39.537 HLR
DR ratio= 148079999529708608.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=1 | Year 2001 | D=9 N=26 pairs=36

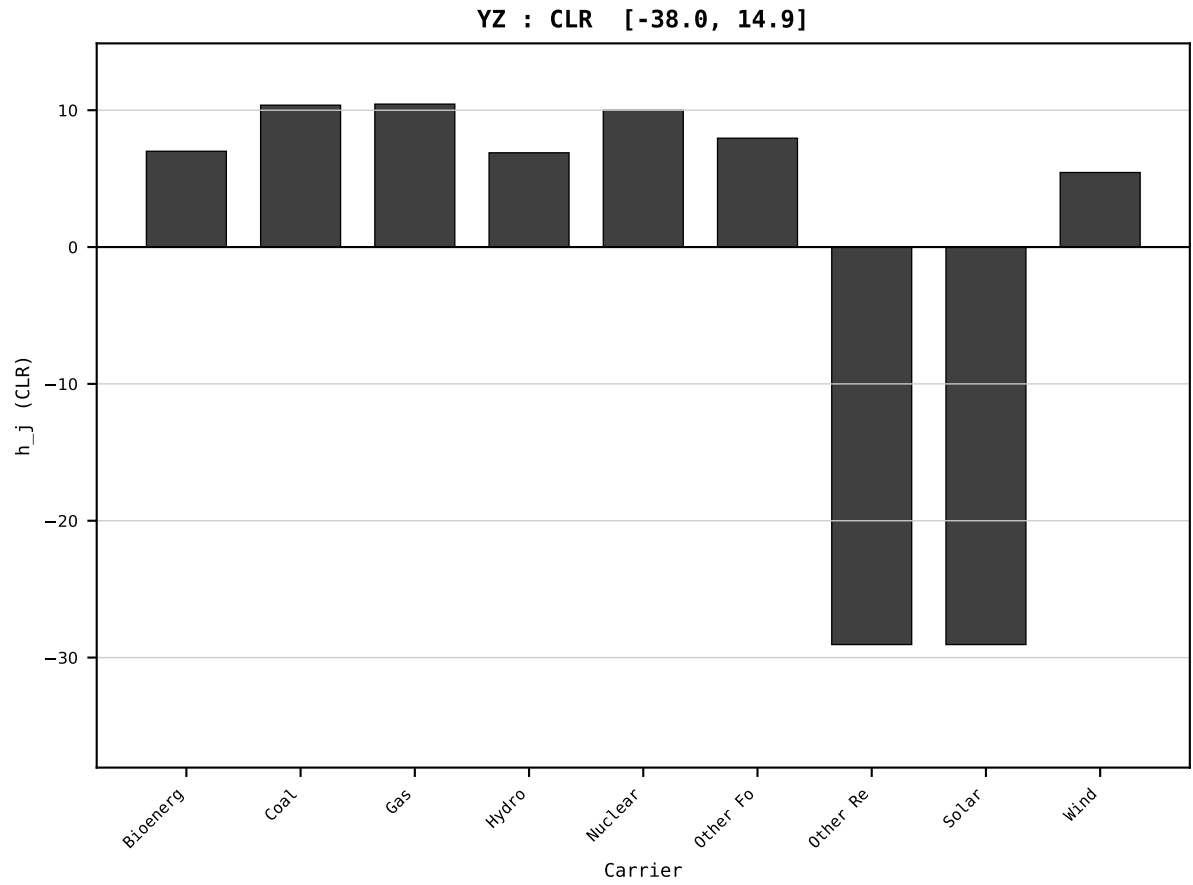
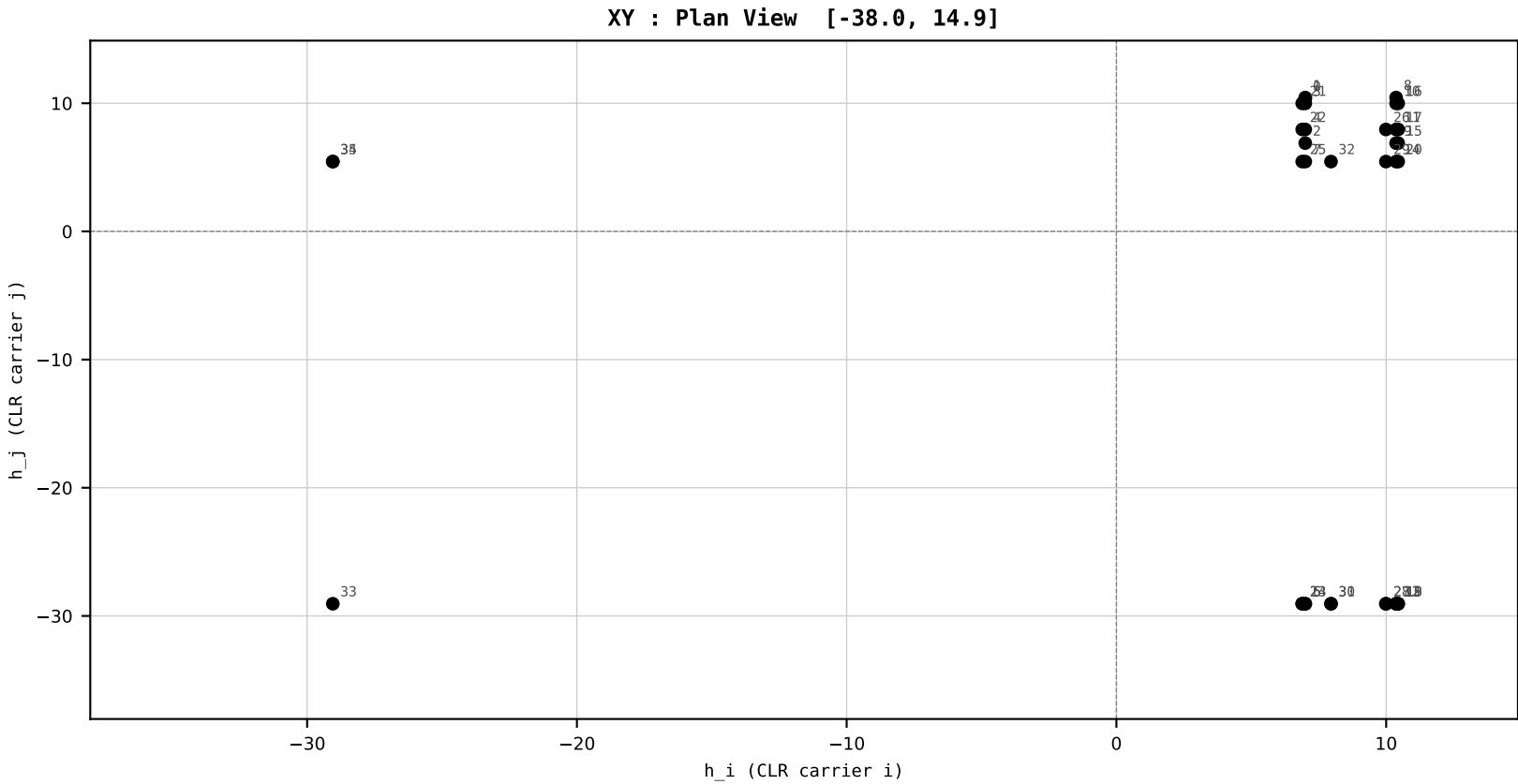
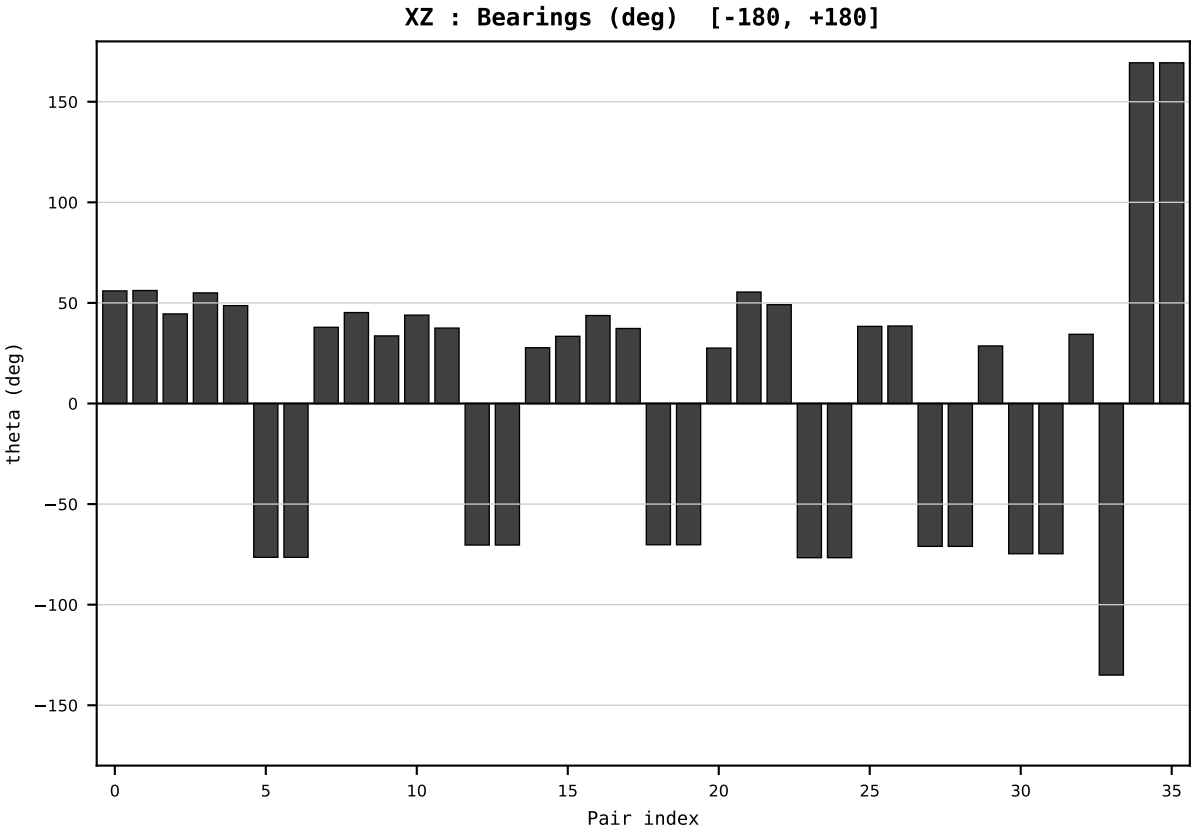
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2001
t : 1 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.409899
Ring = Hs-3
E_metric= 1556.9878
kappa_HS= 14190999999999984.00
omega = 0.4234 deg
d_A = 0.346528
Helm = Hydro
Helm d = -0.214796
DR = 39.494 HLR
DR ratio= 141910000176375712.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=2 | Year 2002 | D=9 N=26 pairs=36

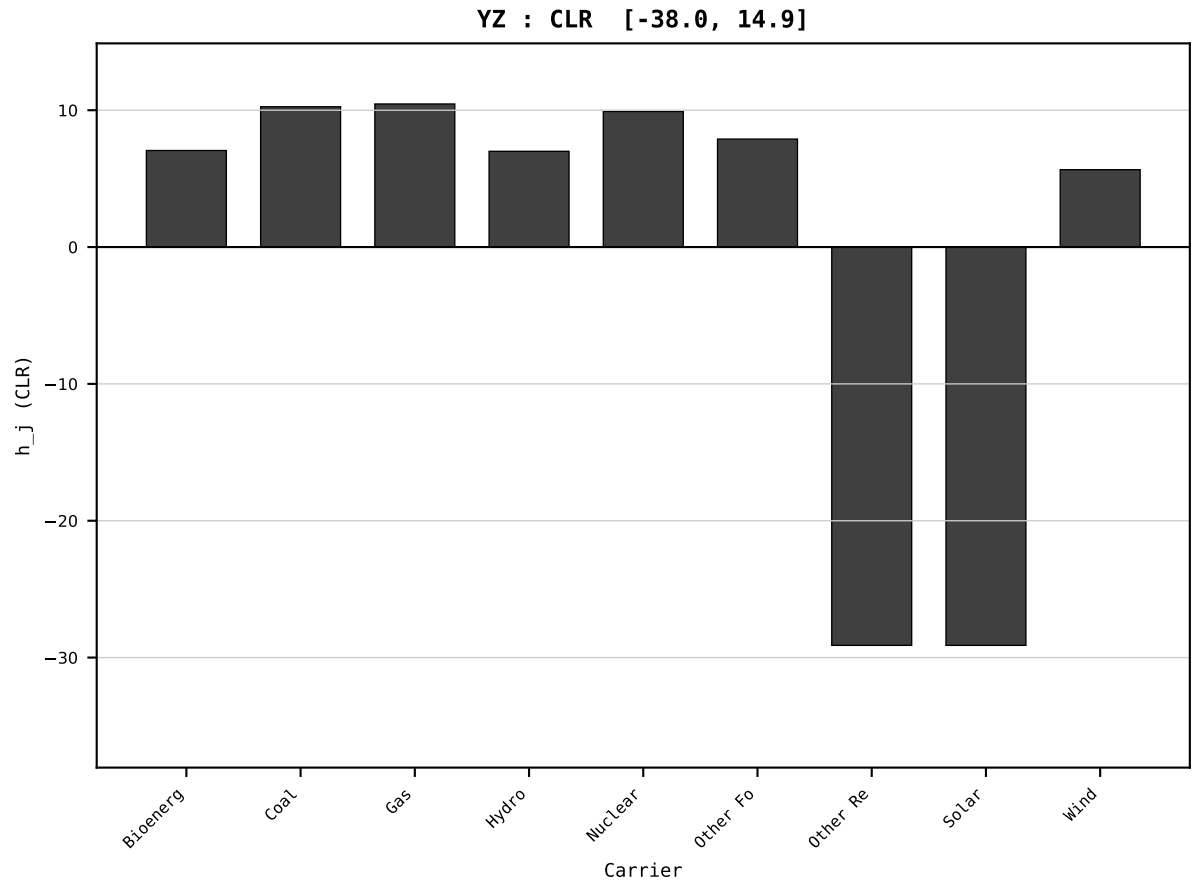
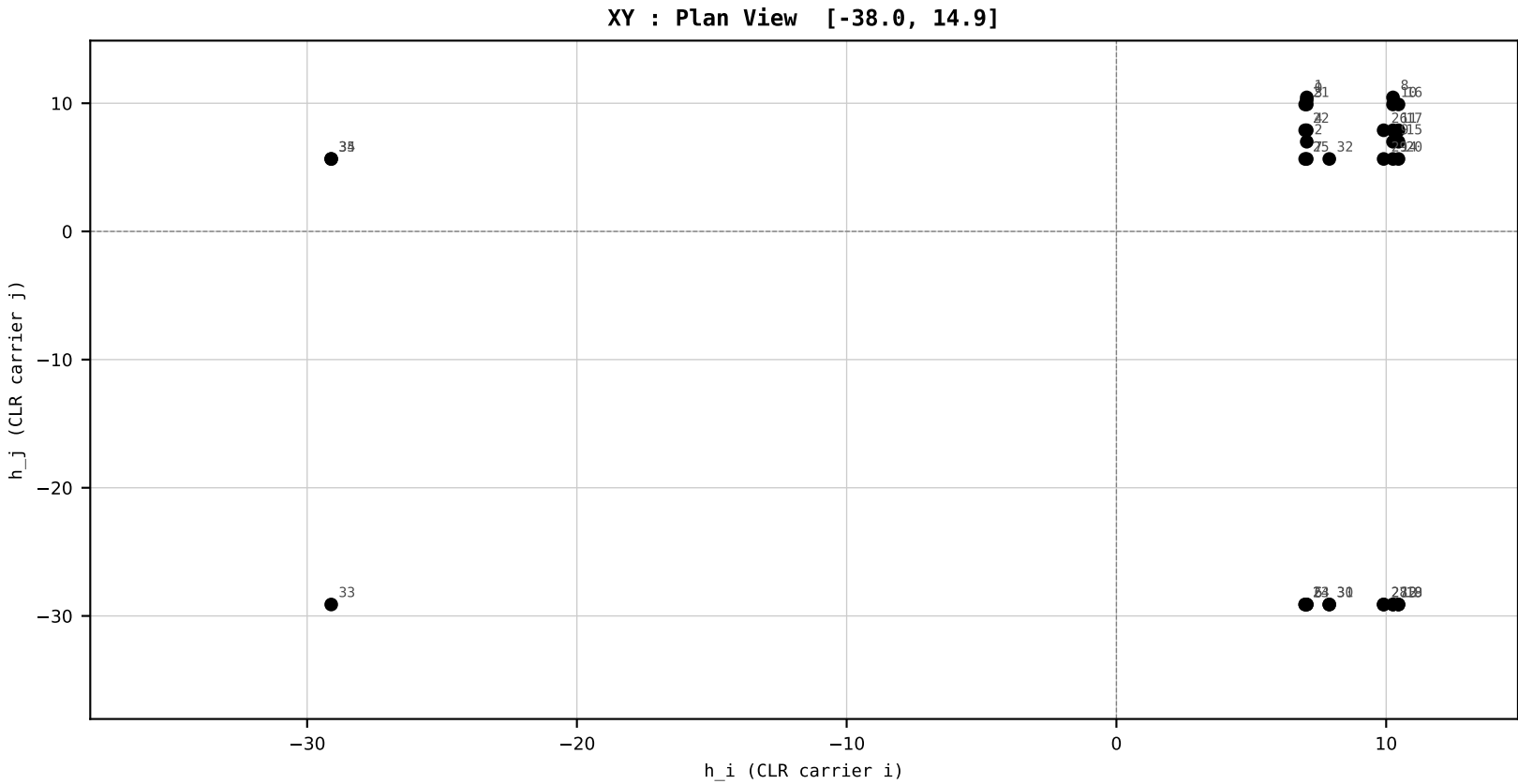
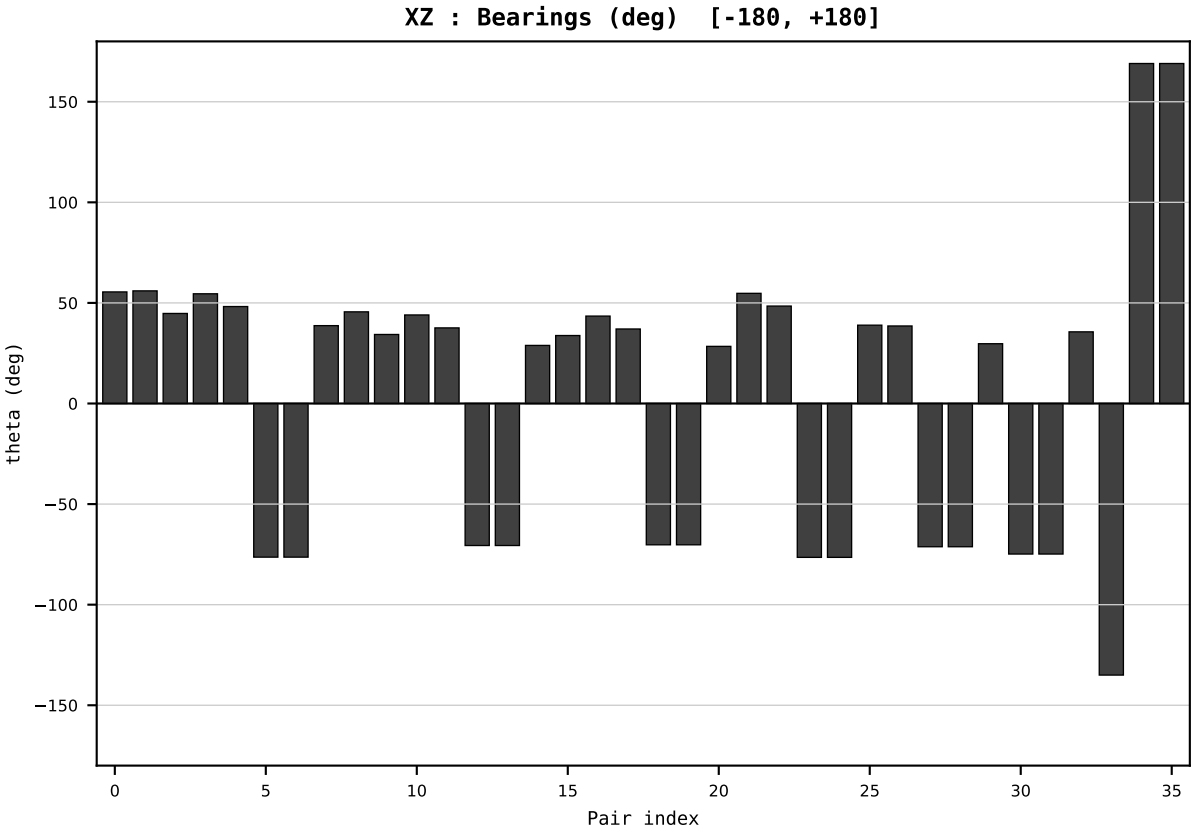
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2002
t : 2 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.406615
Ring = Hs-3
E_metric= 1561.5638
kappa_HS= 15228000000000000.00
omega = 0.3533 deg
d_A = 0.297482
Helm = Wind
Helm d = +0.205028
DR = 39.564 HLR
DR ratio= 152279999021603904.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=3 | Year 2003 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

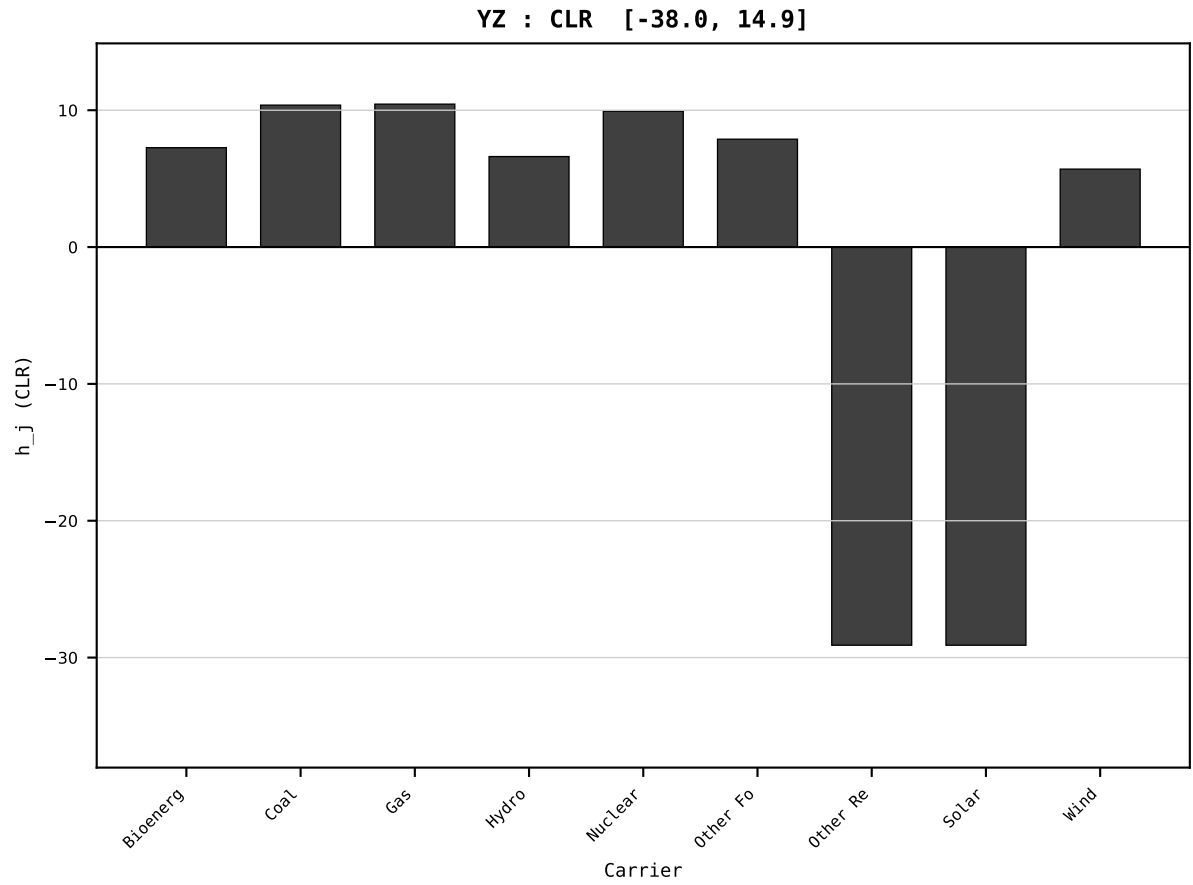
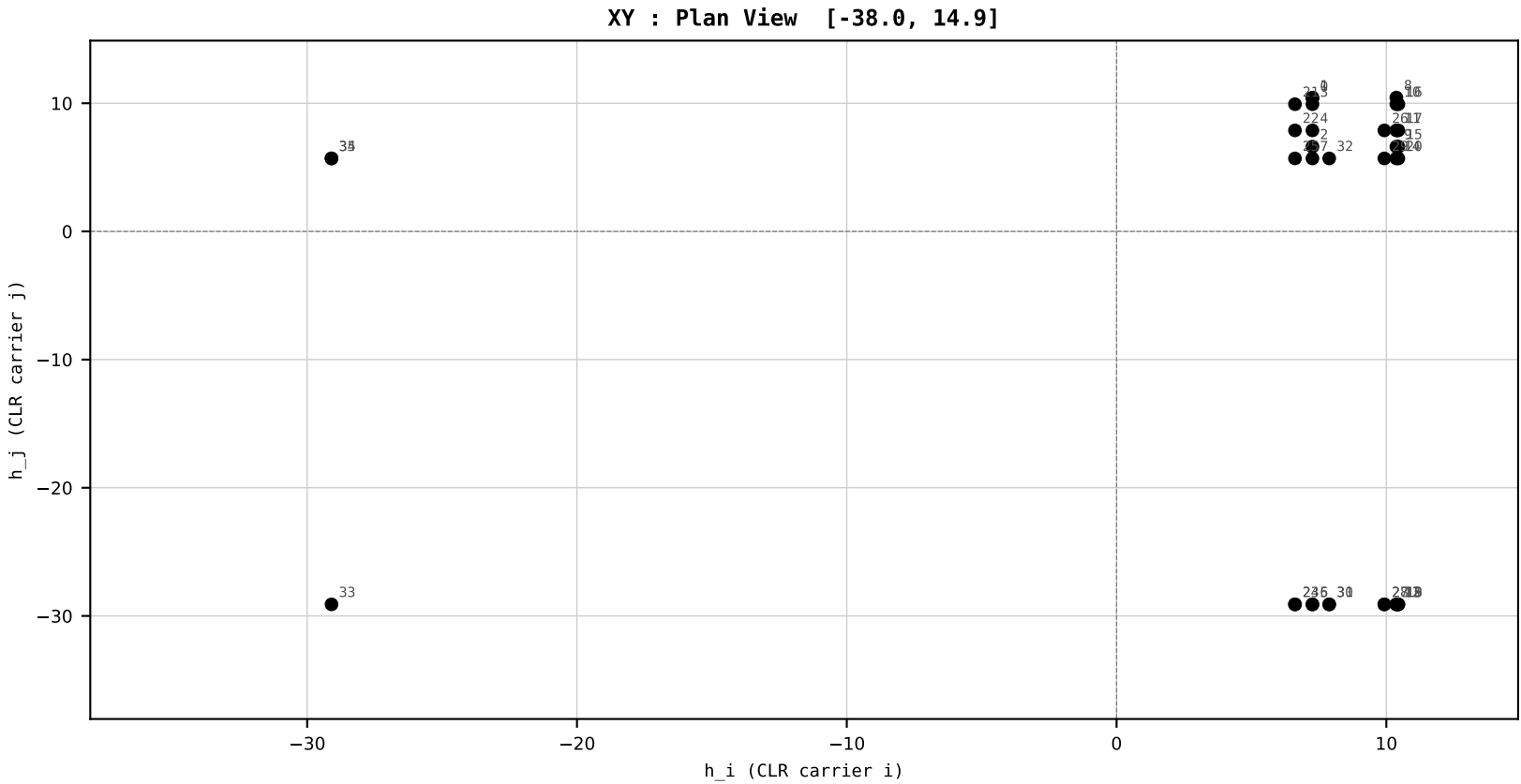
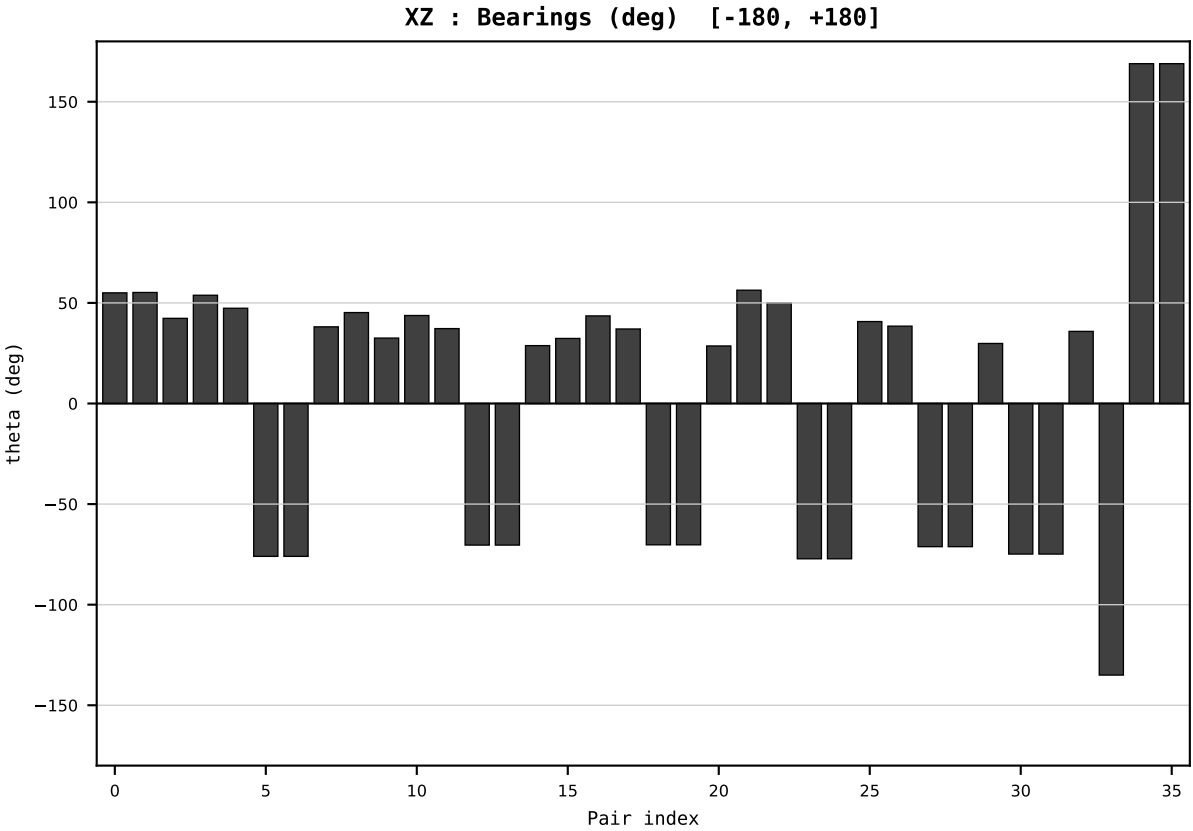
Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2003
t : 3 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.411016
Ring = Hs-3
E_metric= 1558.4988
kappa_HS= 148880000000000000.00
omega = 0.5541 deg
d_A = 0.453589
Helm = Hydro
Helm d = -0.383810
DR = 39.542 HLR
DR ratio= 148880000410211776.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=4 | Year 2004 | D=9 N=26 pairs=36

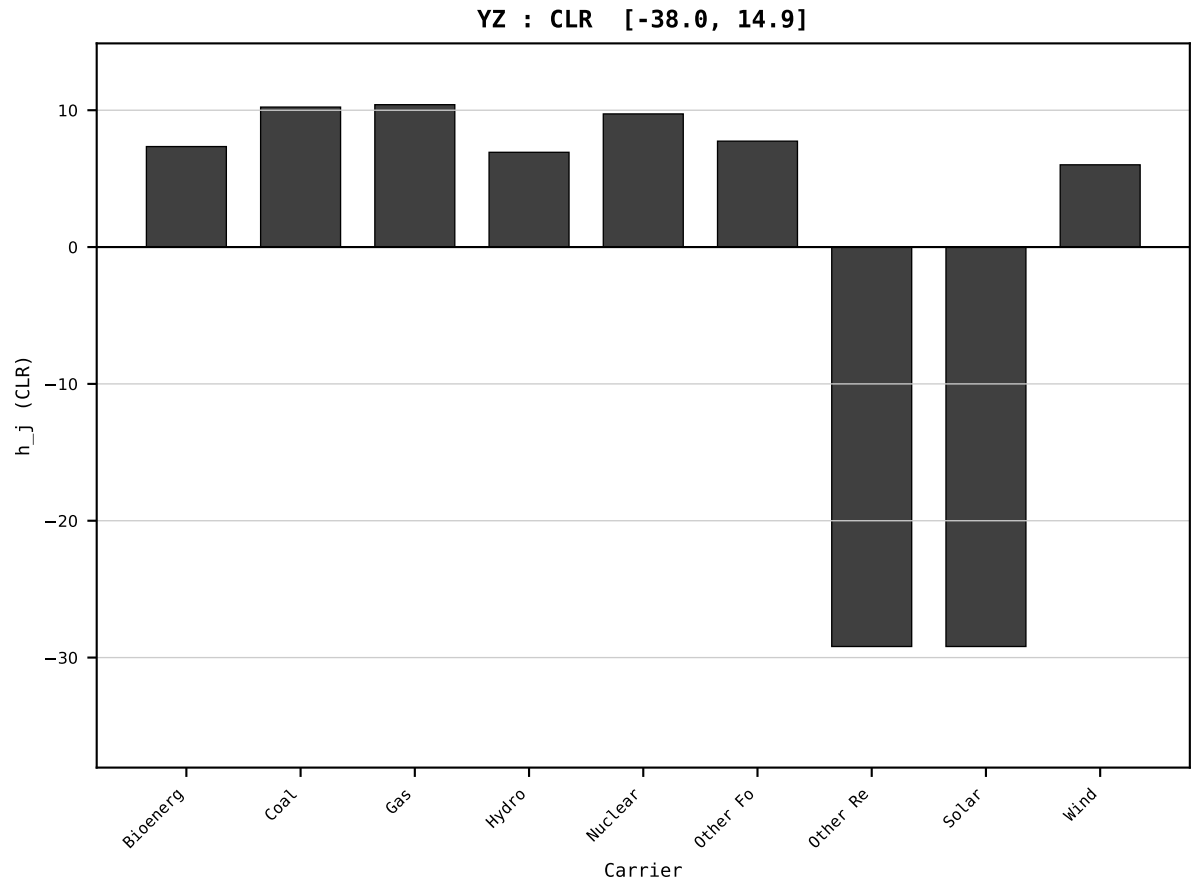
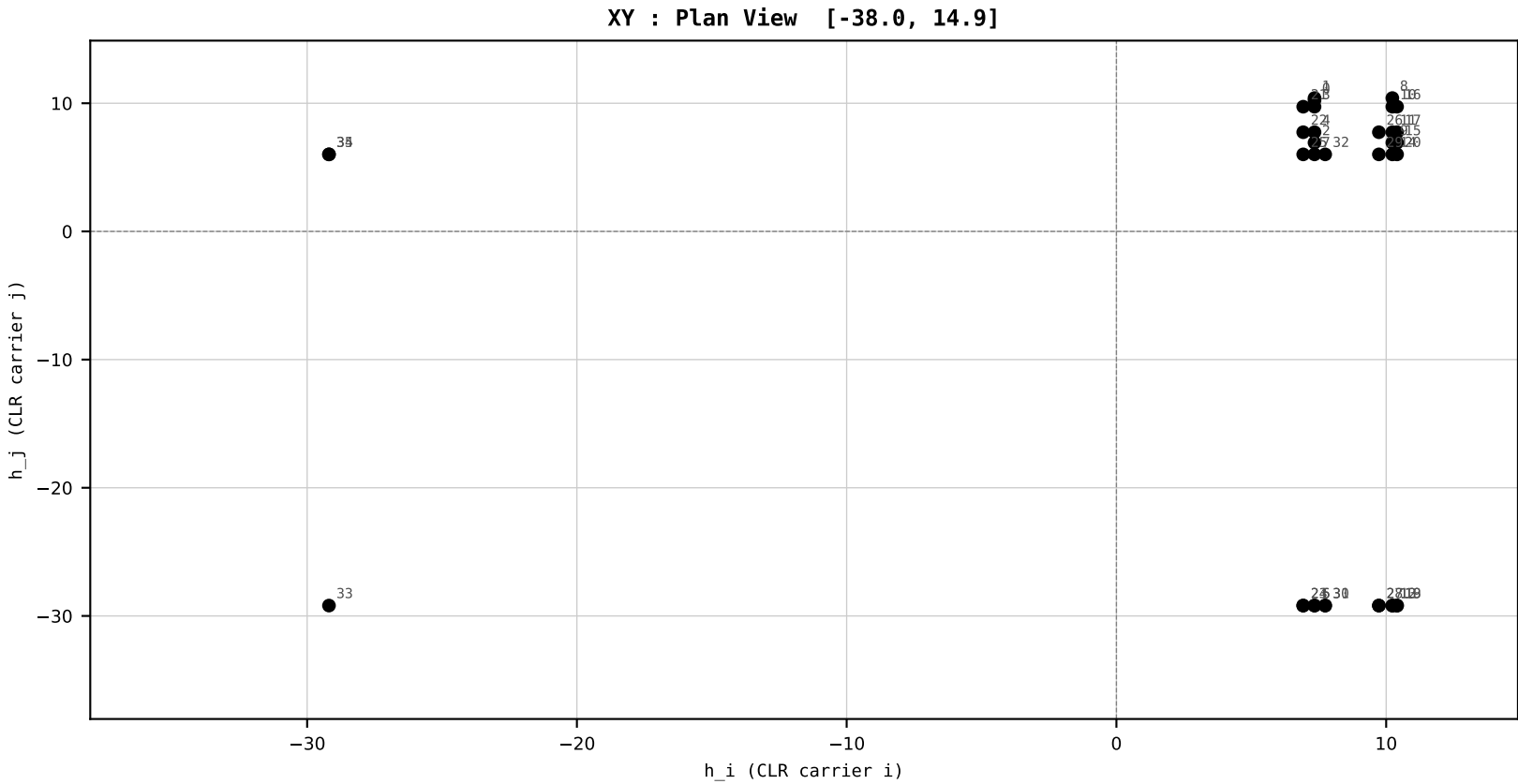
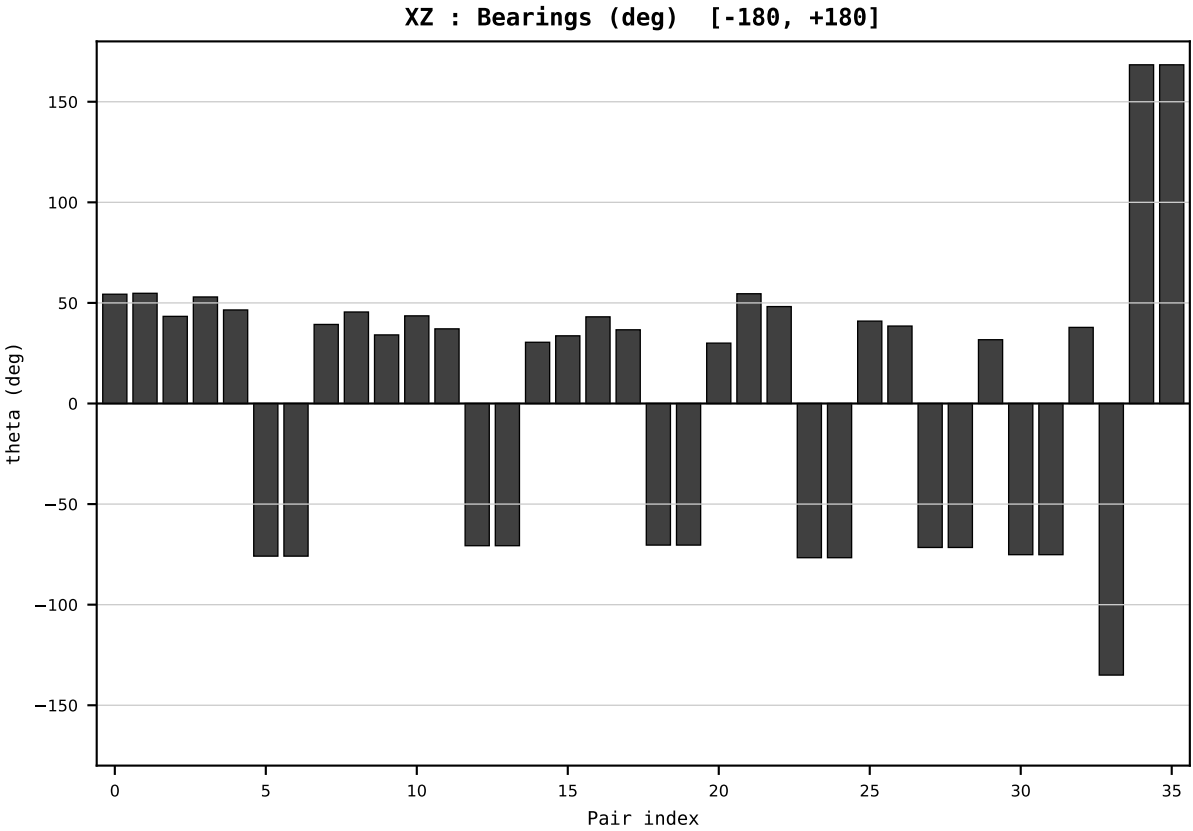
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2004
t : 4 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.403458
Ring = Hs-3
E_metric= 1567.4455
kappa_HS= 15706000000000000.00
omega = 0.6567 deg
d_A = 0.548838
Helm = Wind
Helm d = +0.314727
DR = 39.595 HLR
DR ratio= 157059999551214208.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=5 | Year 2005 | D=9 N=26 pairs=36

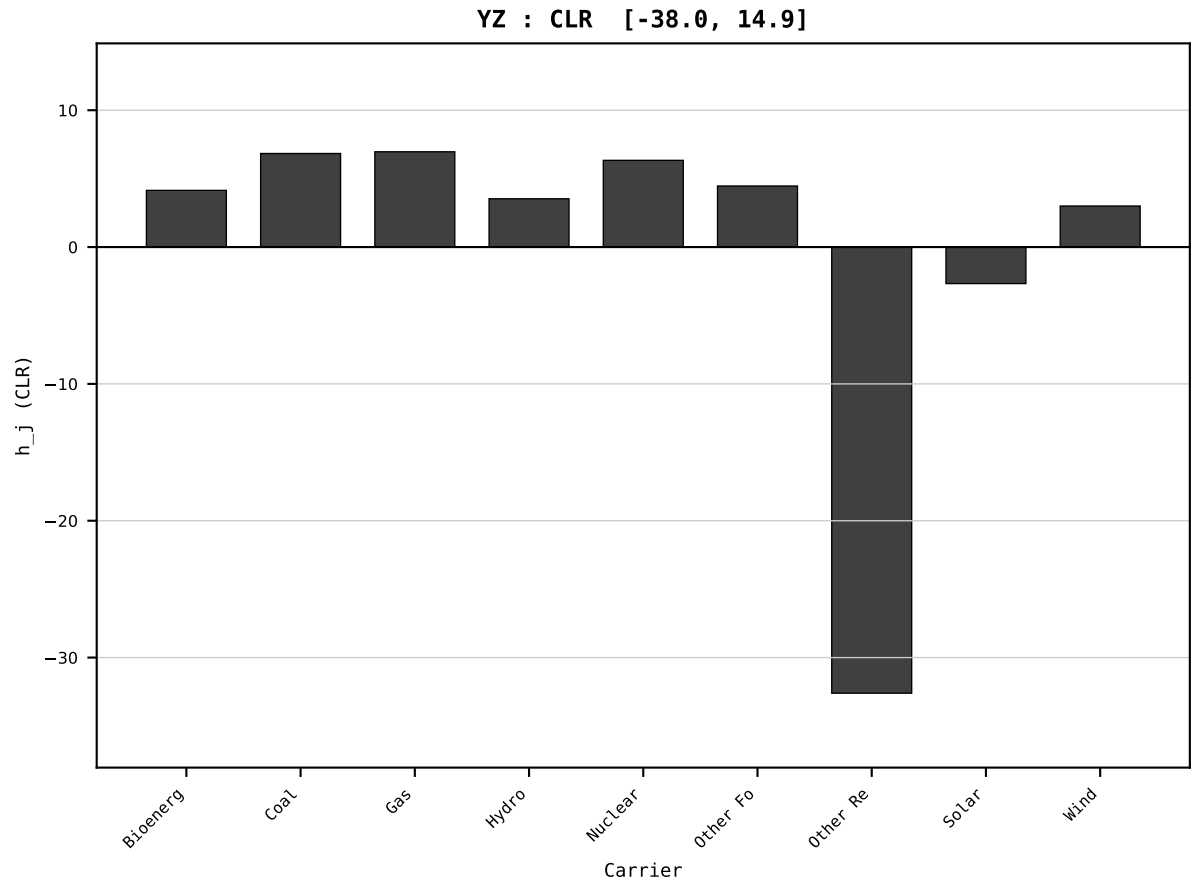
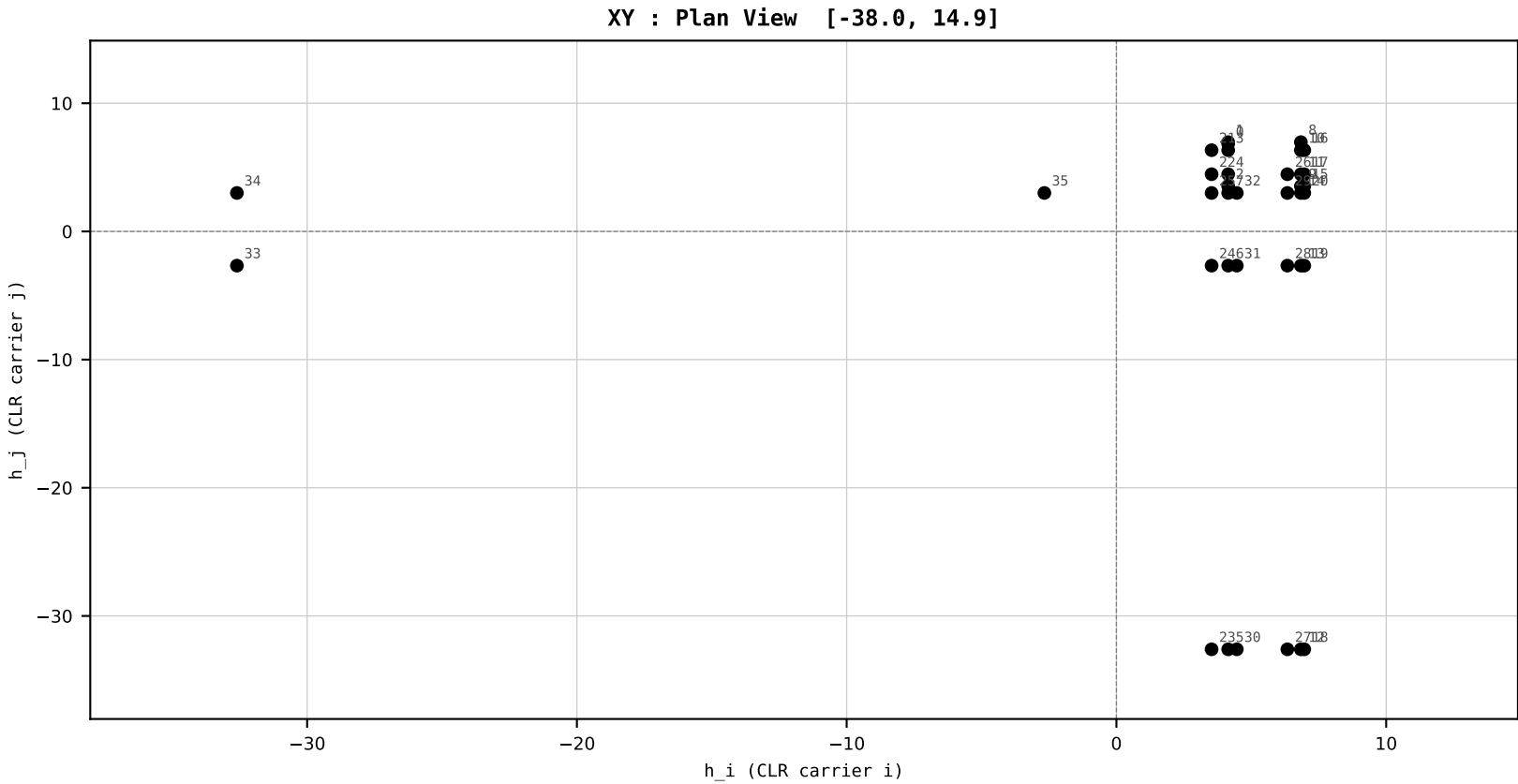
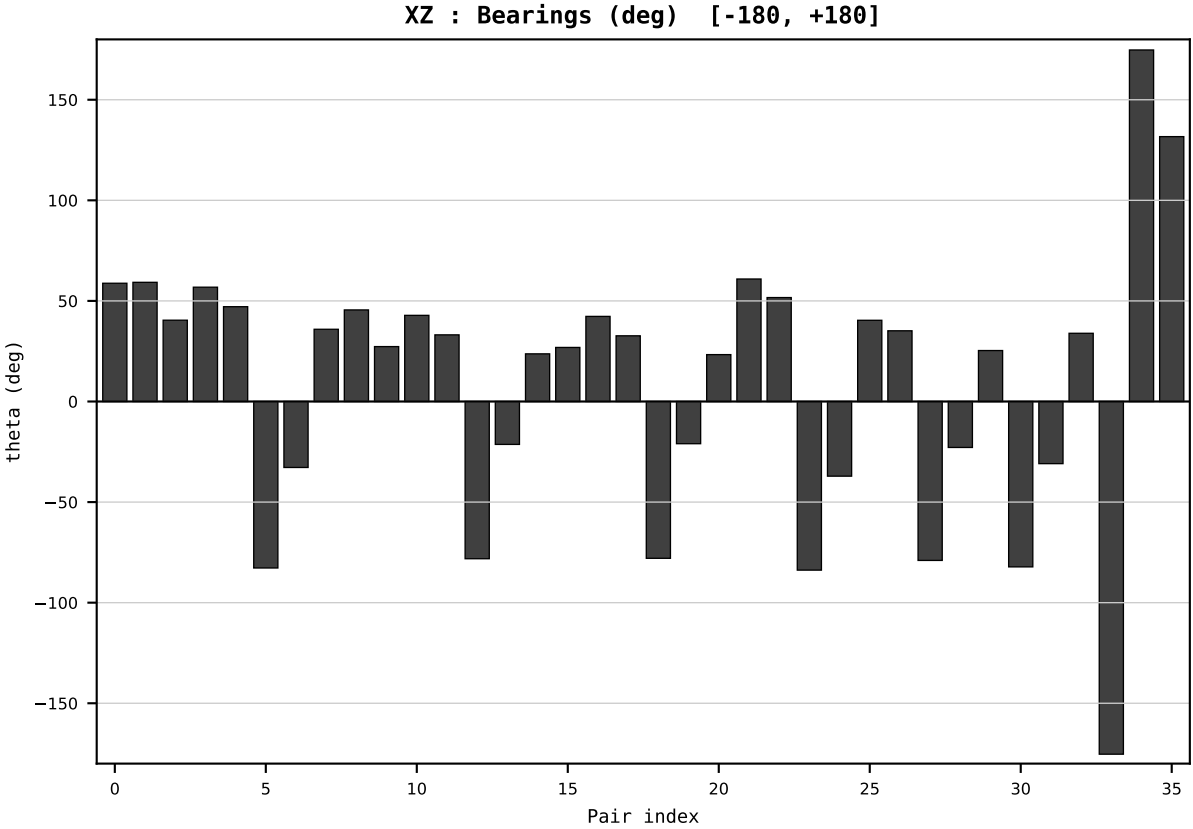
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2005
t : 5 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.388111
Ring = Hs-3
E_metric= 905.5074
kappa_HS= 15264000000000000.00
omega = 36.6332 deg
d_A = 28.132514
Helm = Solar
Helm d = +26.521056
DR = 39.567 HLR
DR ratio= 152640001127908160.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=6 | Year 2006 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

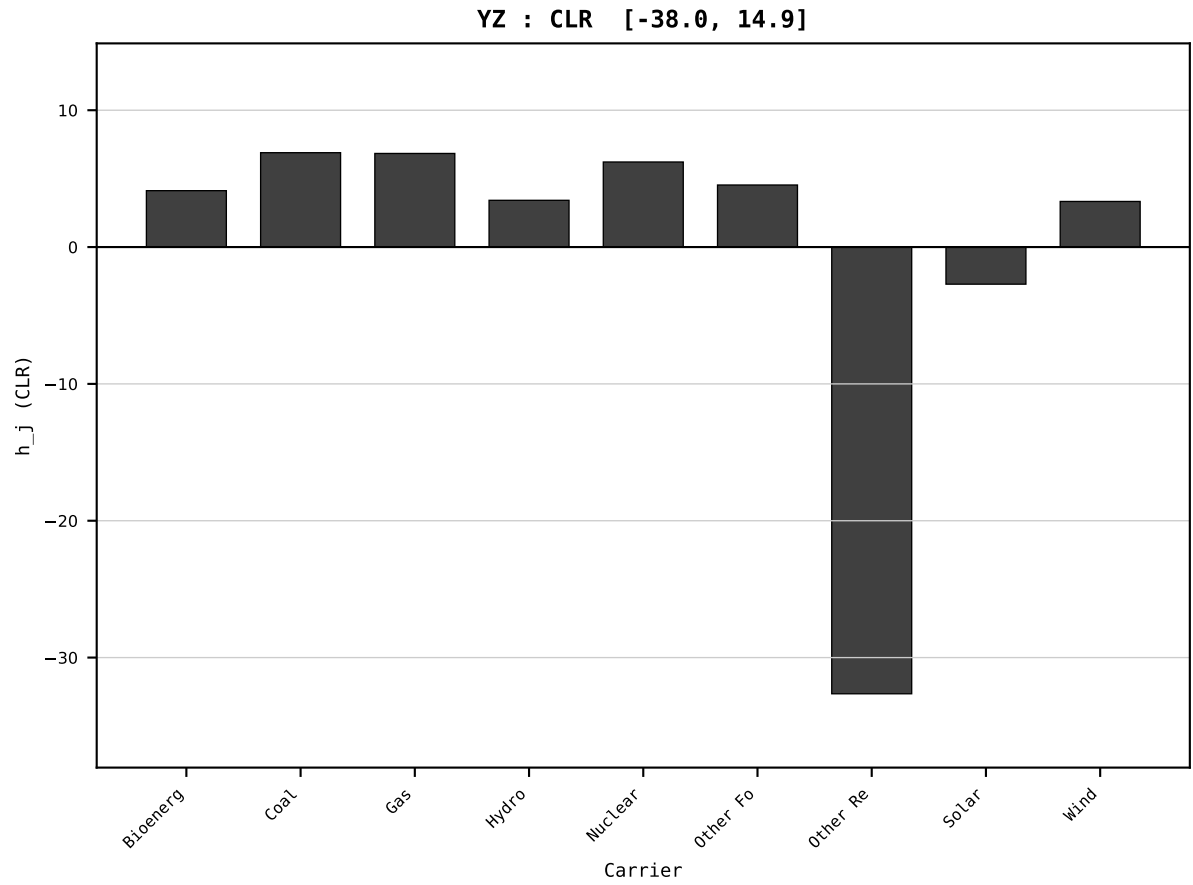
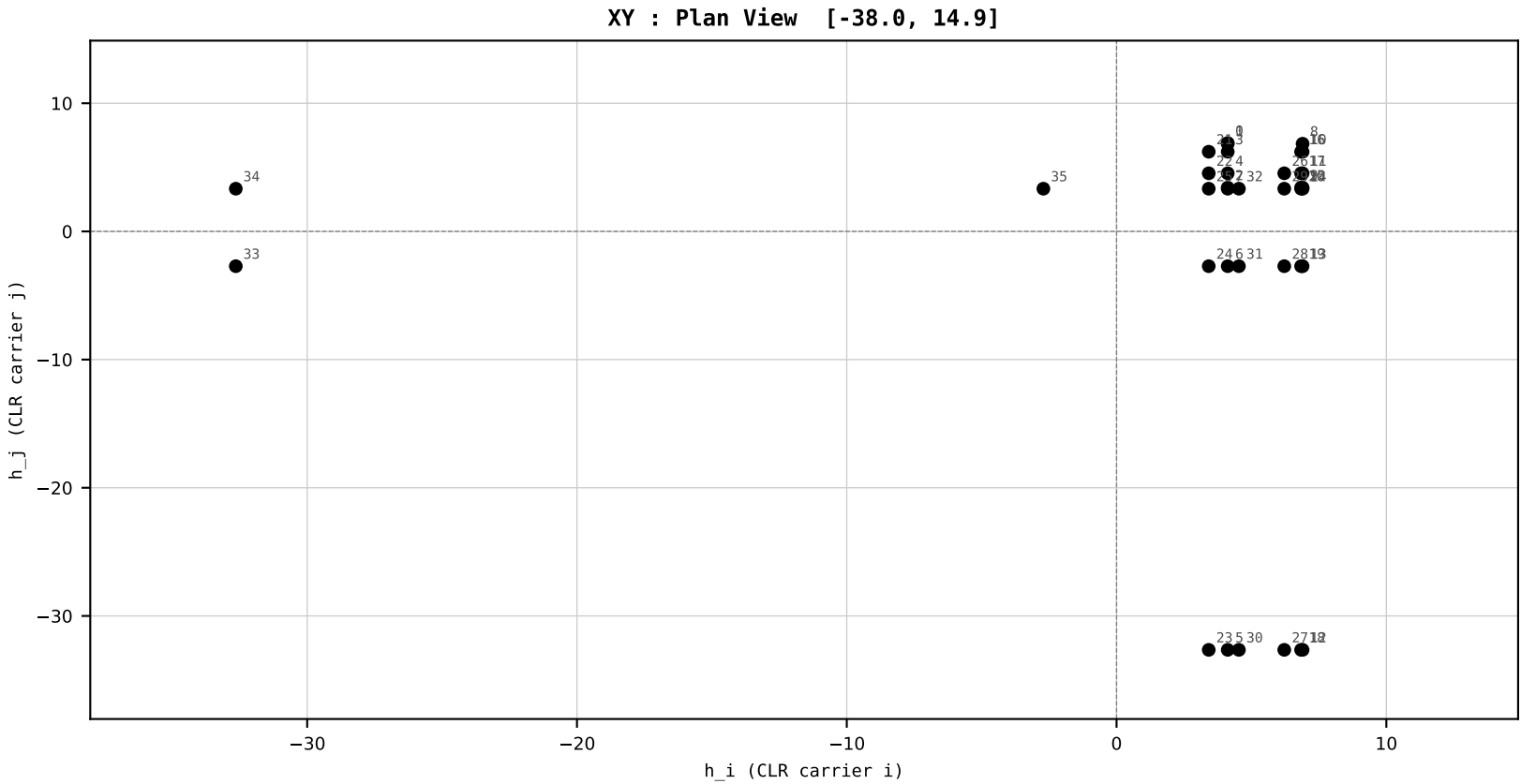
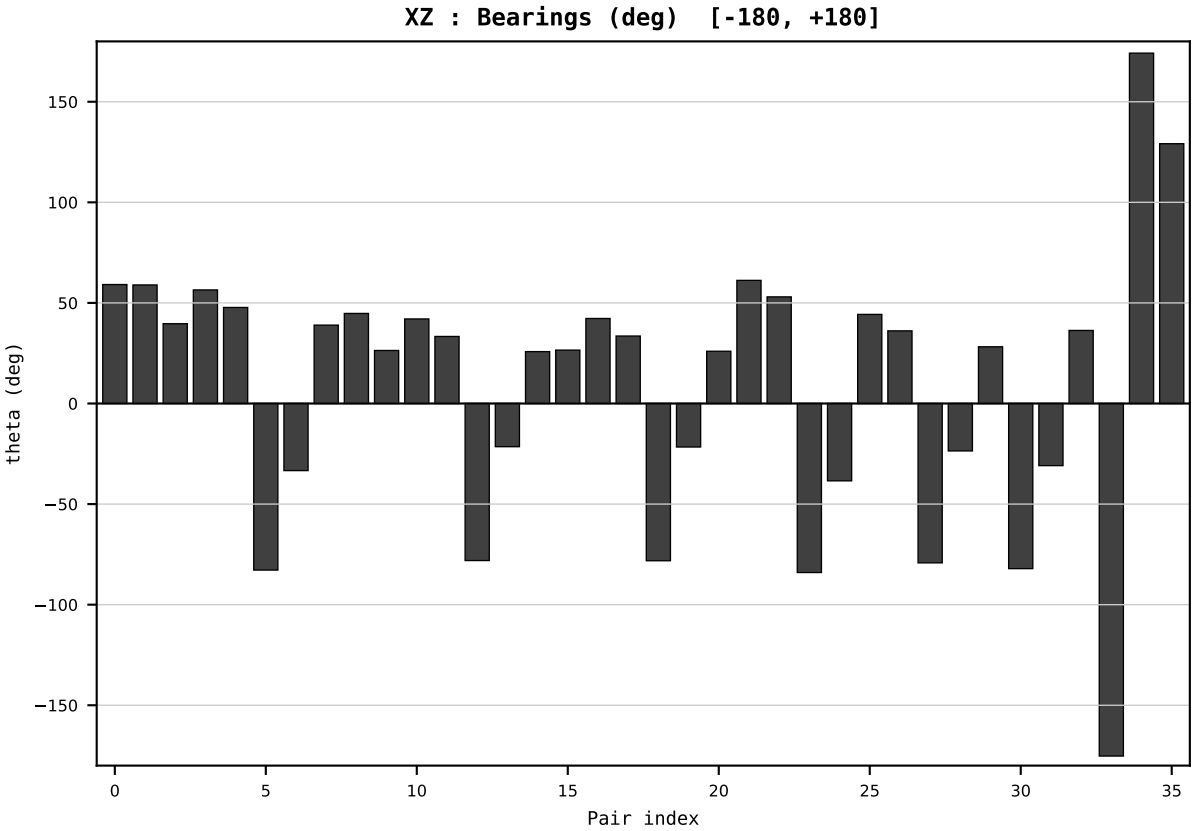
Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2006
t : 6 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.382444
Ring = Hs-3
E_metric= 907.5146
kappa_HS= 14885000000000000.00
omega = 0.6534 deg
d_A = 0.406998
Helm = Wind
Helm d = +0.332784
DR = 39.542 HLR
DR ratio= 148849999646857824.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=7 | Year 2007 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

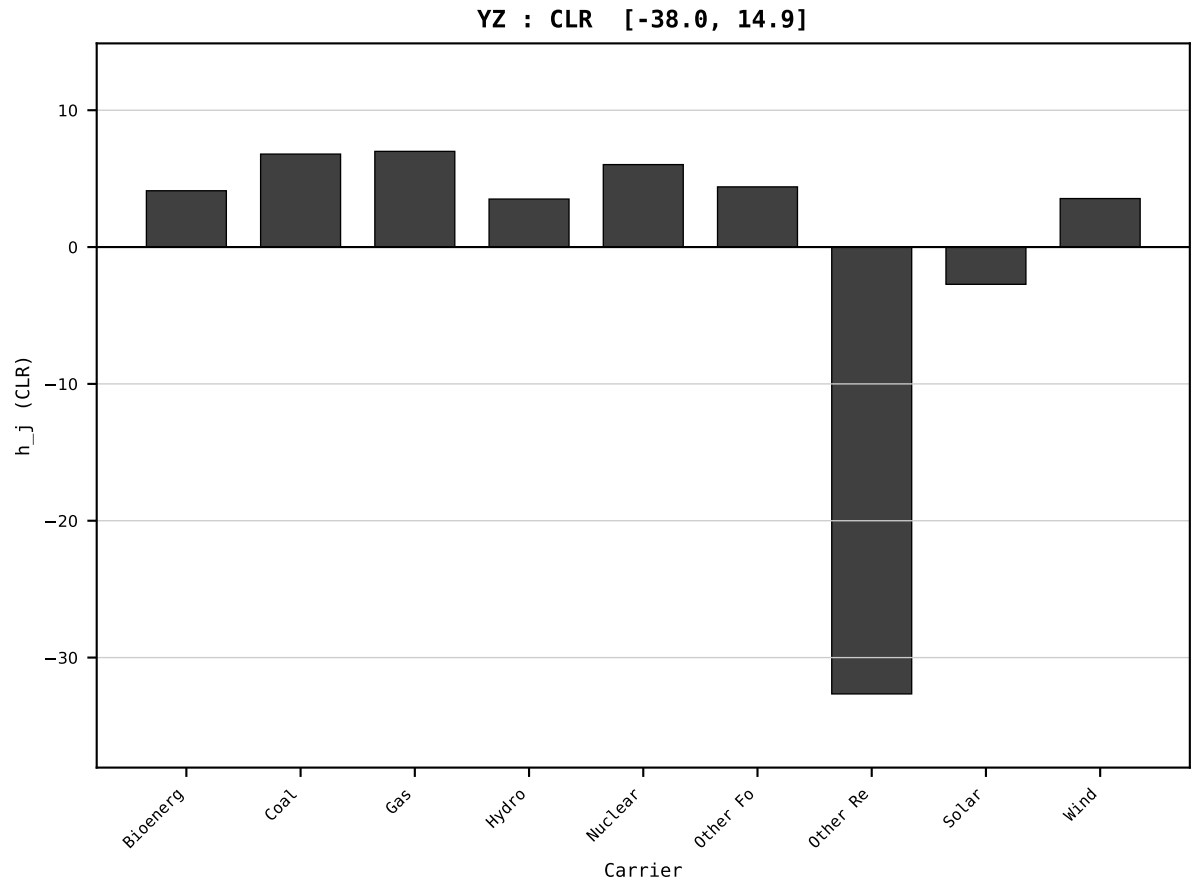
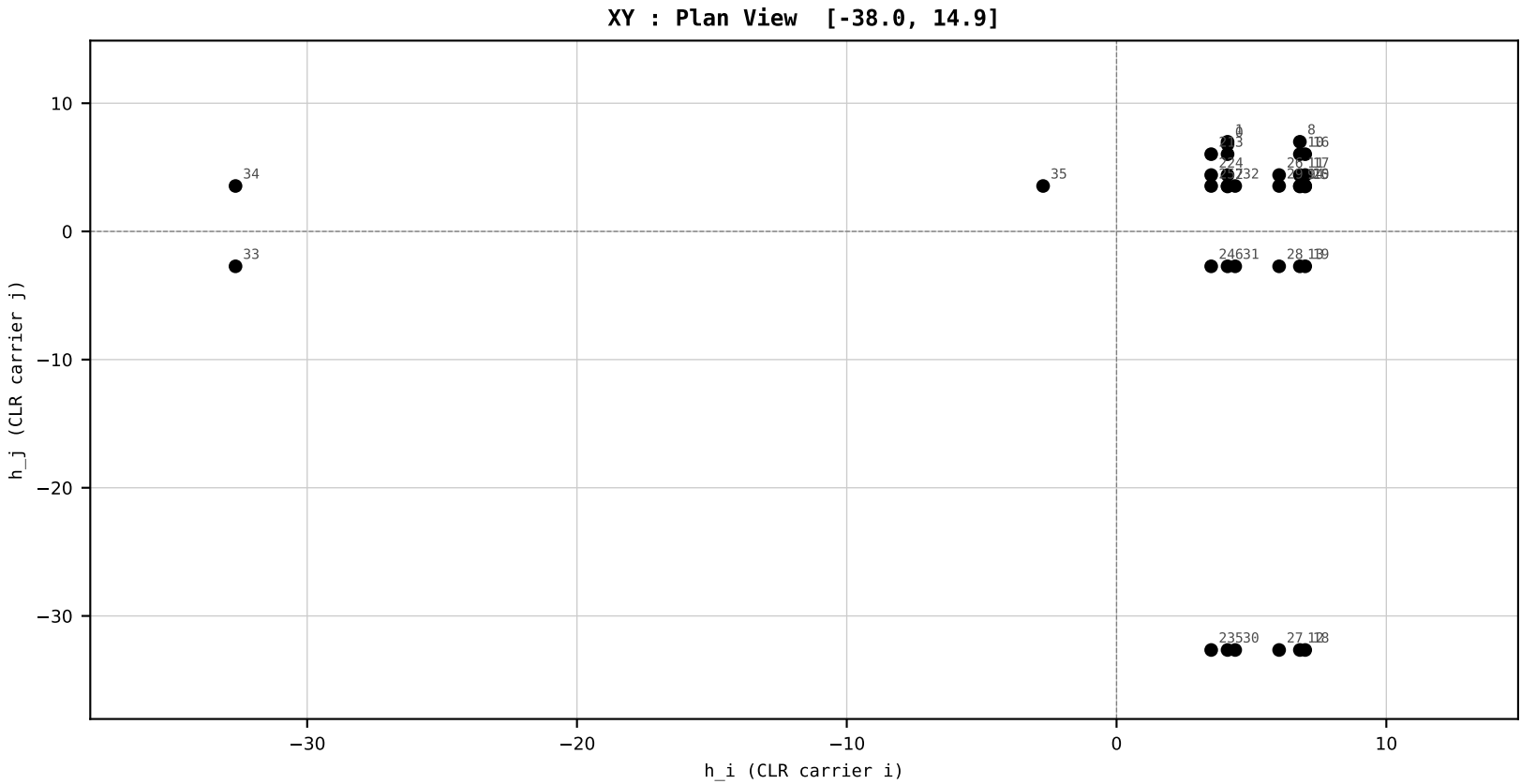
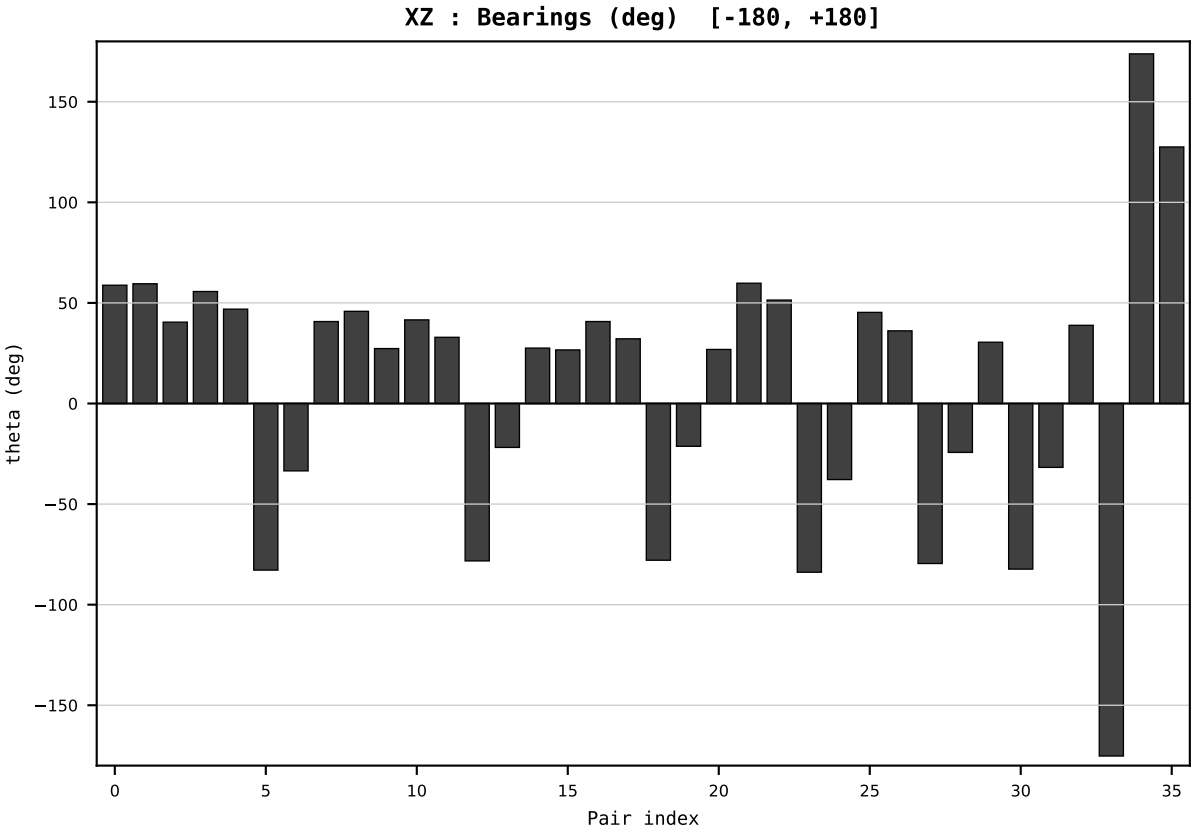
Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2007
t : 7 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.393078
Ring = Hs-3
E_metric= 907.4010
kappa_HS= 16578999999999968.00
omega = 0.6066 deg
d_A = 0.376749
Helm = Wind
Helm d = +0.211765
DR = 39.649 HLR
DR ratio= 165789999641552096.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=8 | Year 2008 | D=9 N=26 pairs=36

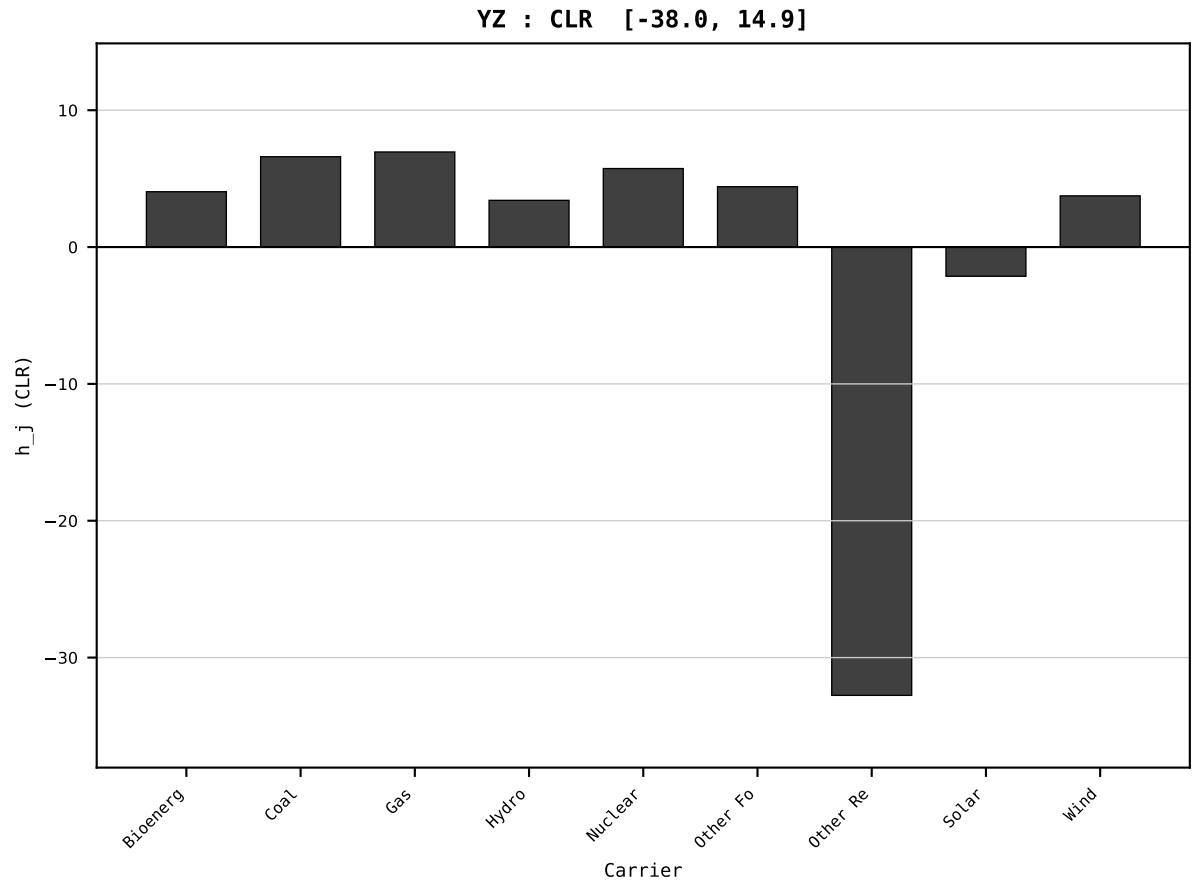
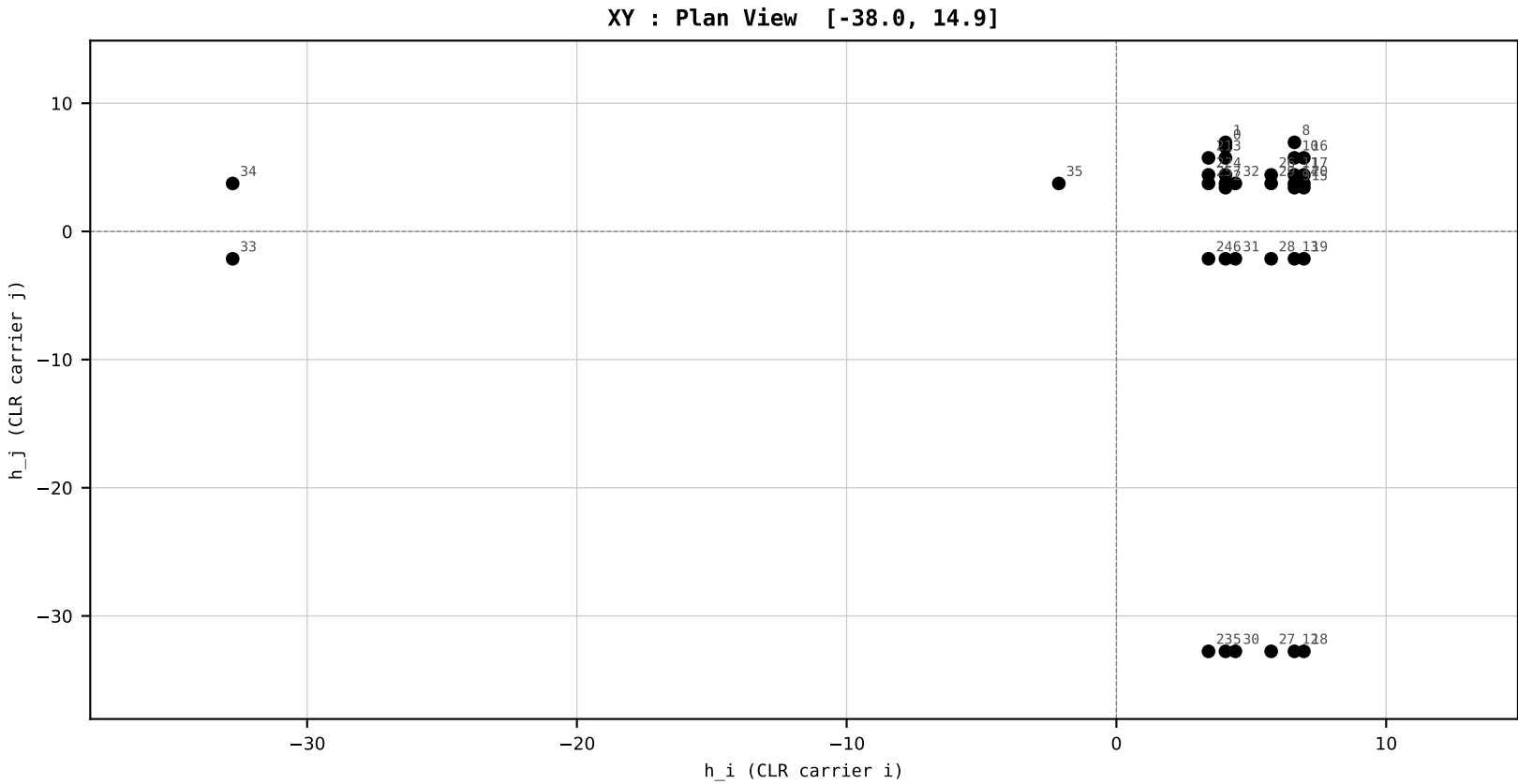
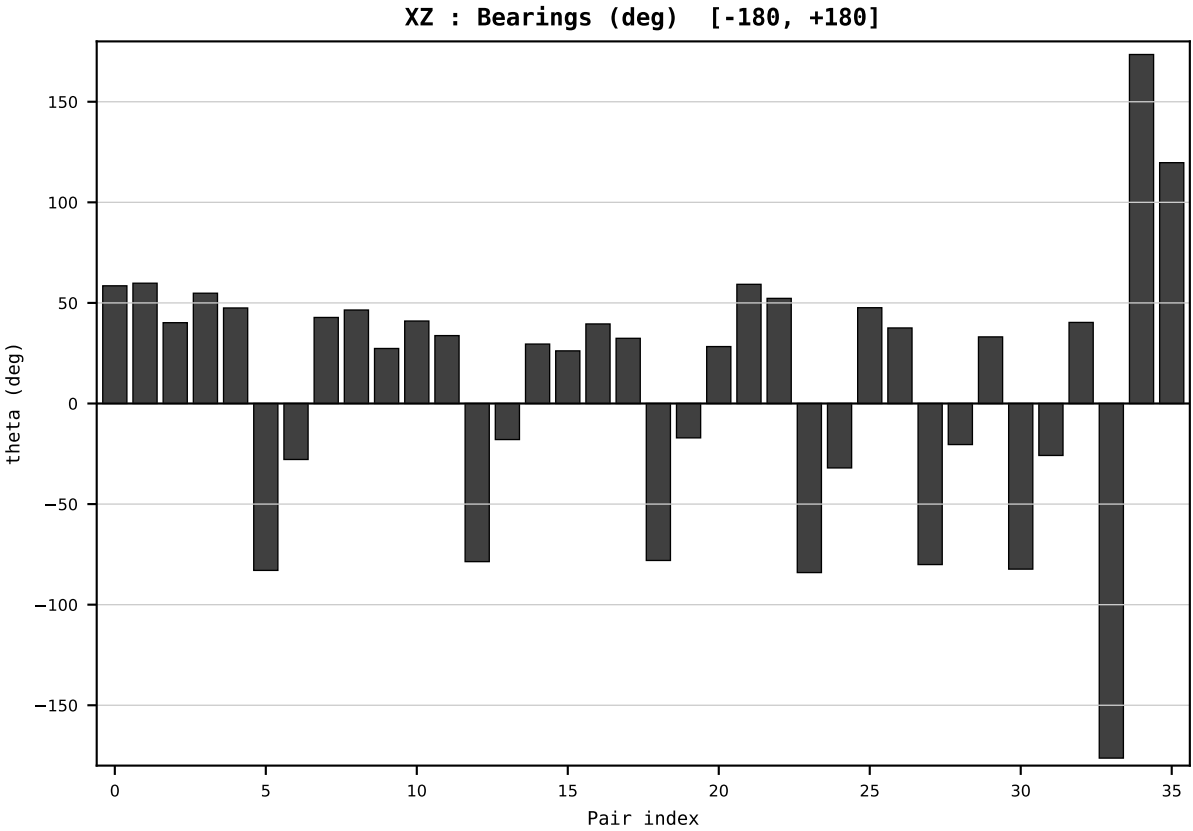
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2008
t : 8 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.392183
Ring = Hs-3
E_metric= 904.9892
kappa_HS= 17622000000000000.00
omega = 1.1739 deg
d_A = 0.729475
Helm = Solar
Helm d = +0.587604
DR = 39.711 HLR
DR ratio= 176220000114672288.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

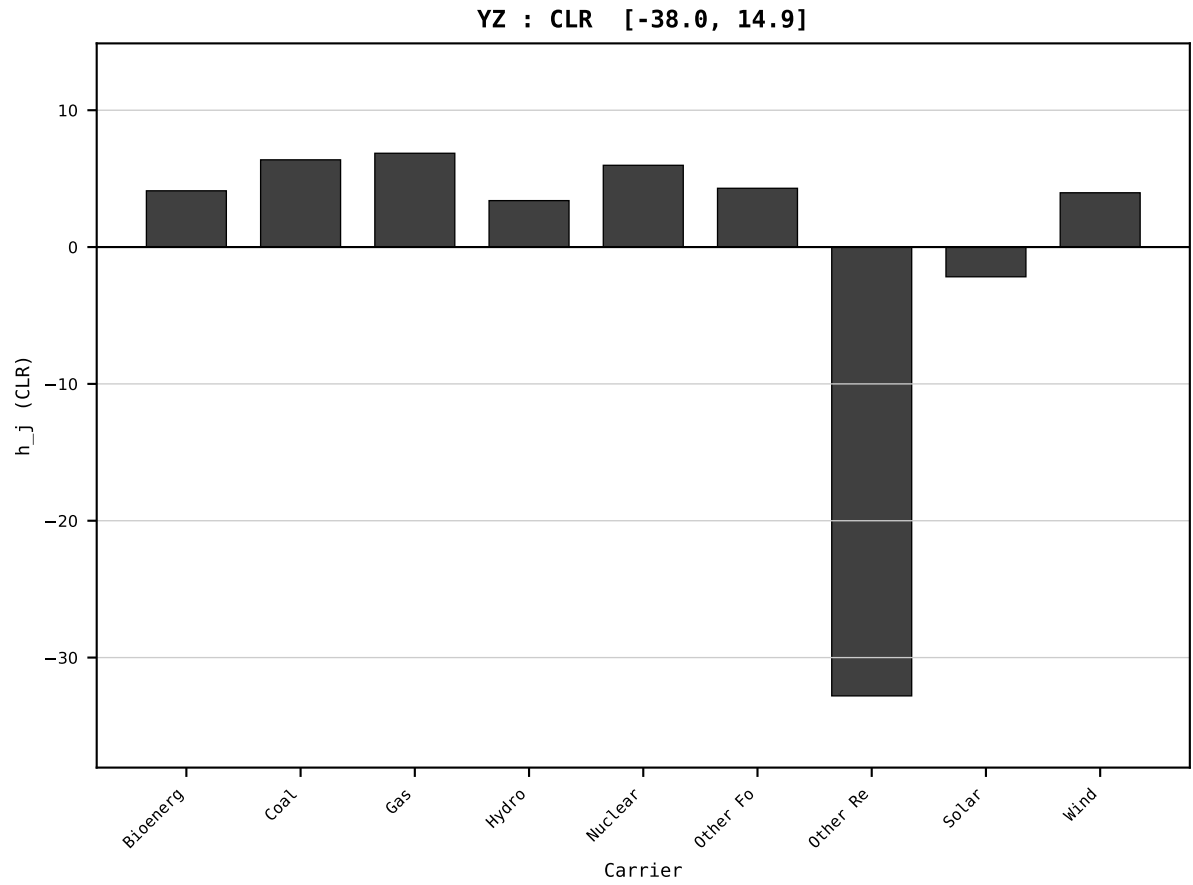
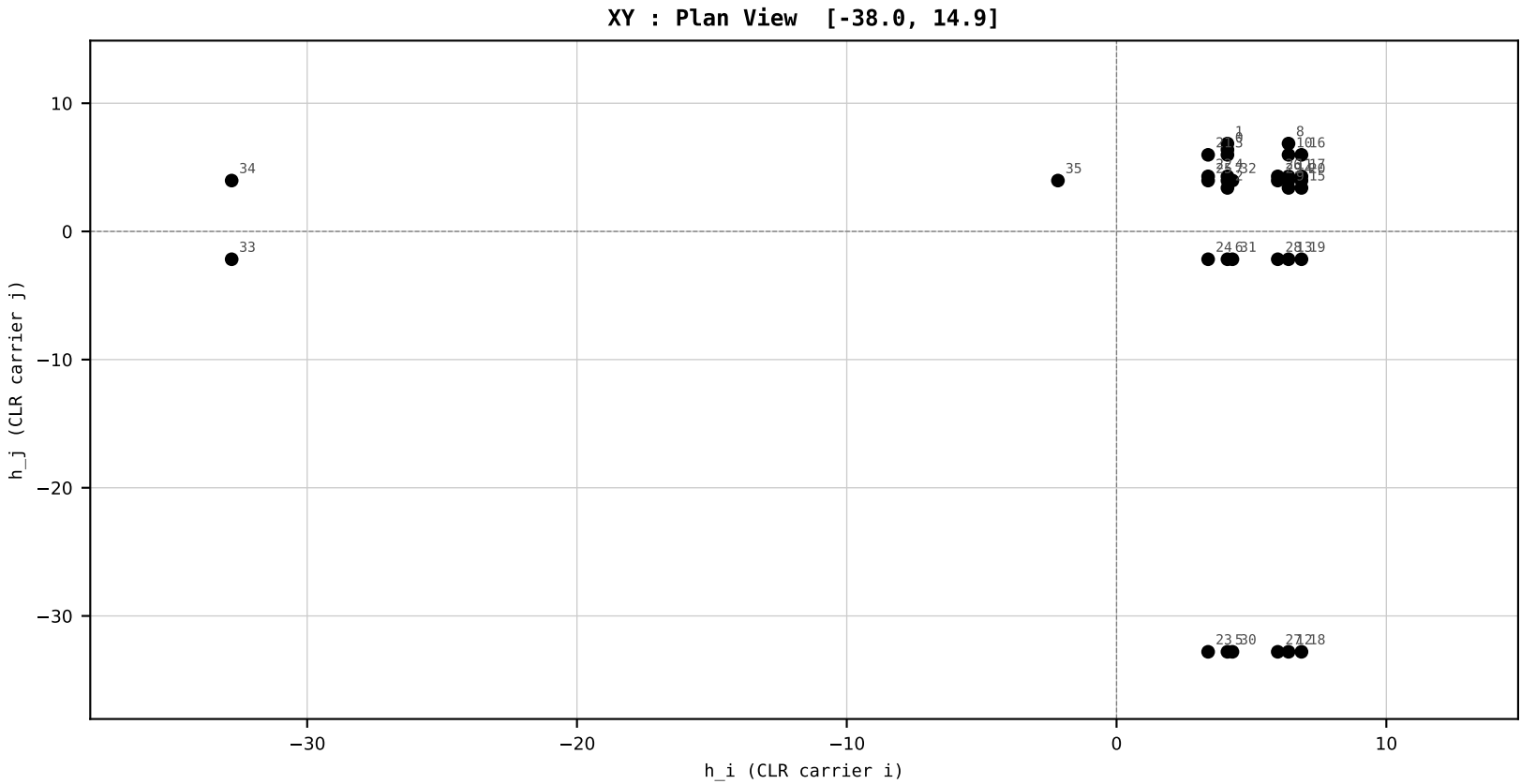
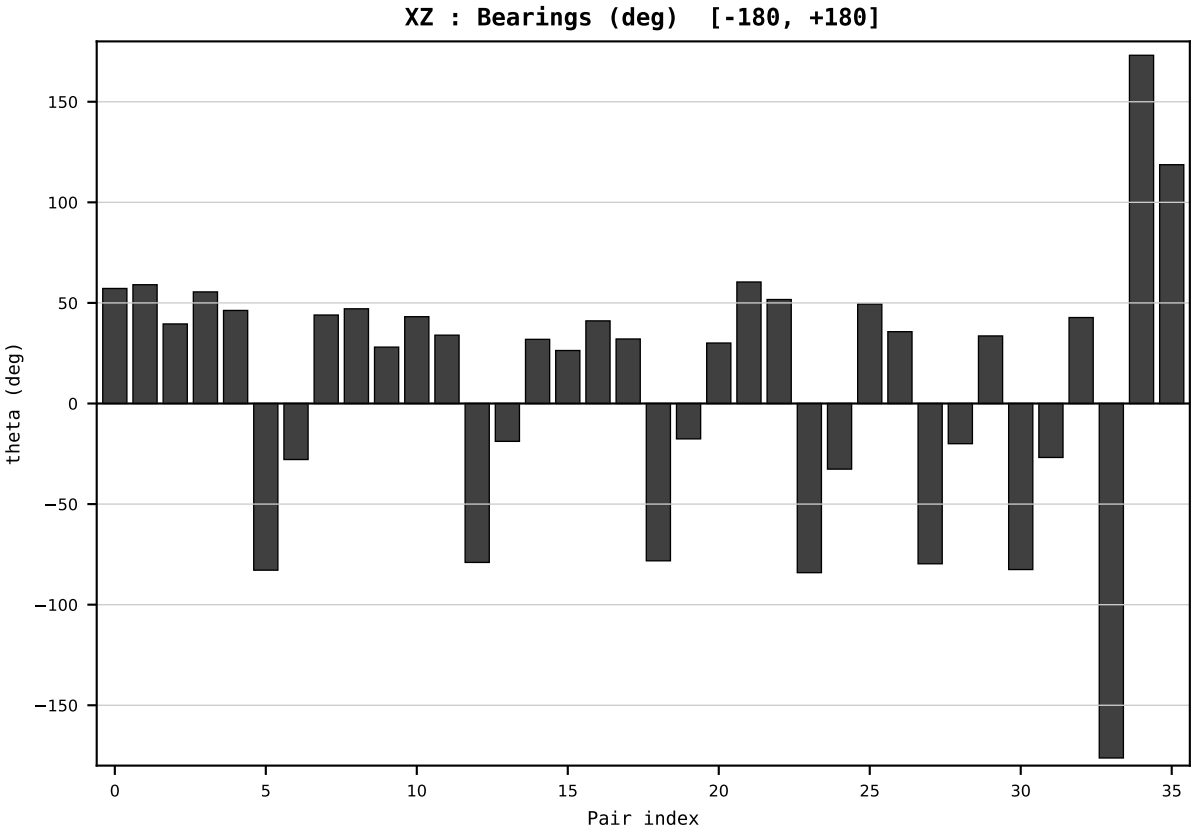
Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2009
t : 9 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.365440
Ring = Hs-3
E_metric= 908.6087
kappa_HS= 16650000000000000.00
omega = 0.6972 deg
d_A = 0.434051
Helm = Nuclear
Helm d = +0.237518
DR = 39.654 HLR
DR ratio= 166499999278857920.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=10 | Year 2010 | D=9 N=26 pairs=36

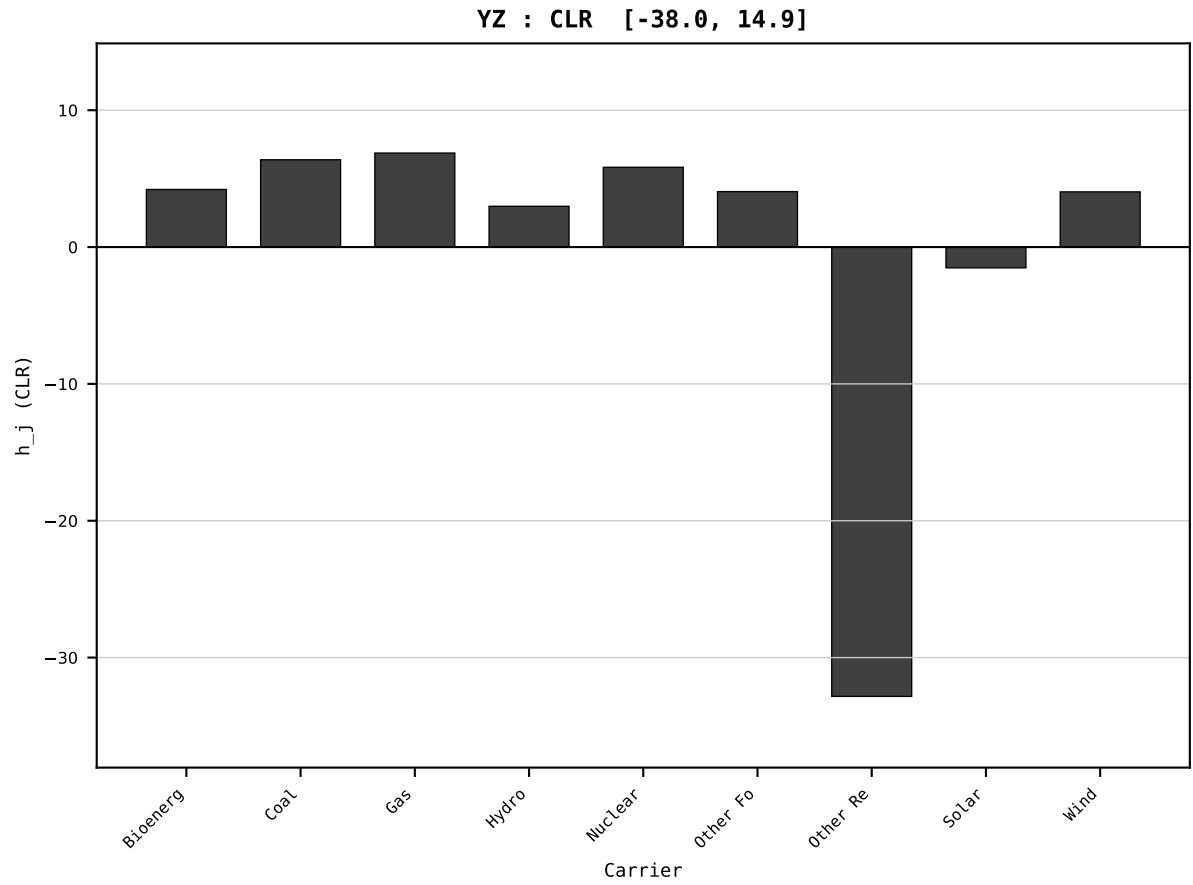
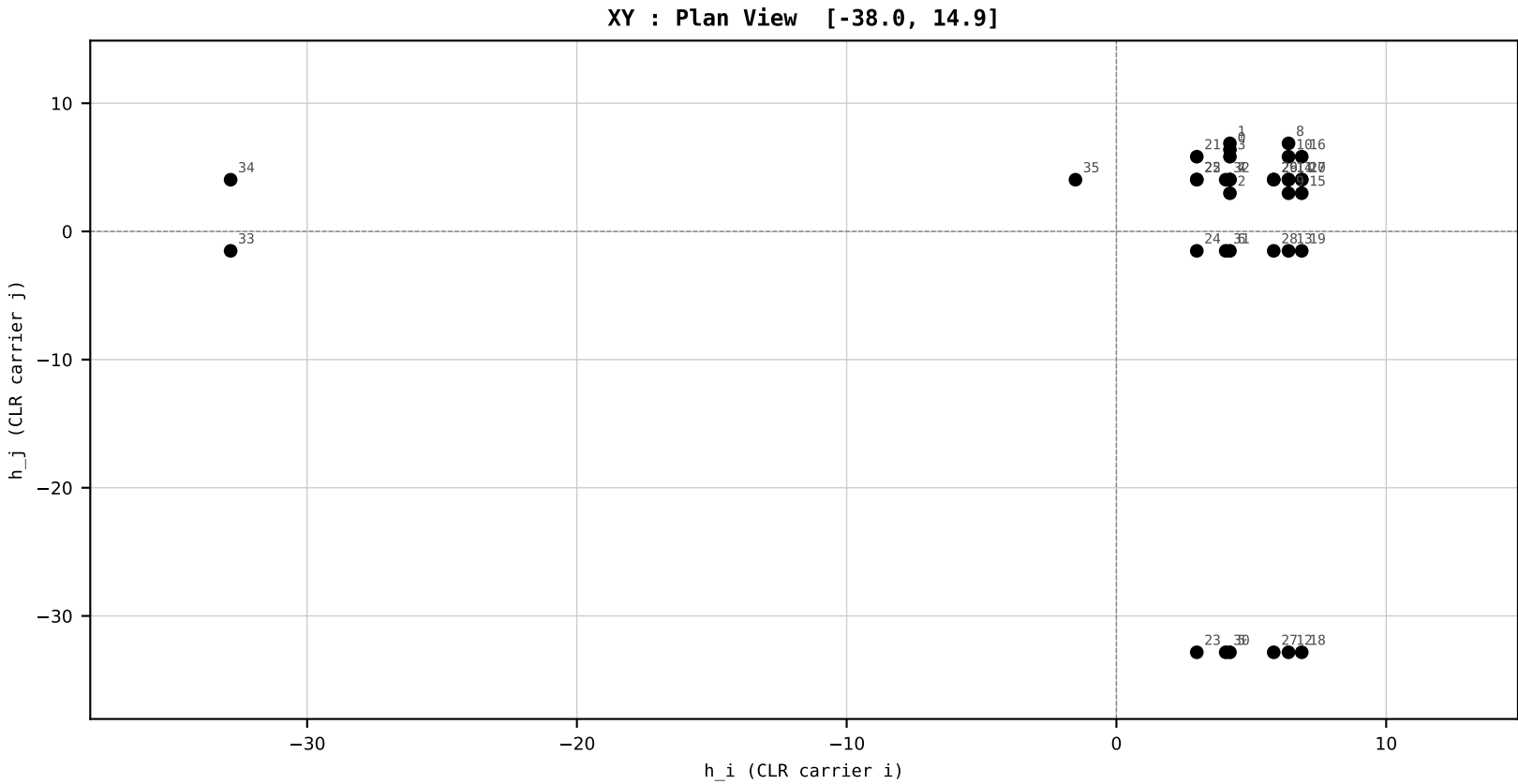
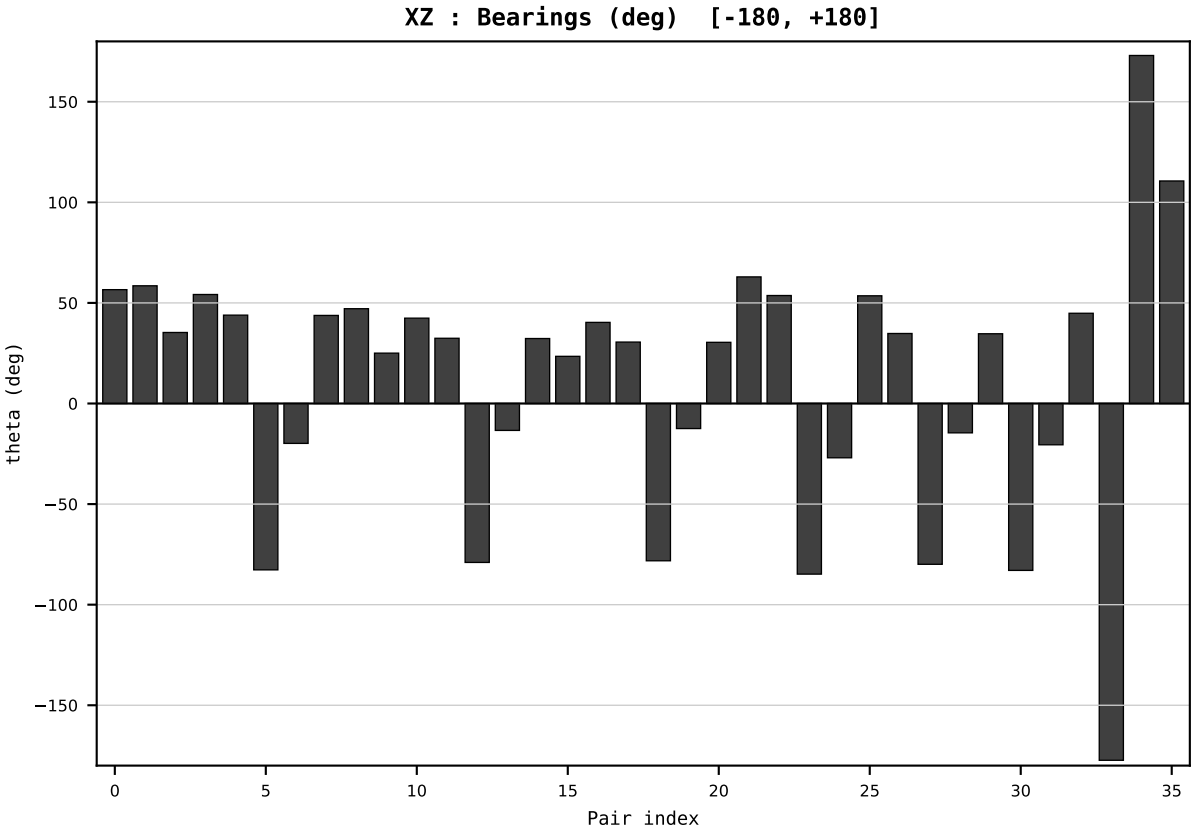
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2010
t : 10 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.380664
Ring = Hs-3
E_metric= 902.7101
kappa_HS= 17565000000000000.00
omega = 1.3400 deg
d_A = 0.834110
Helm = Solar
Helm d = +0.655191
DR = 39.707 HLR
DR ratio= 175649999363704736.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=11 | Year 2011 | D=9 N=26 pairs=36

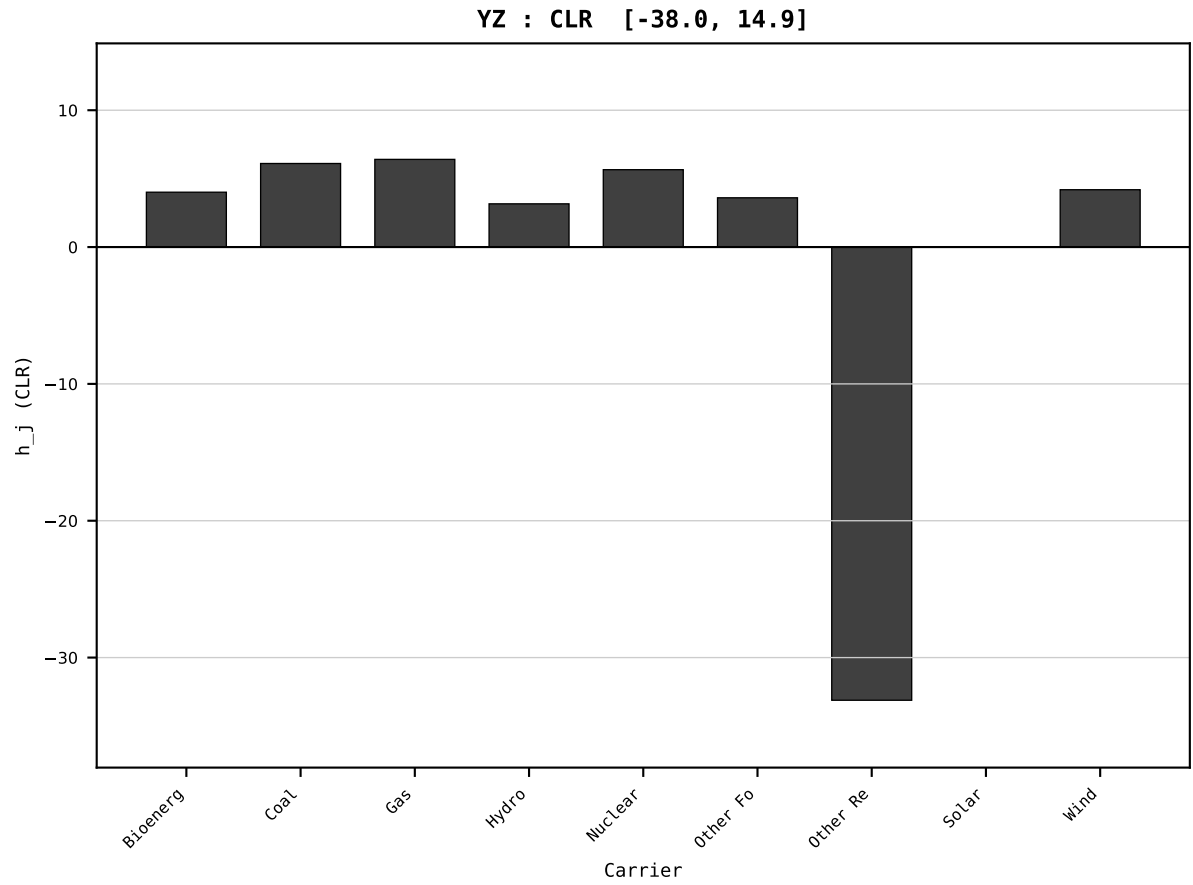
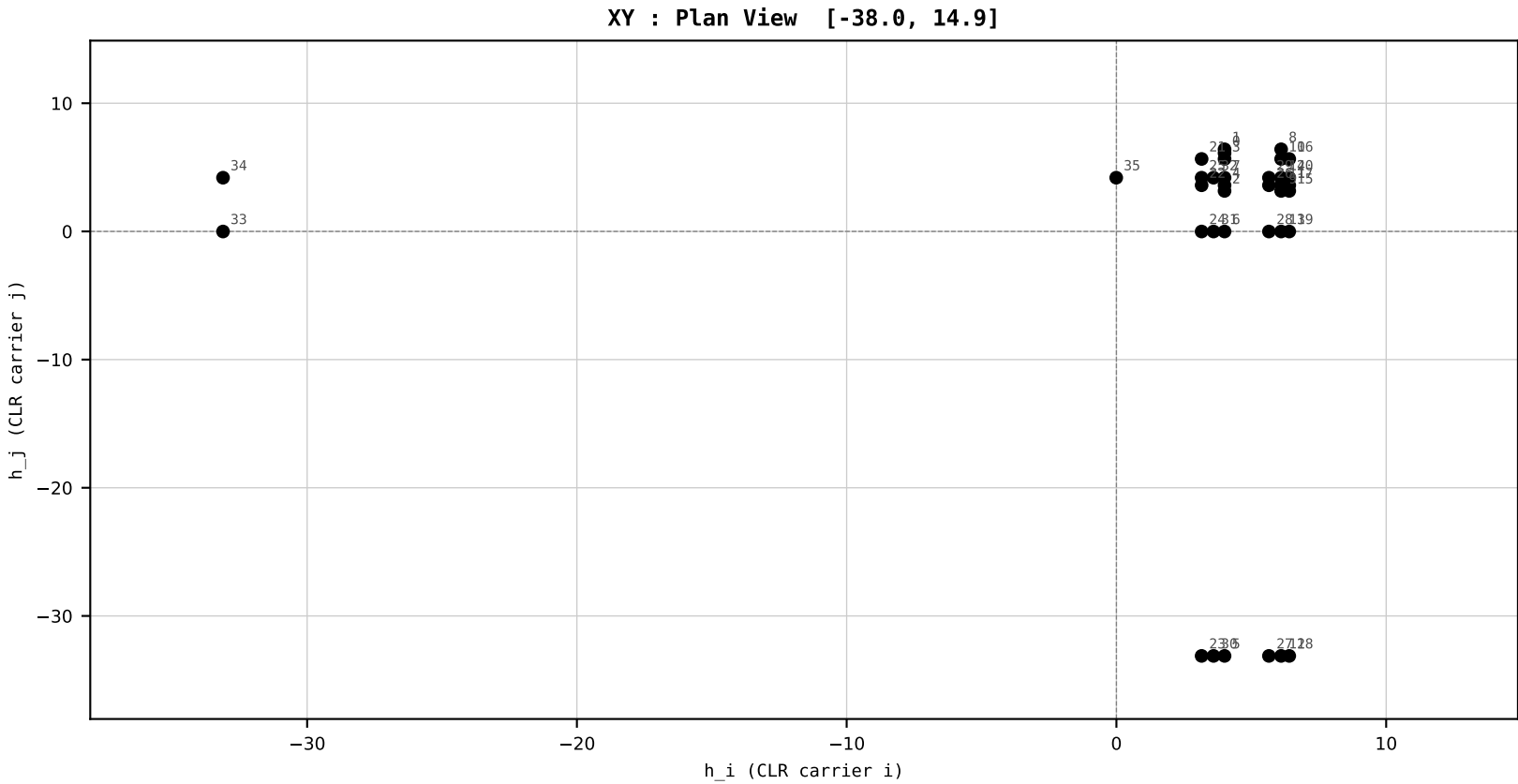
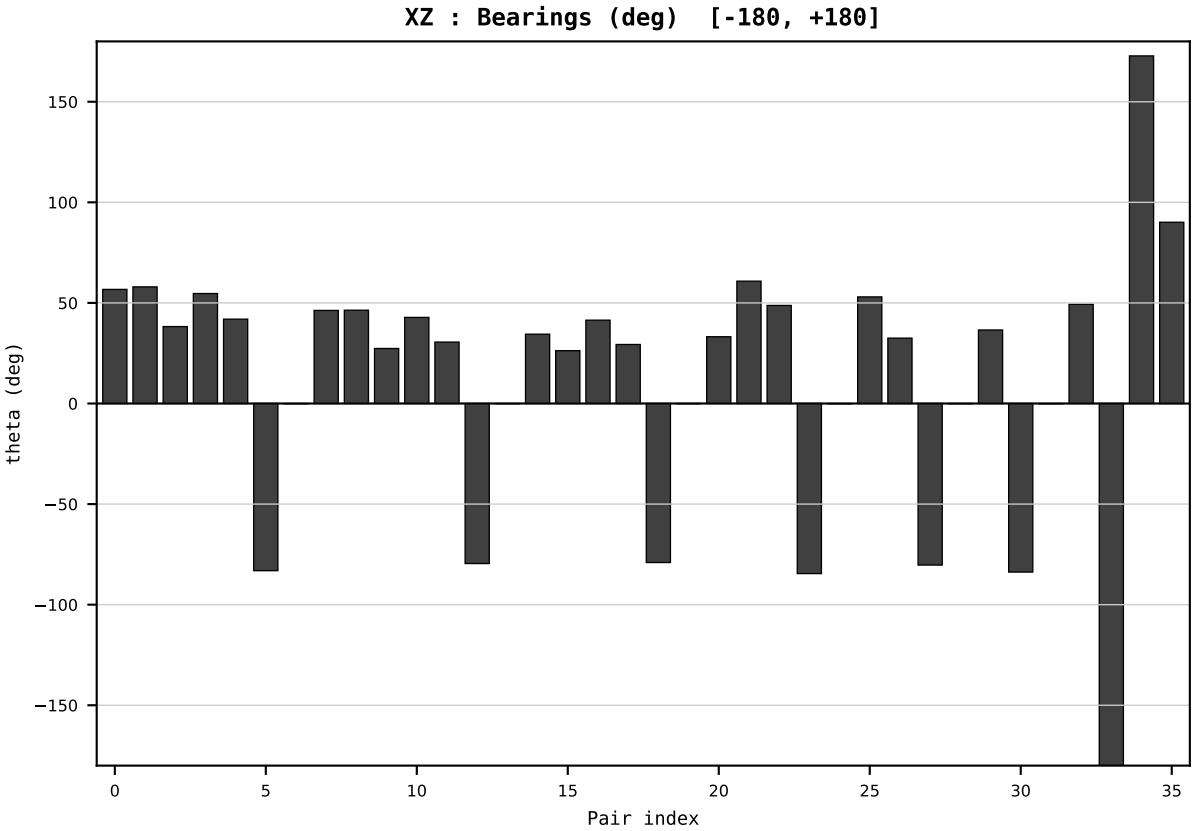
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2011
t : 11 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.337503
Ring = Hs-3
E_metric= 903.4163
kappa_HS= 14649999999999968.00
omega = 2.7842 deg
d_A = 1.726842
Helm = Solar
Helm d = +1.510285
DR = 39.526 HLR
DR ratio= 146499999506615968.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=12 | Year 2012 | D=9 N=26 pairs=36

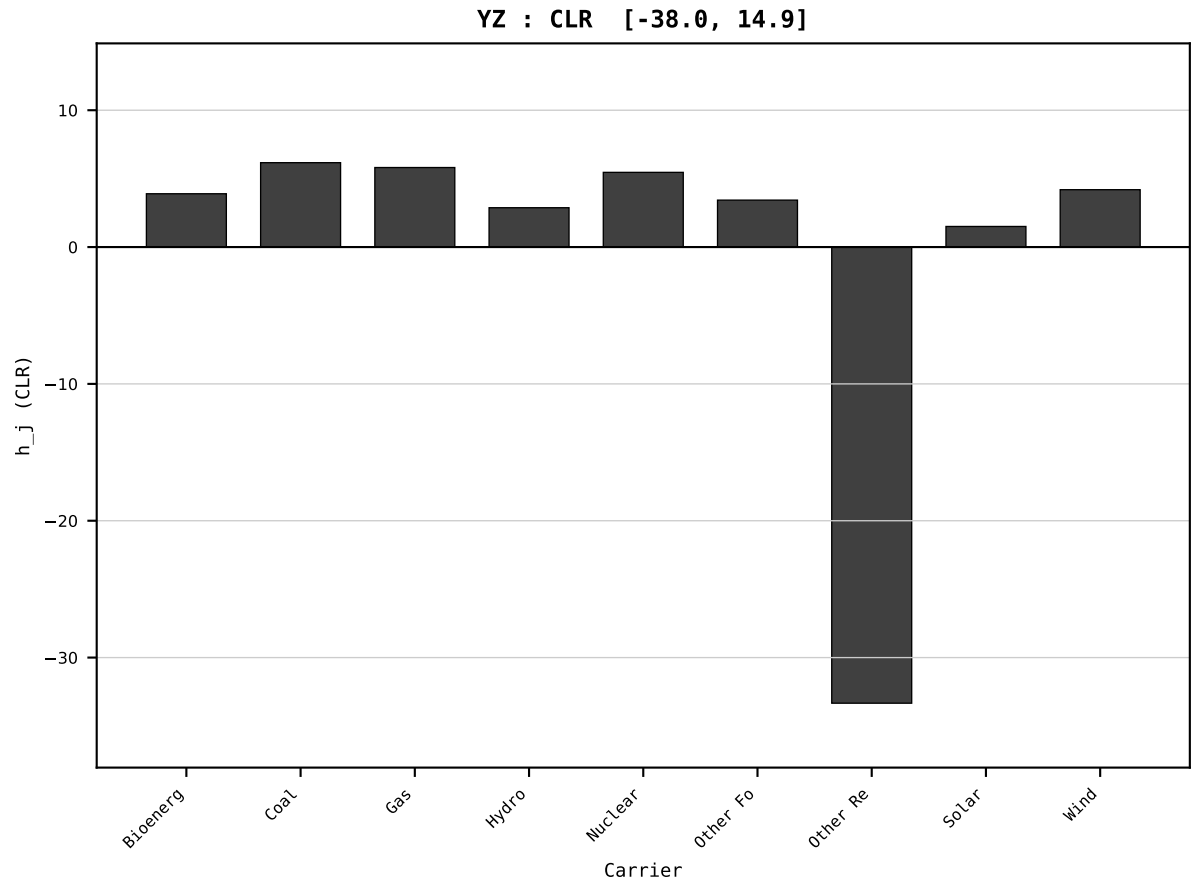
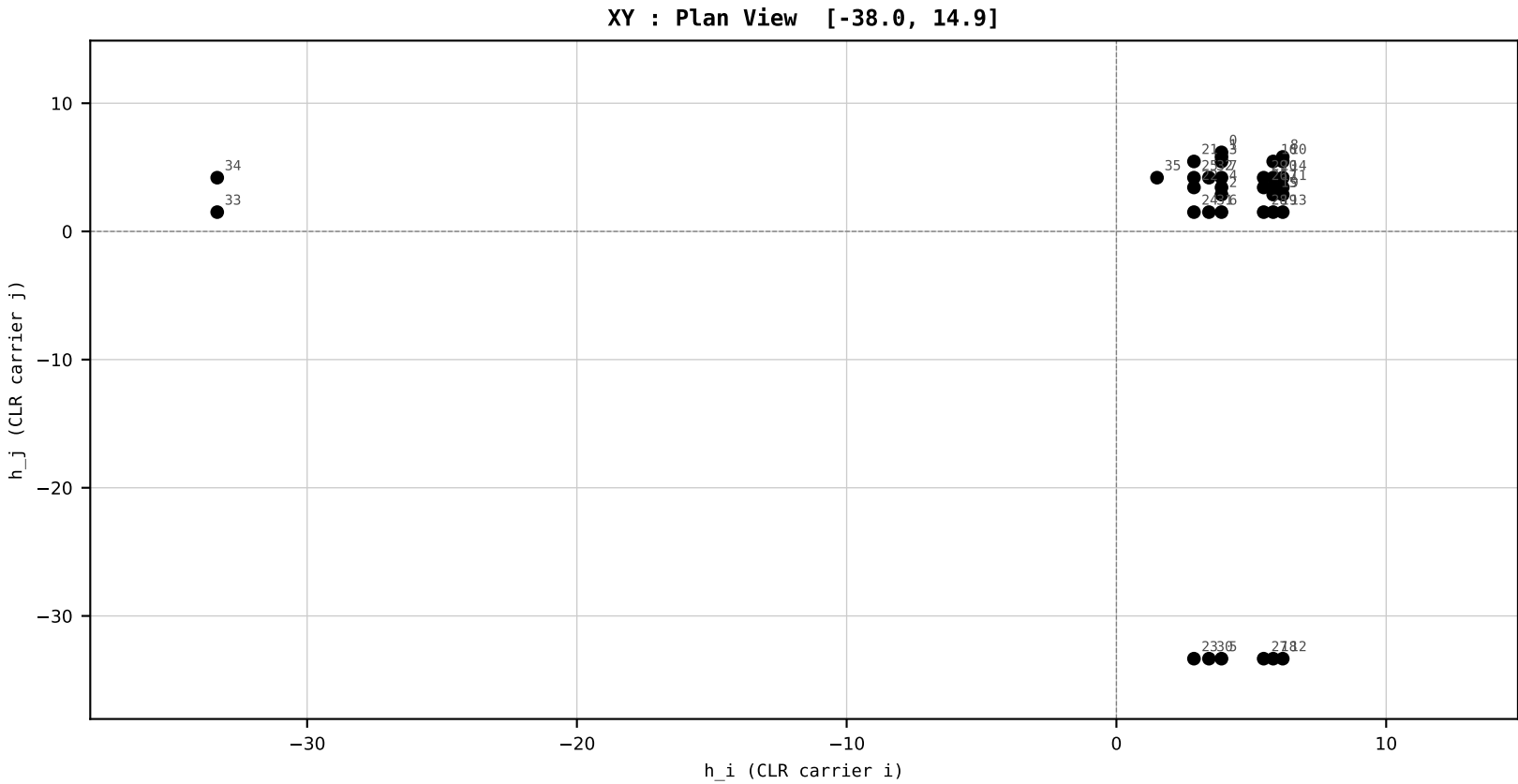
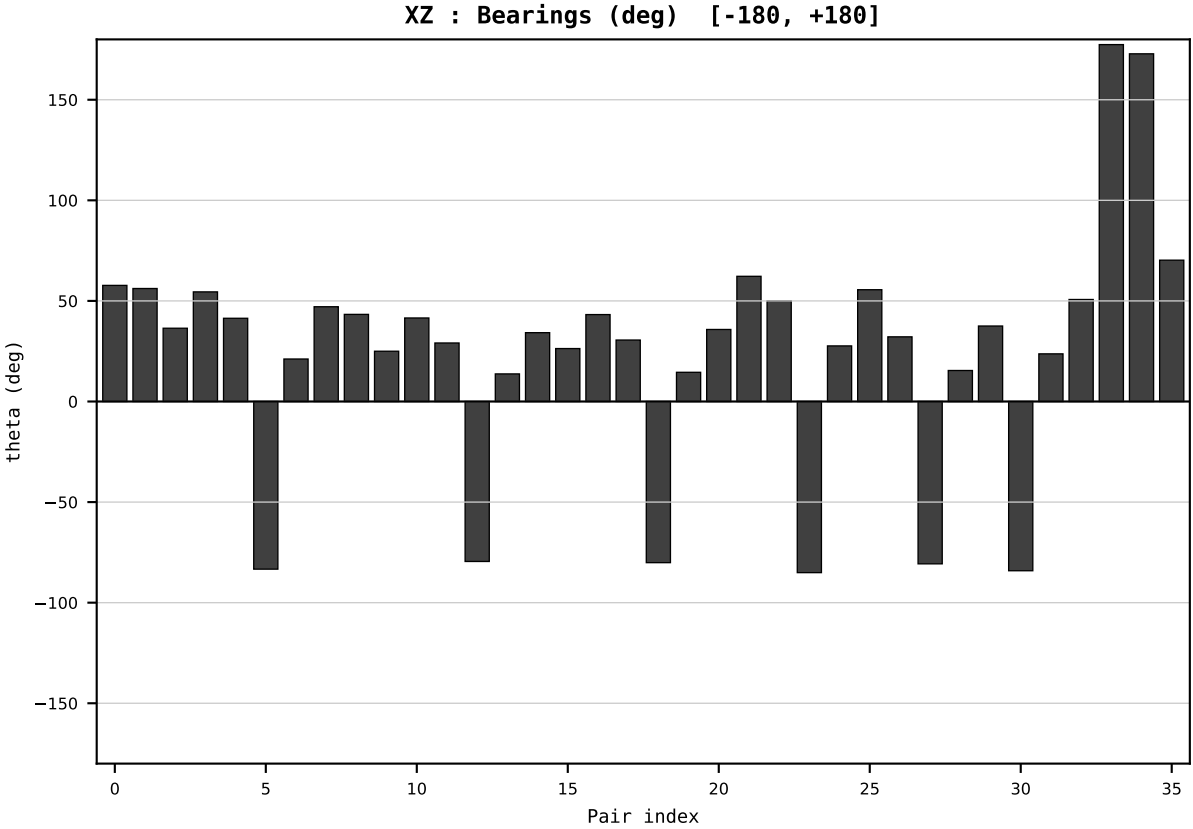
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2012
t : 12 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.315326
Ring = Hs-3
E_metric= 905.5799
kappa_HS= 142789999999999984.00
omega = 2.7179 deg
d_A = 1.688485
Helm = Solar
Helm d = +1.511822
DR = 39.500 HLR
DR ratio= 142789999378451088.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=13 | Year 2013 | D=9 N=26 pairs=36

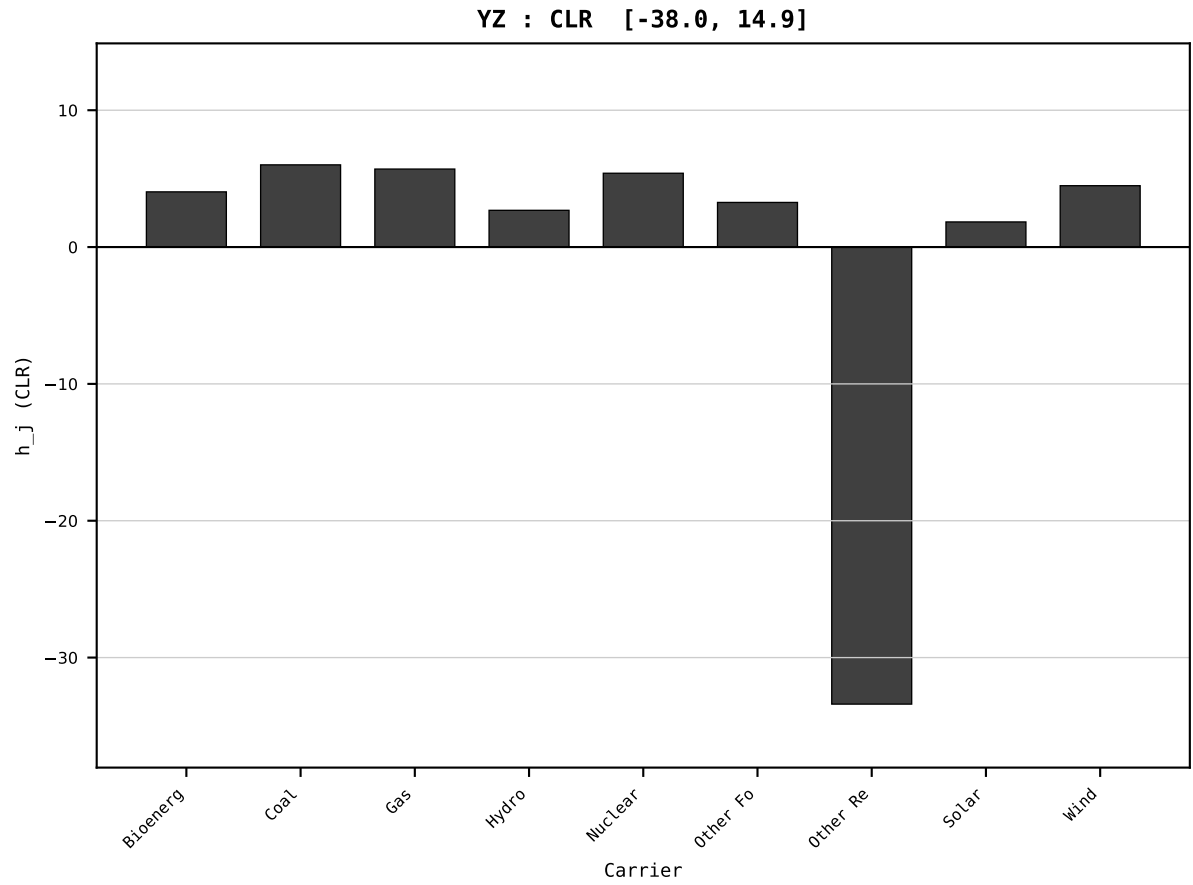
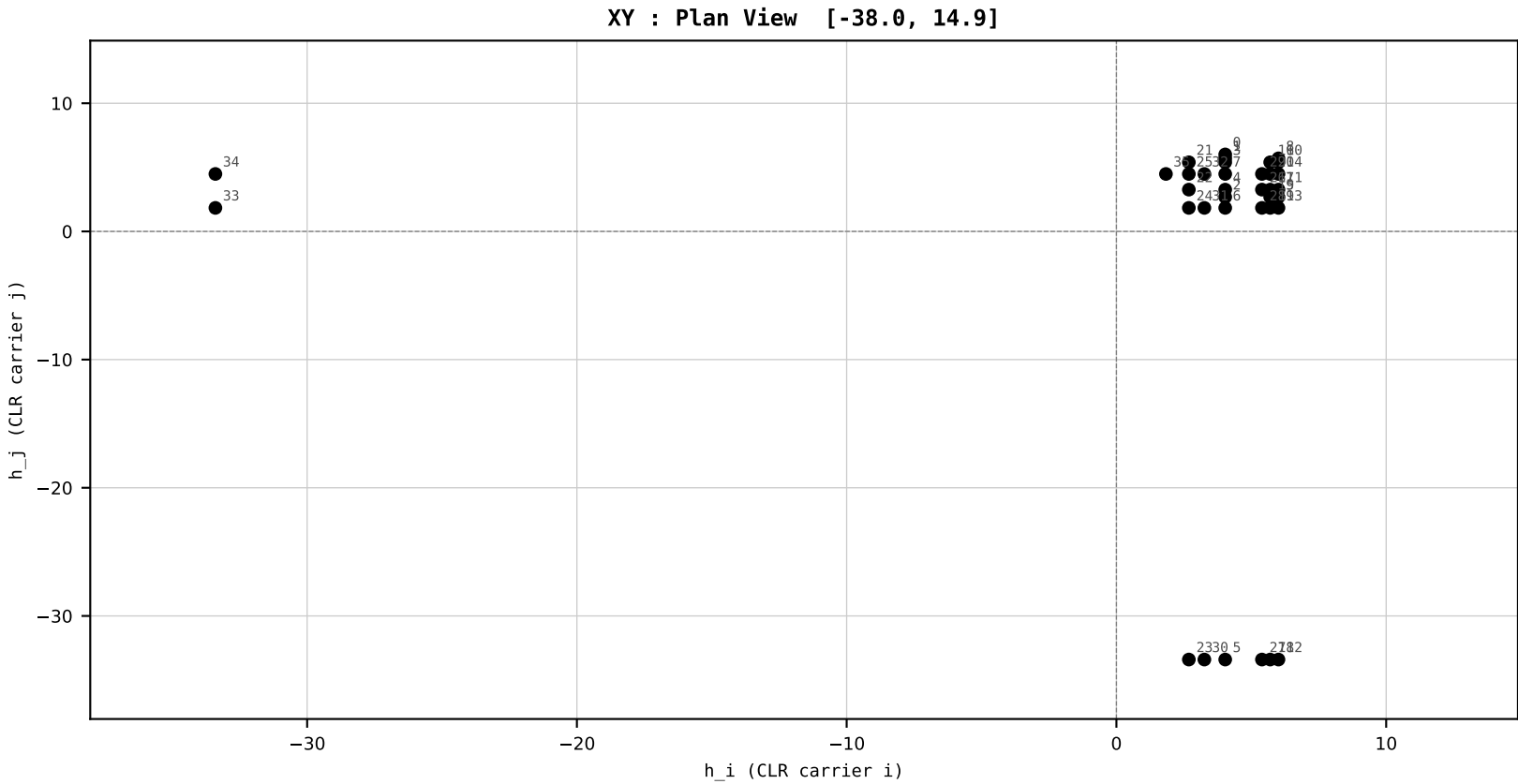
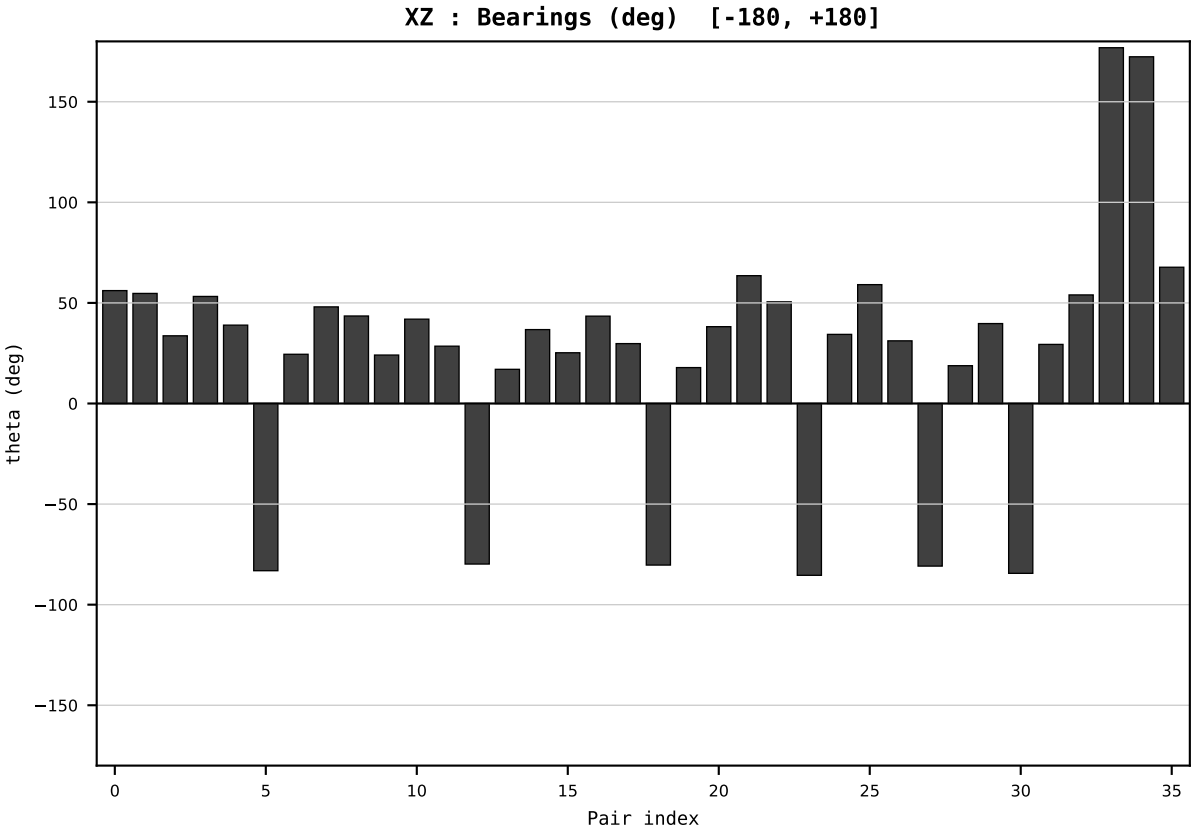
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2013
t : 13 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.287261
Ring = Hs-2
E_metric= 908.7712
kappa_HS= 130259999999999984.00
omega = 0.9132 deg
d_A = 0.569543
Helm = Solar
Helm d = +0.330853
DR = 39.408 HLR
DR ratio= 130260000256749840.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=14 | Year 2014 | D=9 N=26 pairs=36

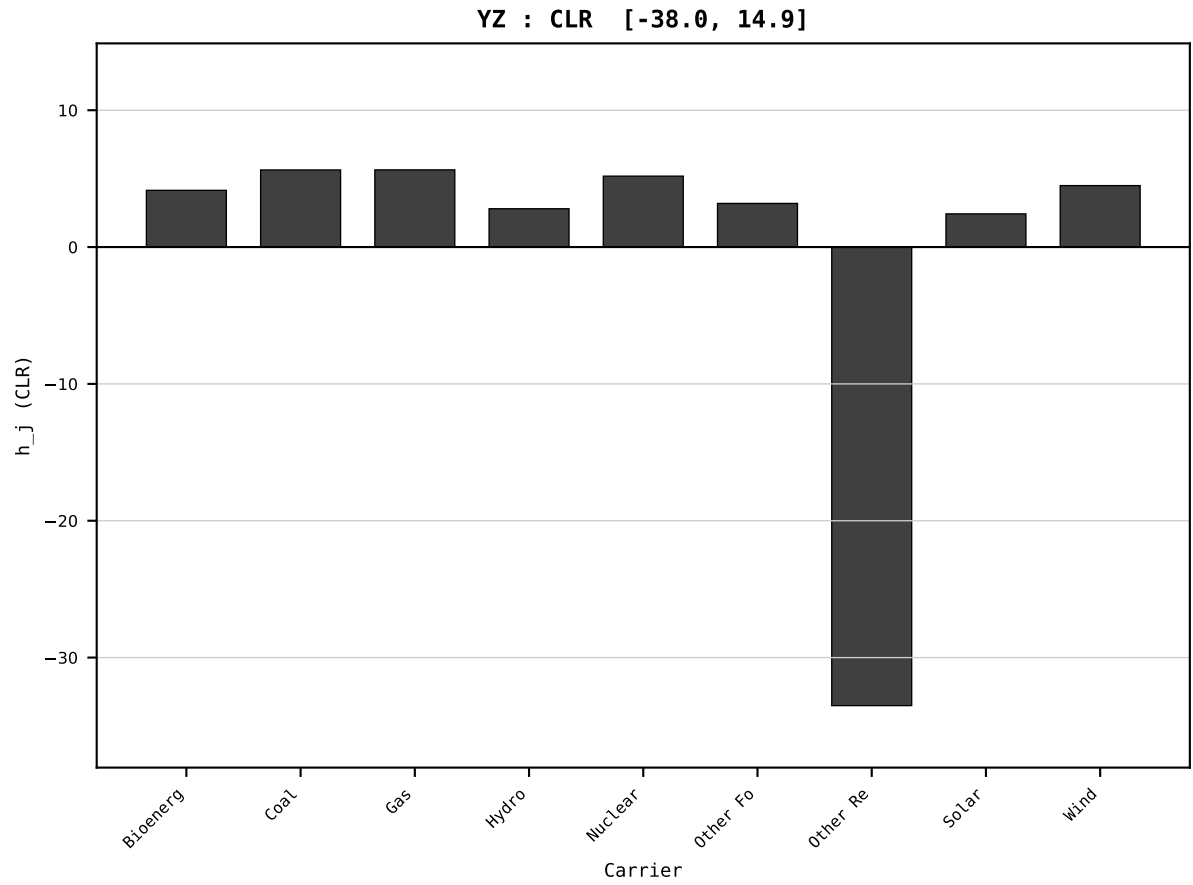
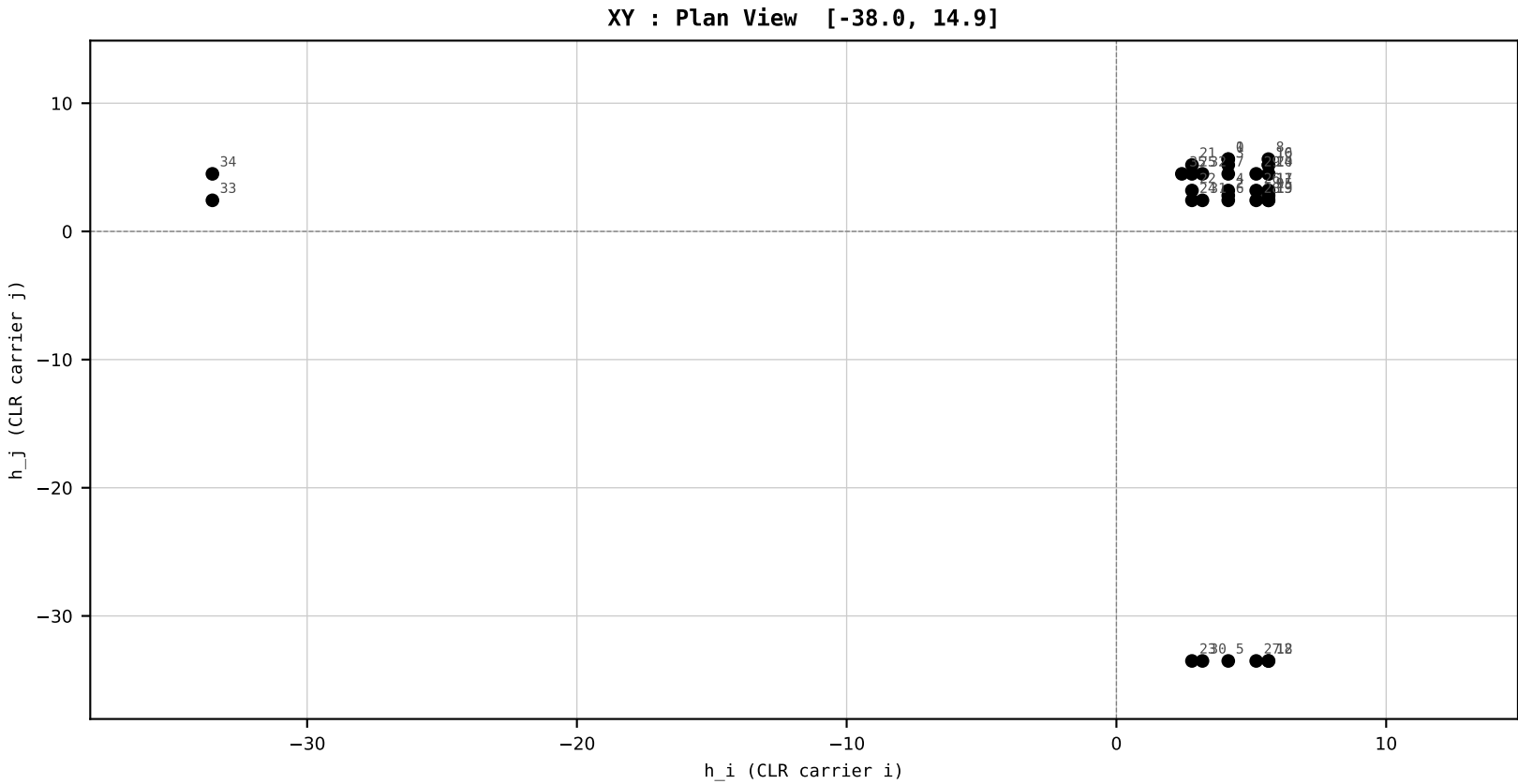
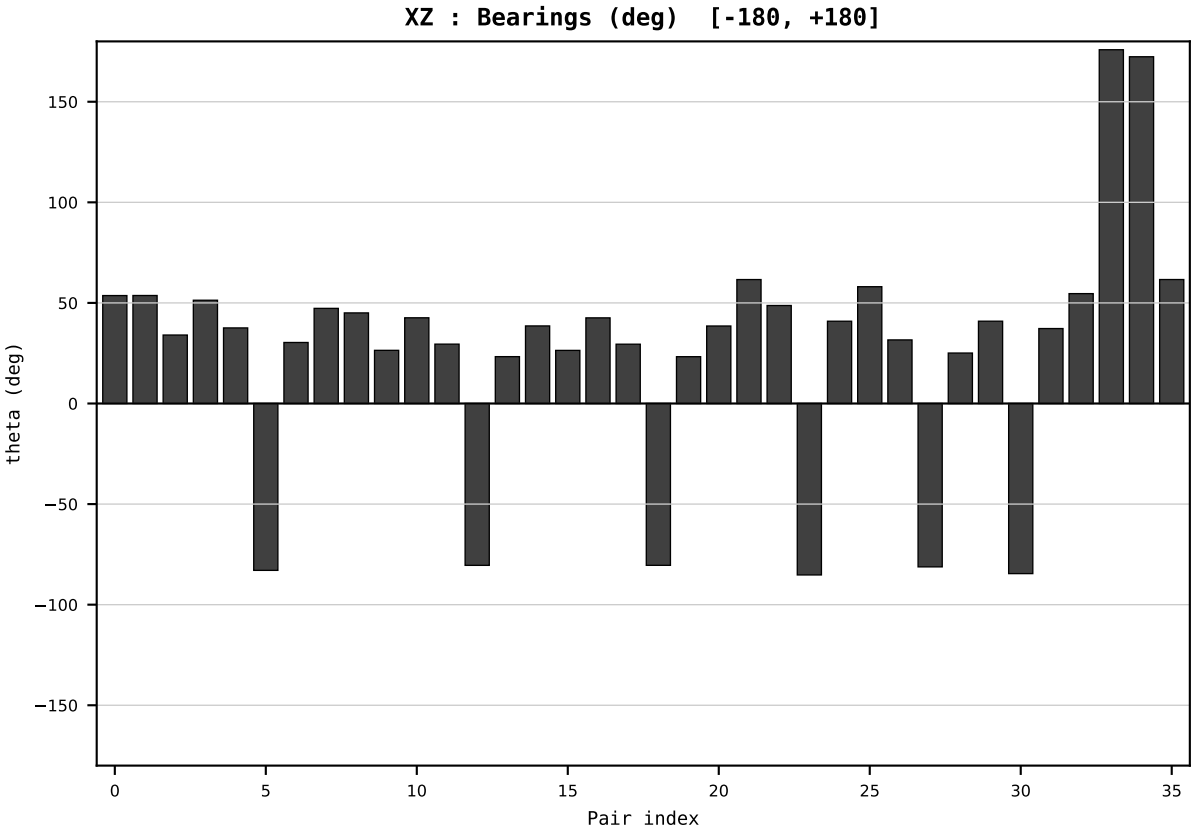
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2014
t : 14 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.245653
Ring = Hs-2
E_metric= 914.5648
kappa_HS= 100889999999999984.00
omega = 1.2190 deg
d_A = 0.760916
Helm = Solar
Helm d = +0.590262
DR = 39.153 HLR
DR ratio= 1008900000067498000.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=15 | Year 2015 | D=9 N=26 pairs=36

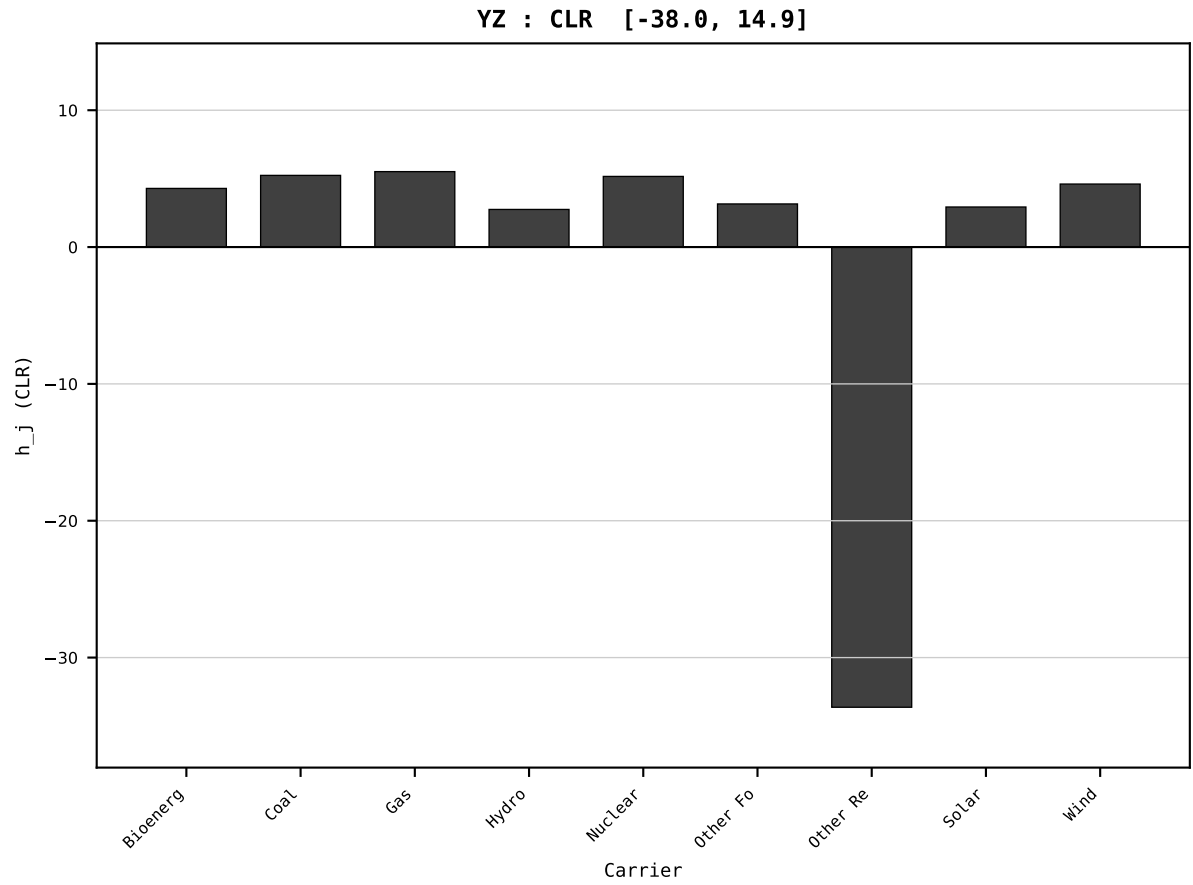
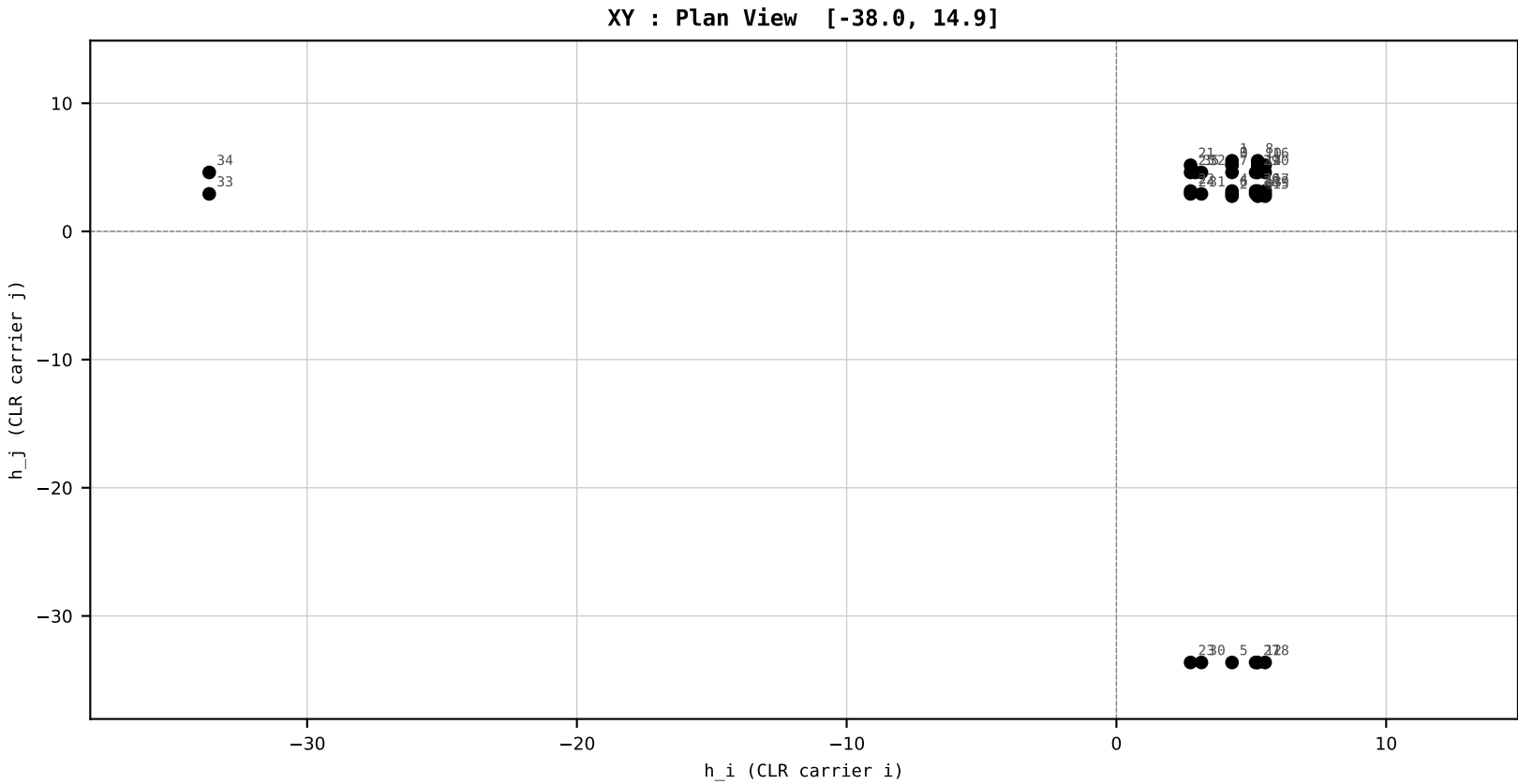
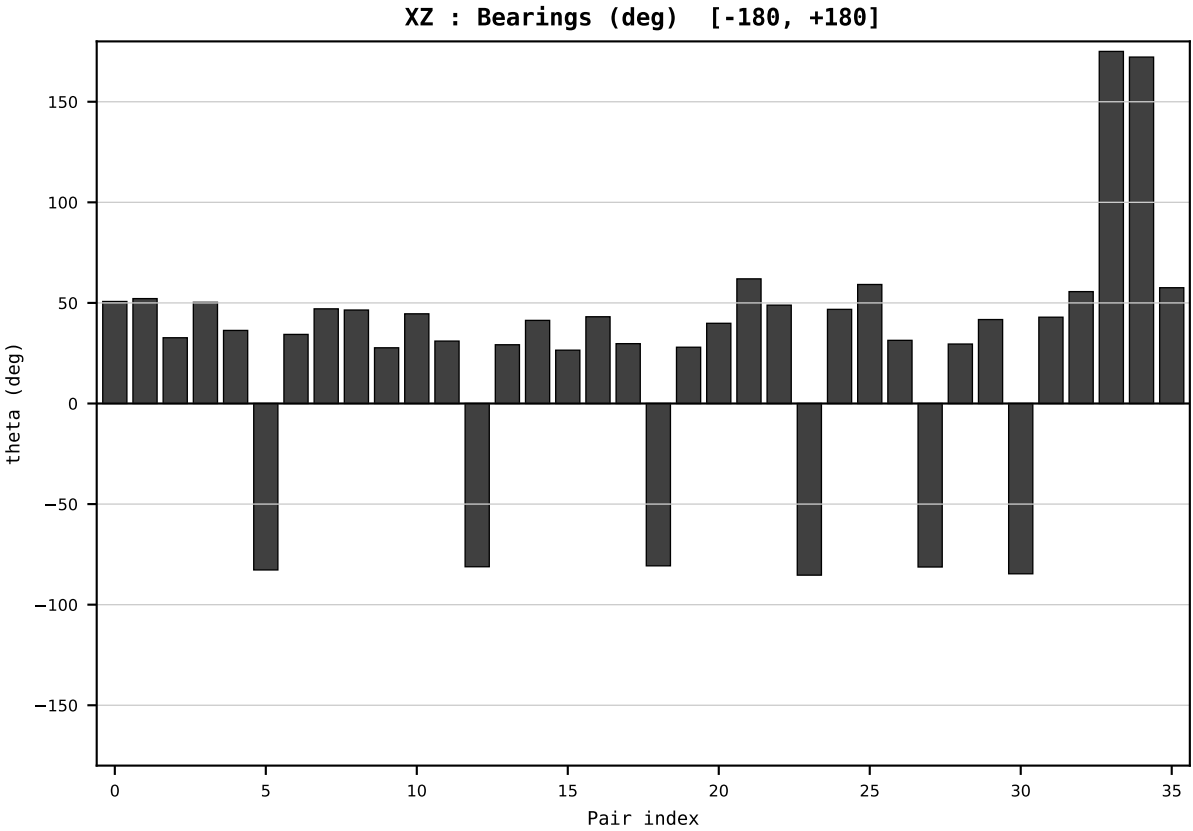
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2015
t : 15 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.206119
Ring = Hs-2
E_metric= 919.4045
kappa_HS= 9987999999999984.00
omega = 1.0971 deg
d_A = 0.690200
Helm = Solar
Helm d = +0.501757
DR = 39.143 HLR
DR ratio= 99879999967813408.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=16 | Year 2016 | D=9 N=26 pairs=36

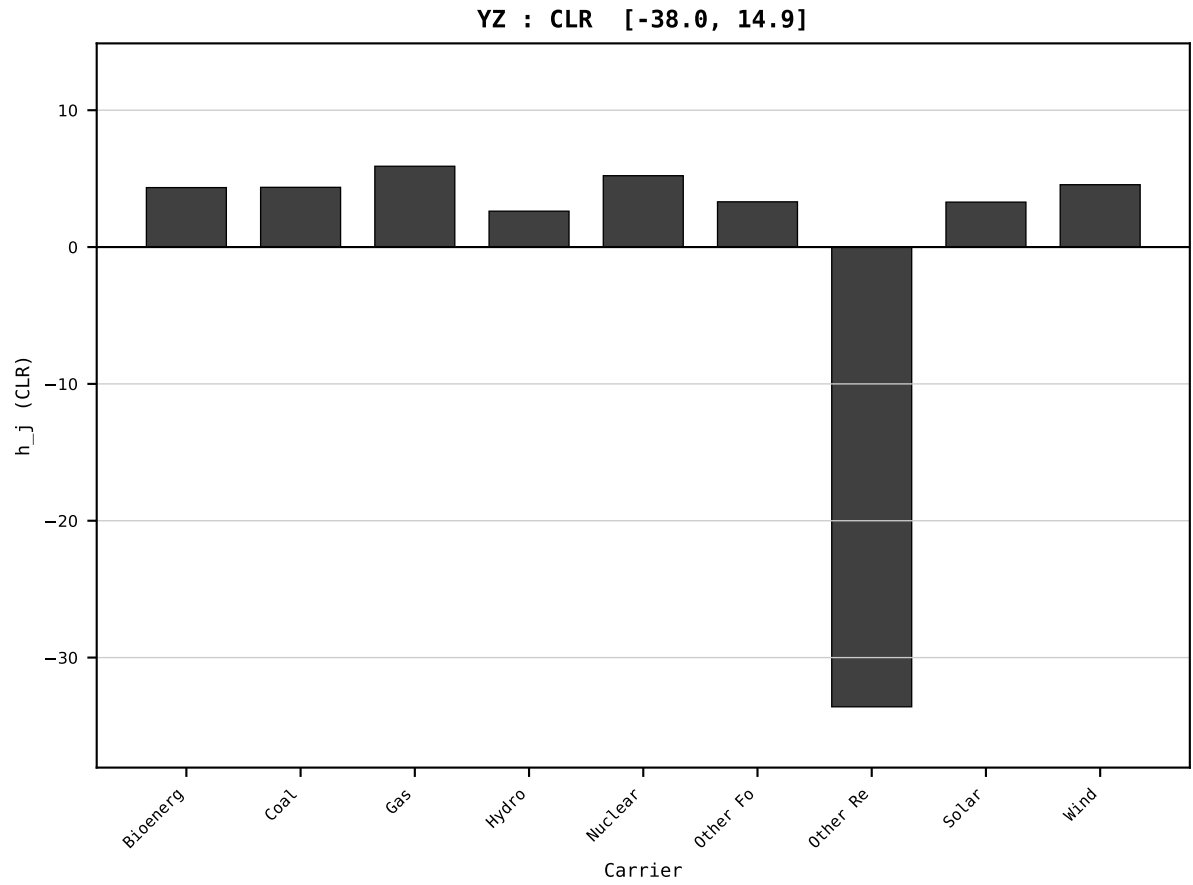
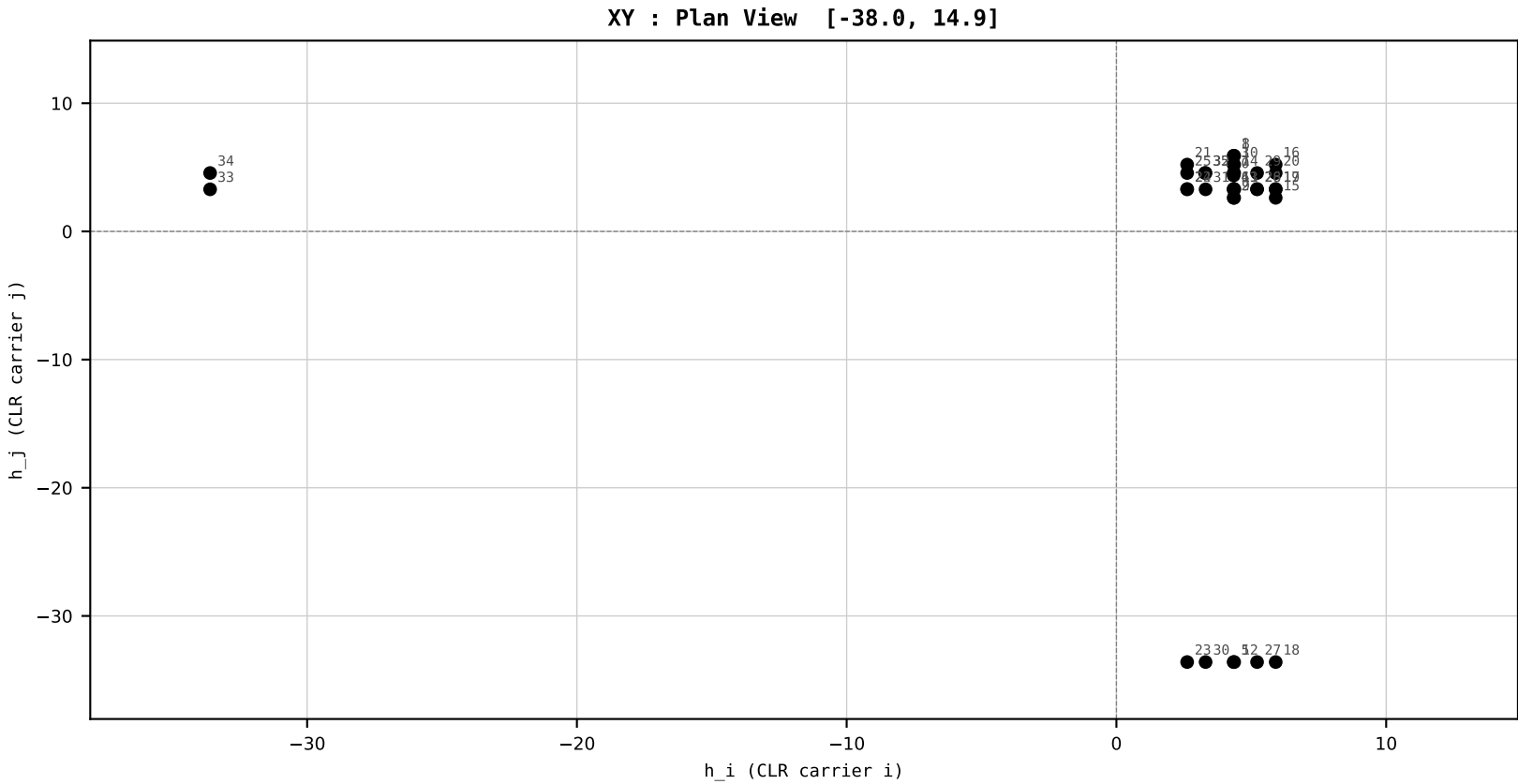
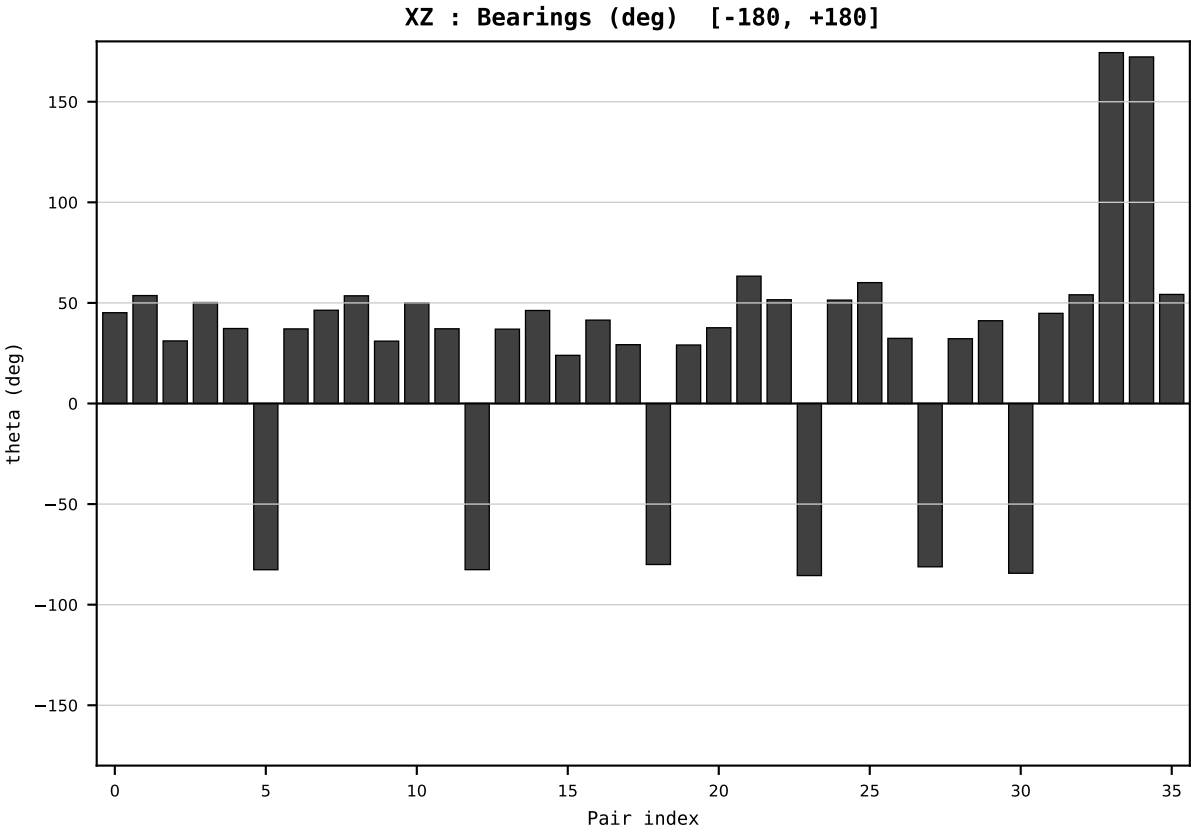
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2016
t : 16 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.249877
Ring = Hs-2
E_metric= 916.7901
kappa_HS= 143129999999999984.00
omega = 1.6738 deg
d_A = 1.045760
Helm = Coal
Helm d = -0.873359
DR = 39.503 HLR
DR ratio= 143129999555321712.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=17 | Year 2017 | D=9 N=26 pairs=36

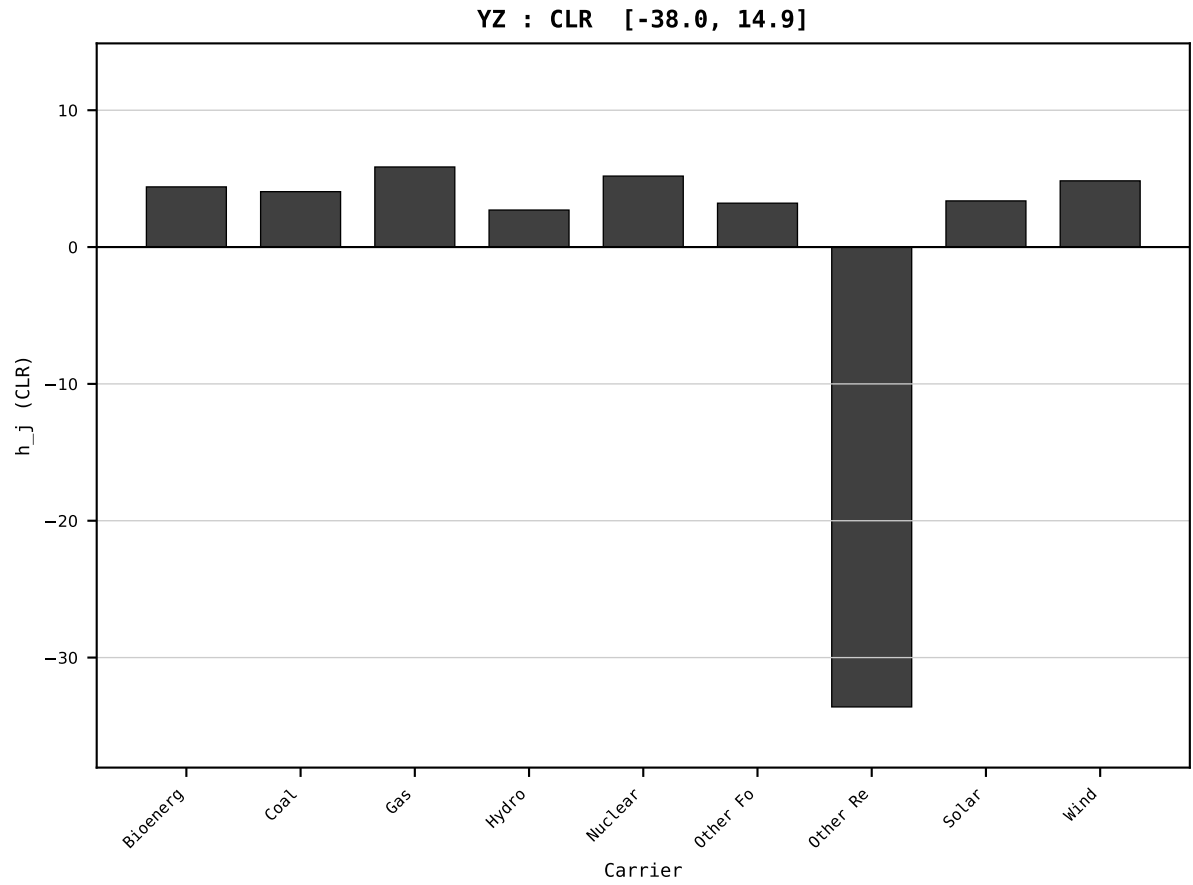
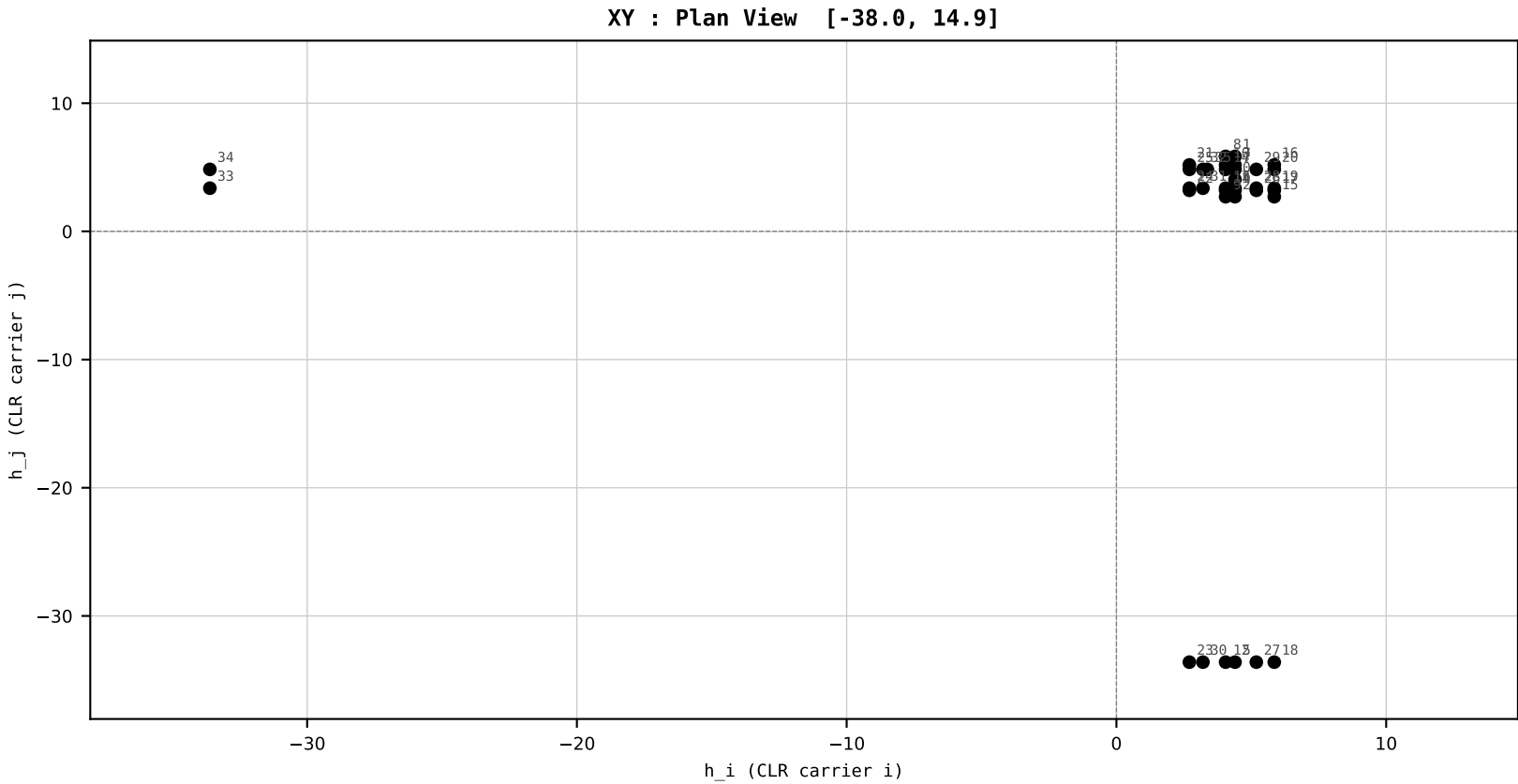
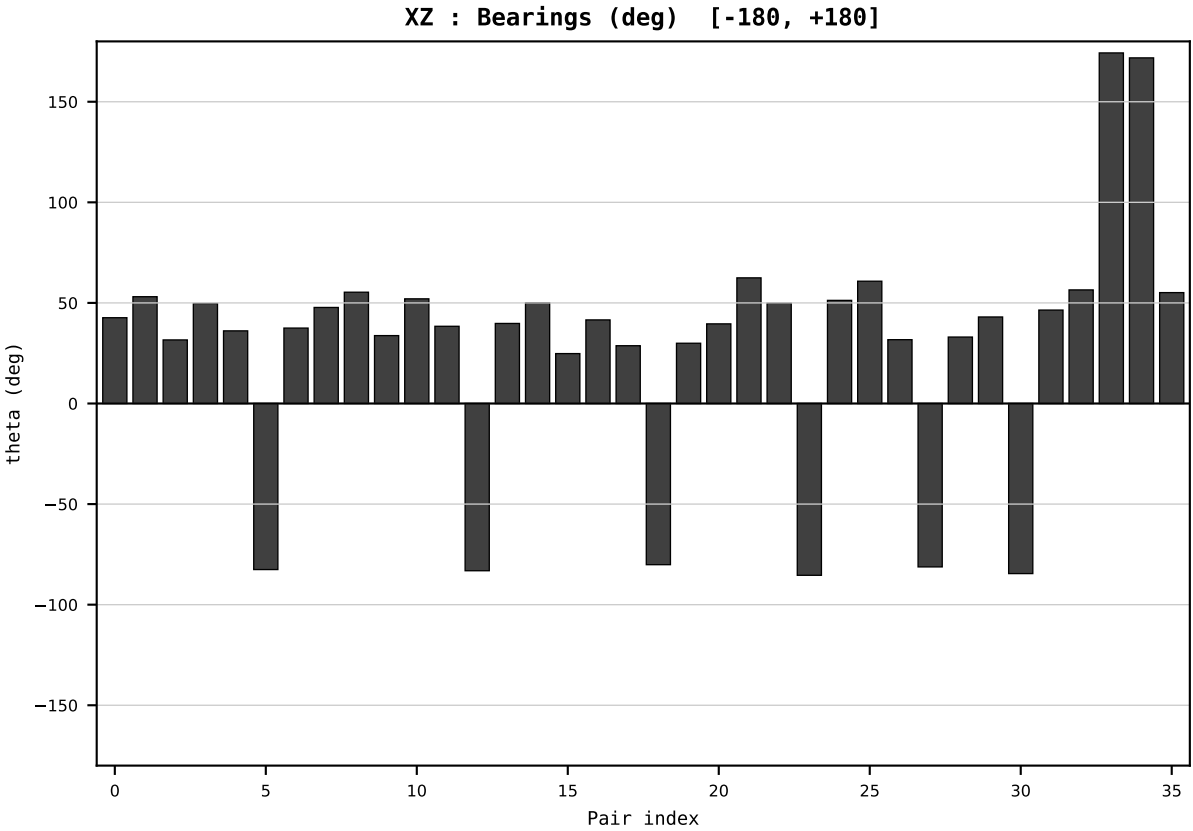
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2017
t : 17 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.242414
Ring = Hs-2
E_metric= 917.2914
kappa_HS= 13674999999999984.00
omega = 0.7349 deg
d_A = 0.458615
Helm = Coal
Helm d = -0.316282
DR = 39.457 HLR
DR ratio= 136749999150403536.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=18 | Year 2018 | D=9 N=26 pairs=36

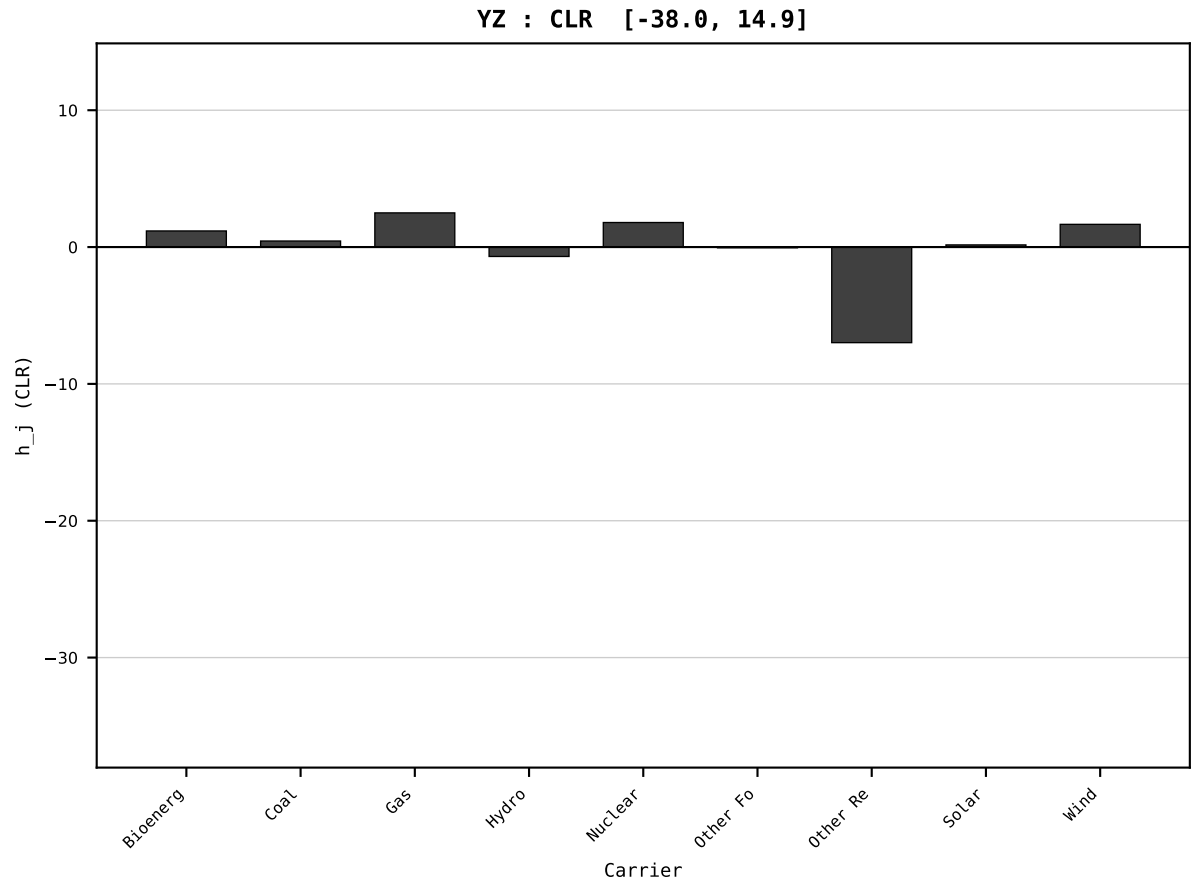
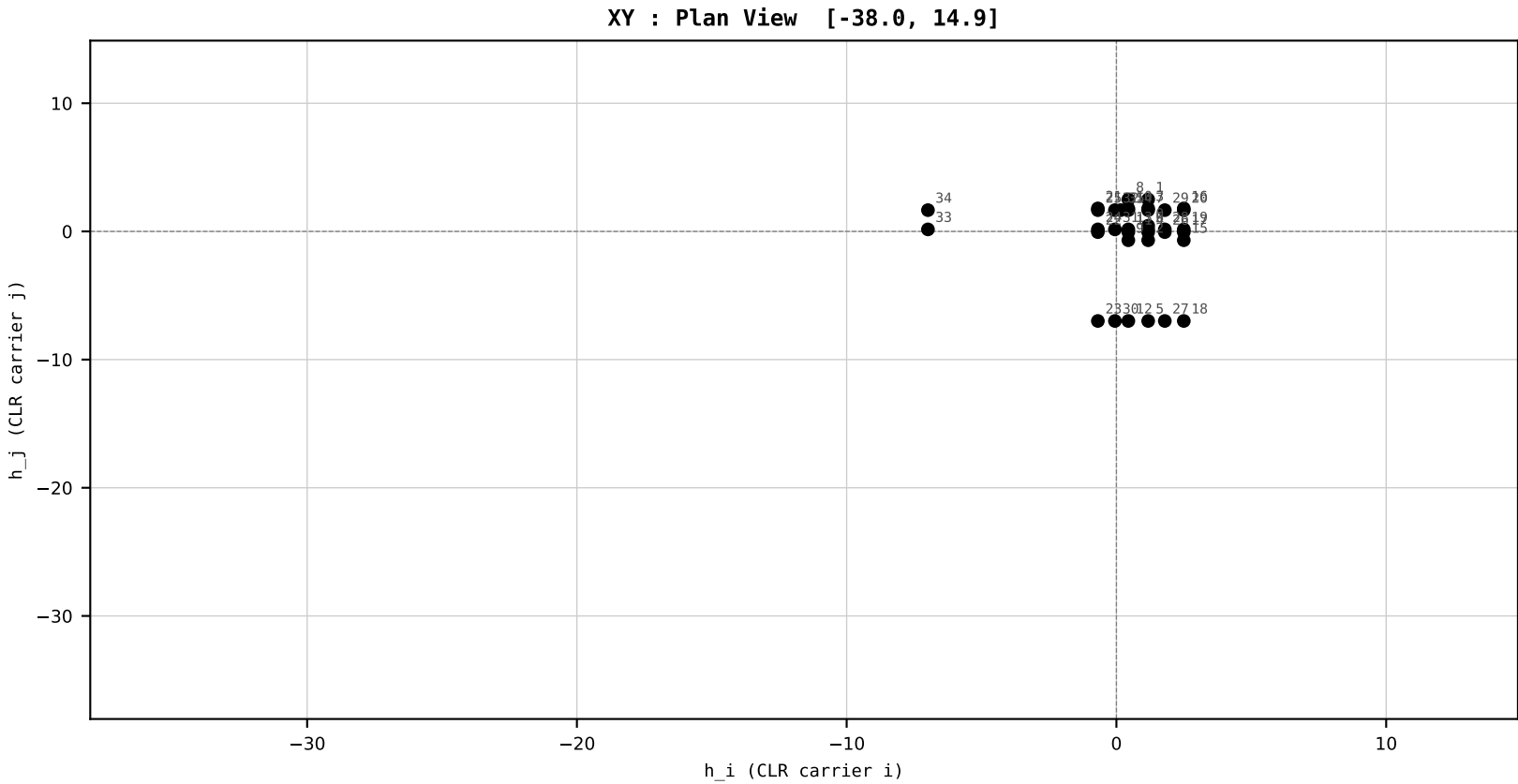
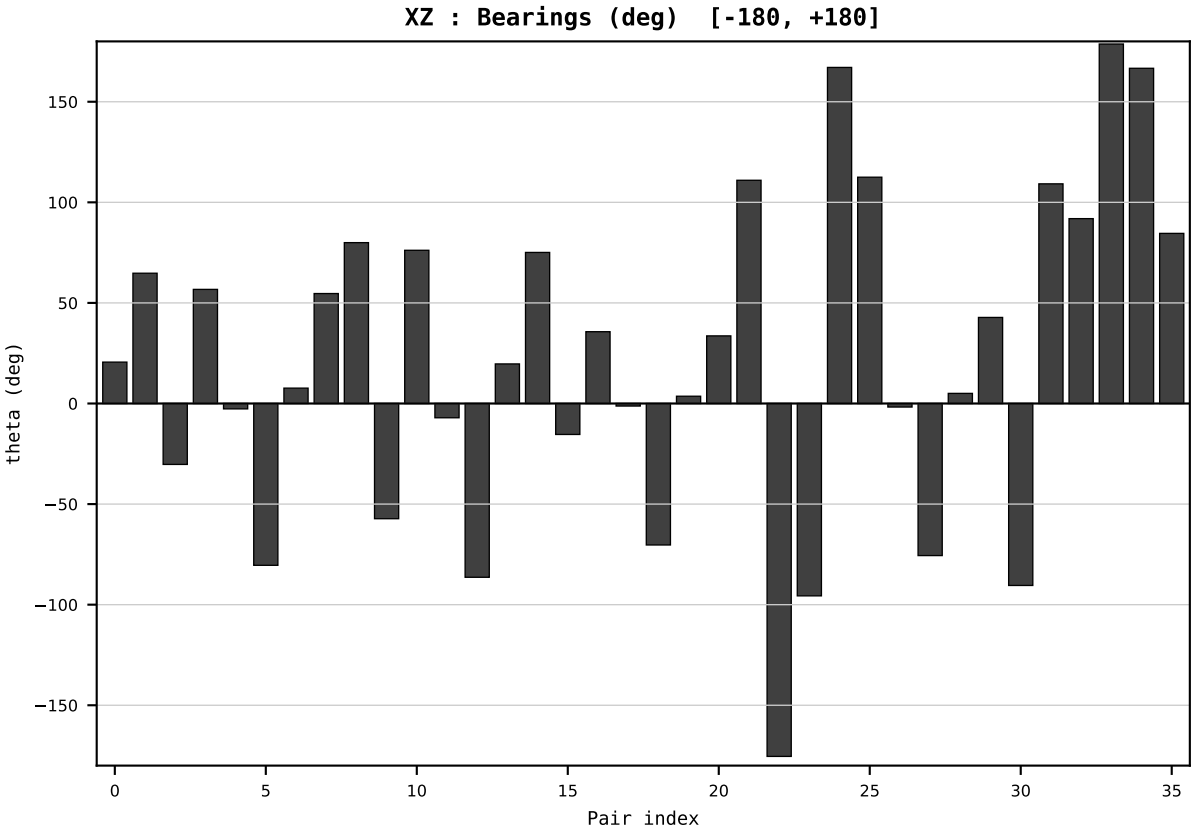
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2018
t : 18 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.238427
Ring = Hs-2
E_metric= 63.1052
kappa_HS= 13149.00
omega = 16.6103 deg
d_A = 28.236007
Helm = Other Renewables
Helm d = +26.618855
DR = 9.484 HLR
DR ratio= 13149.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=19 | Year 2019 | D=9 N=26 pairs=36

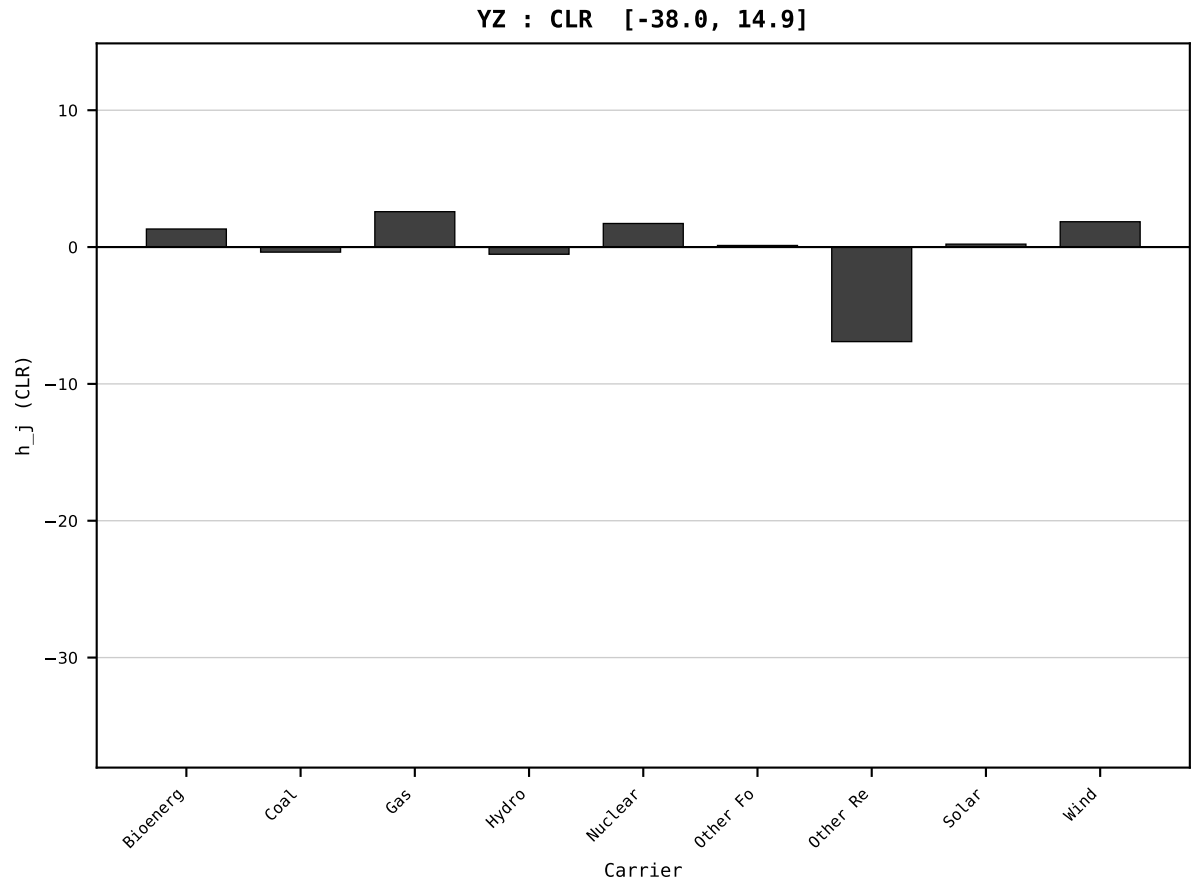
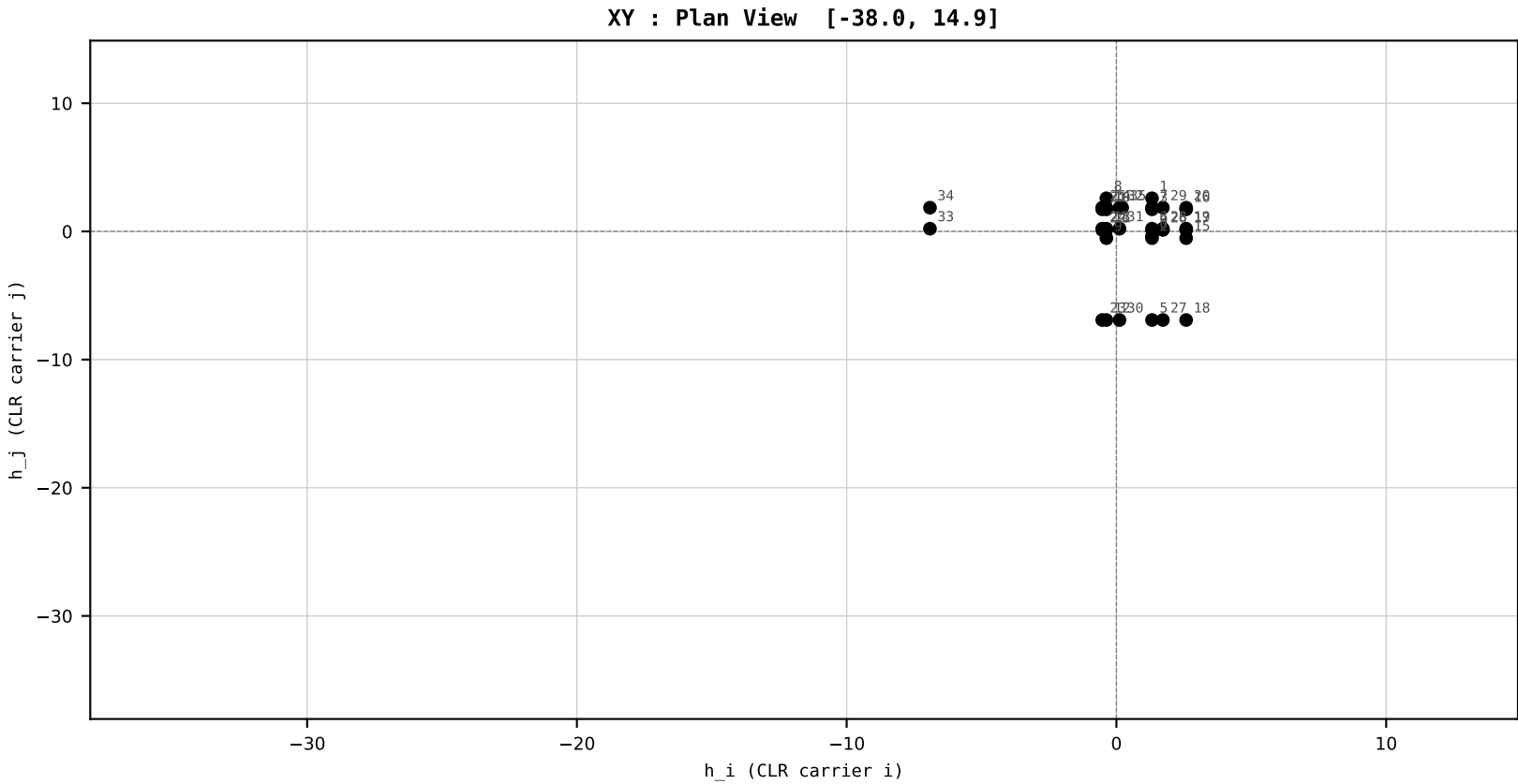
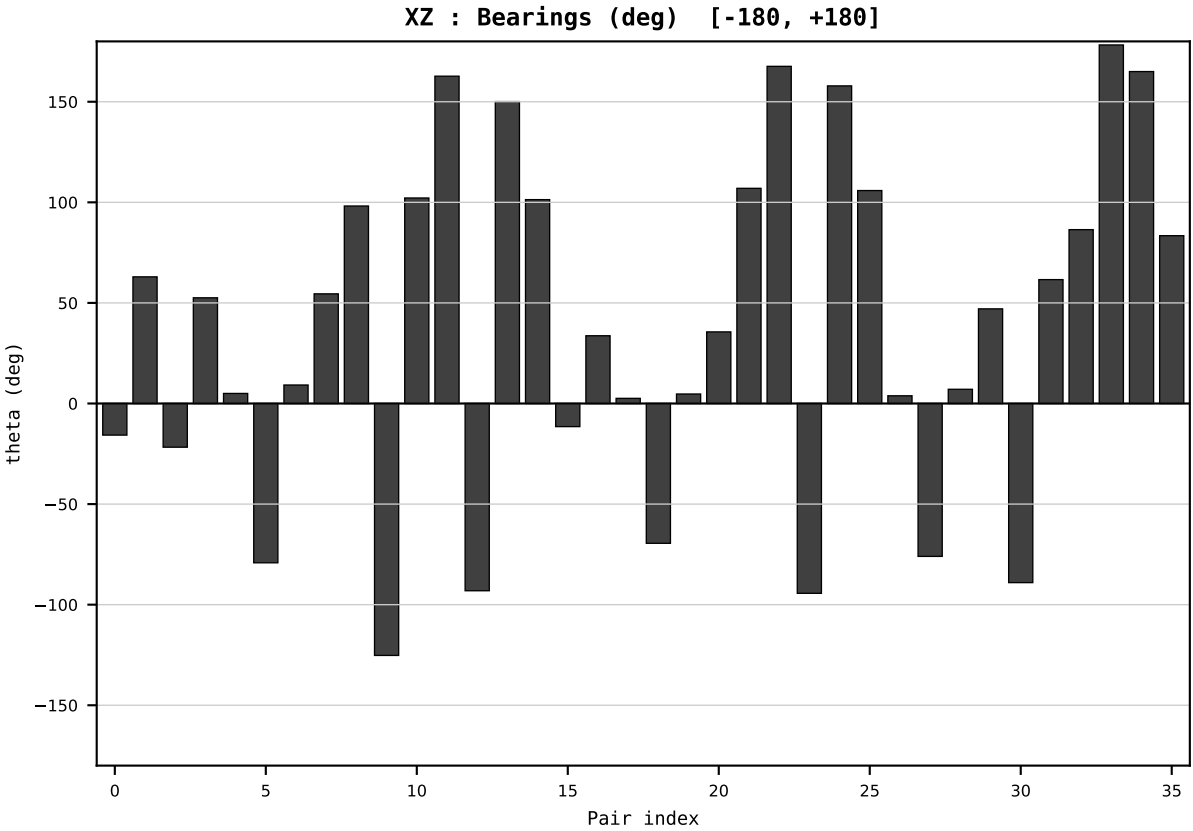
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2019
t : 19 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.257979
Ring = Hs-2
E_metric= 63.0521
kappa_HS= 13309.00
omega = 6.4343 deg
d_A = 0.891439
Helm = Coal
Helm d = -0.813158
DR = 9.496 HLR
DR ratio= 13309.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=20 | Year 2020 | D=9 N=26 pairs=36

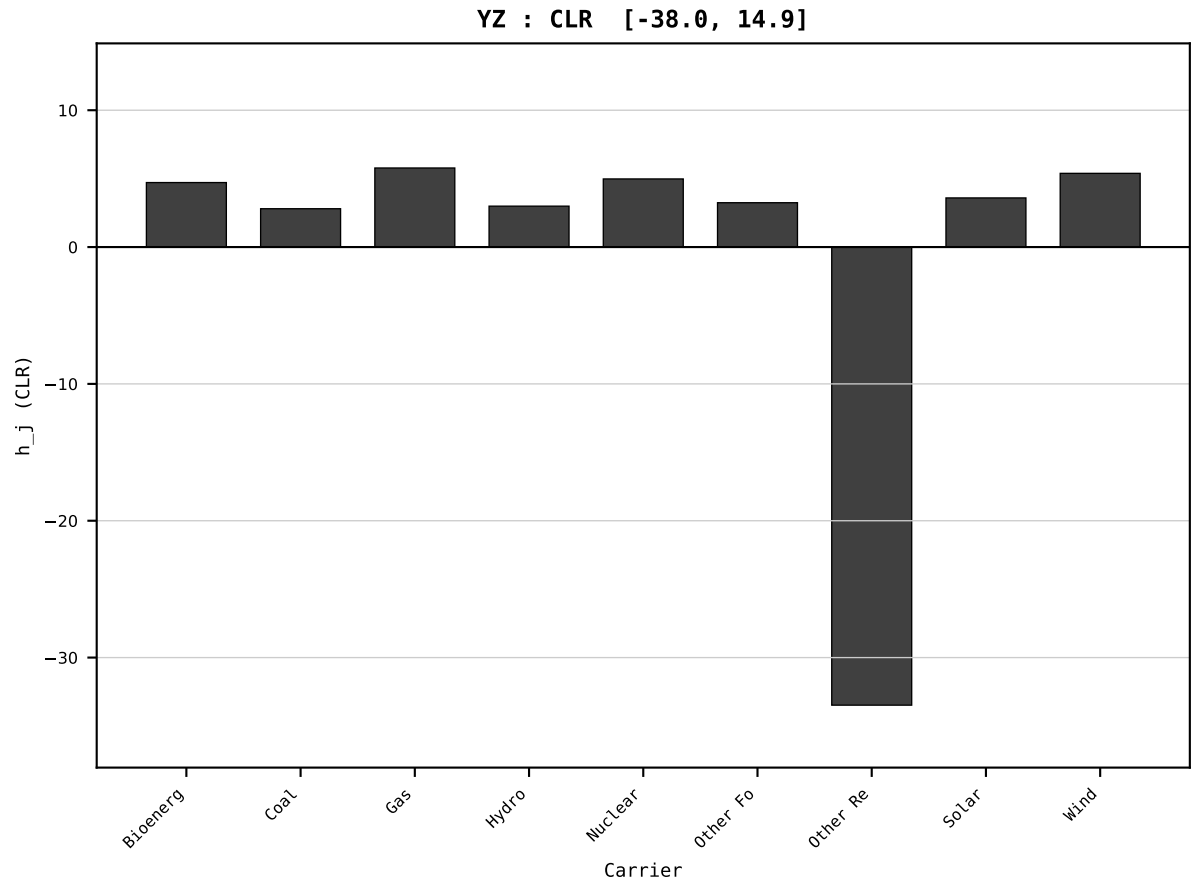
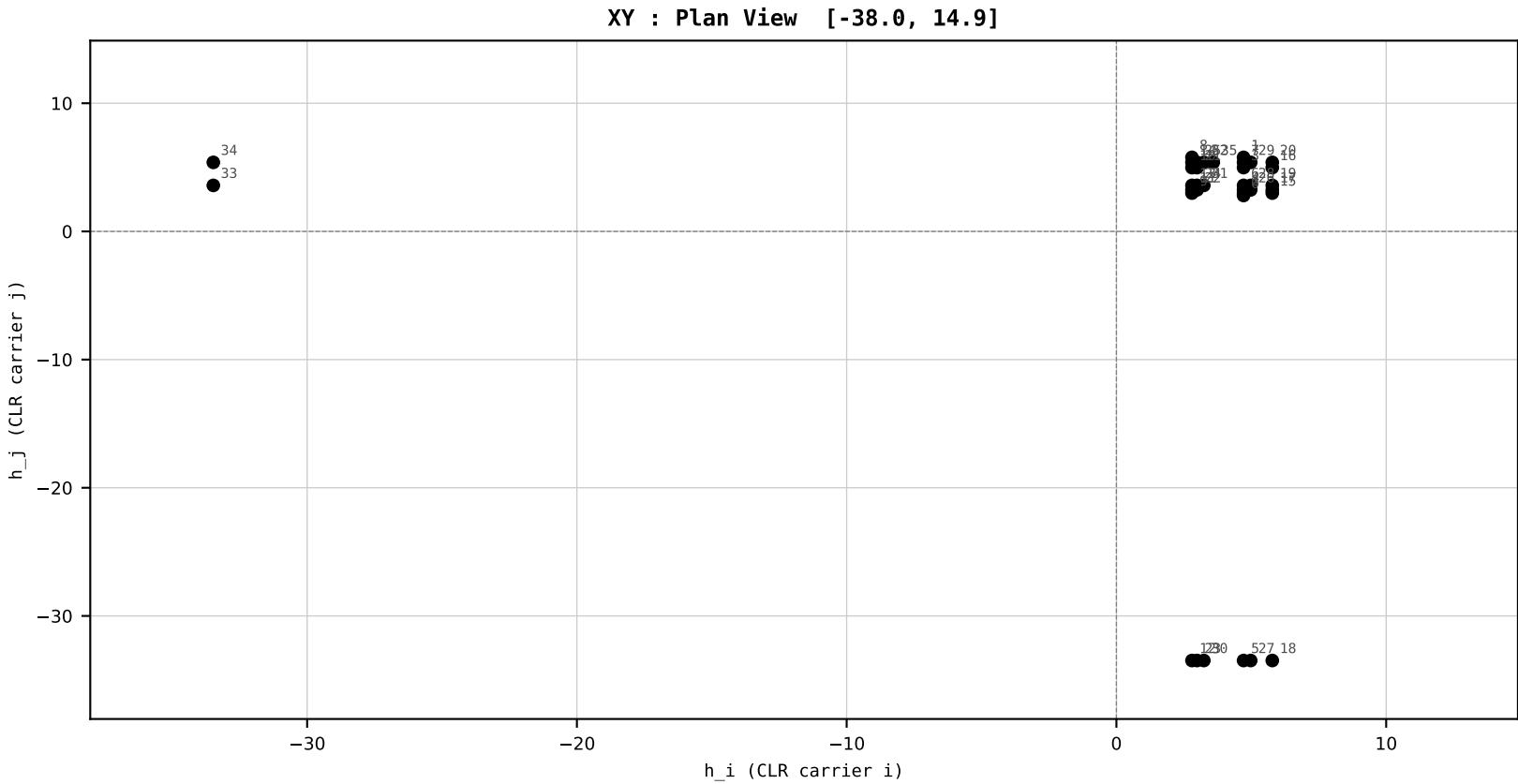
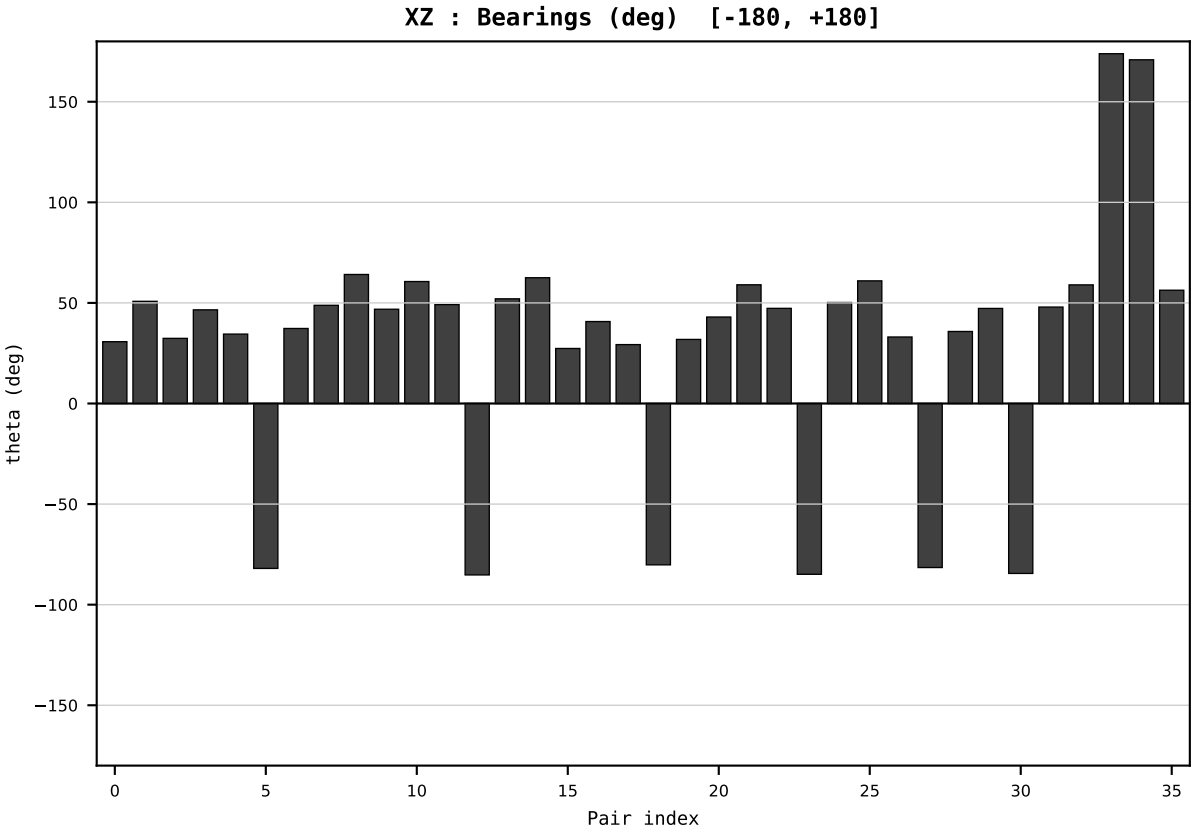
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2020
t : 20 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.246847
Ring = Hs-2
E_metric= 915.4463
kappa_HS= 11188999999999984.00
omega = 17.7152 deg
d_A = 28.181681
Helm = Other Renewables
Helm d = -26.567038
DR = 39.256 HLR
DR ratio= 111889999918284832.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=21 | Year 2021 | D=9 N=26 pairs=36

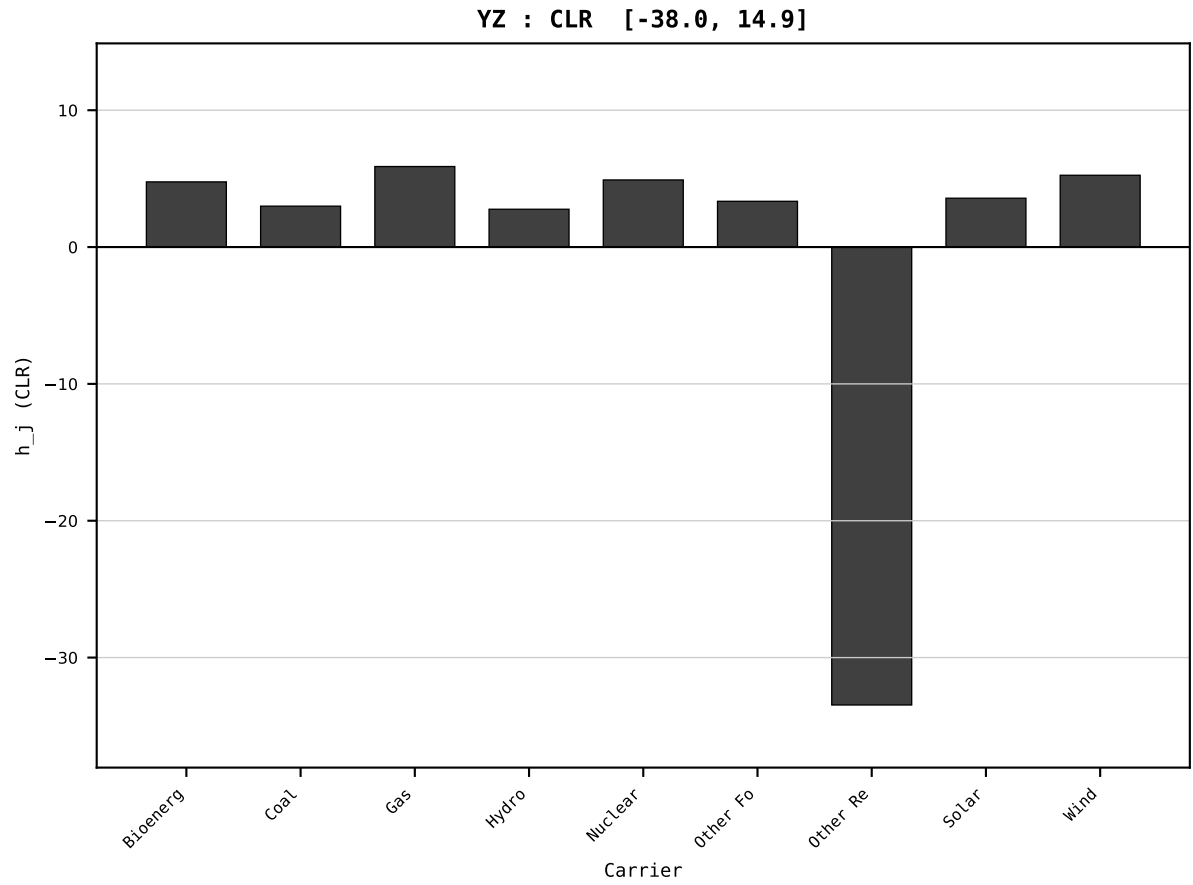
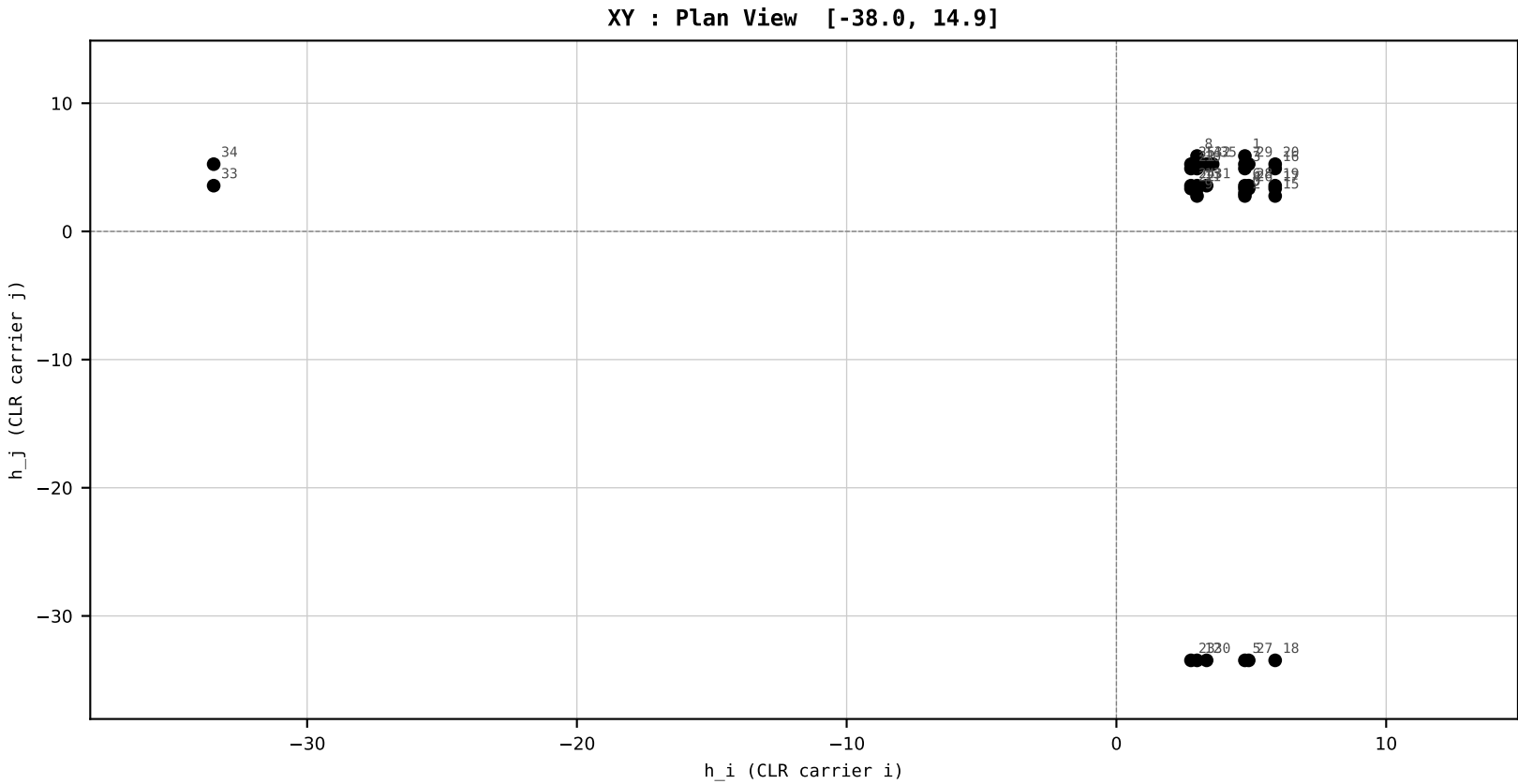
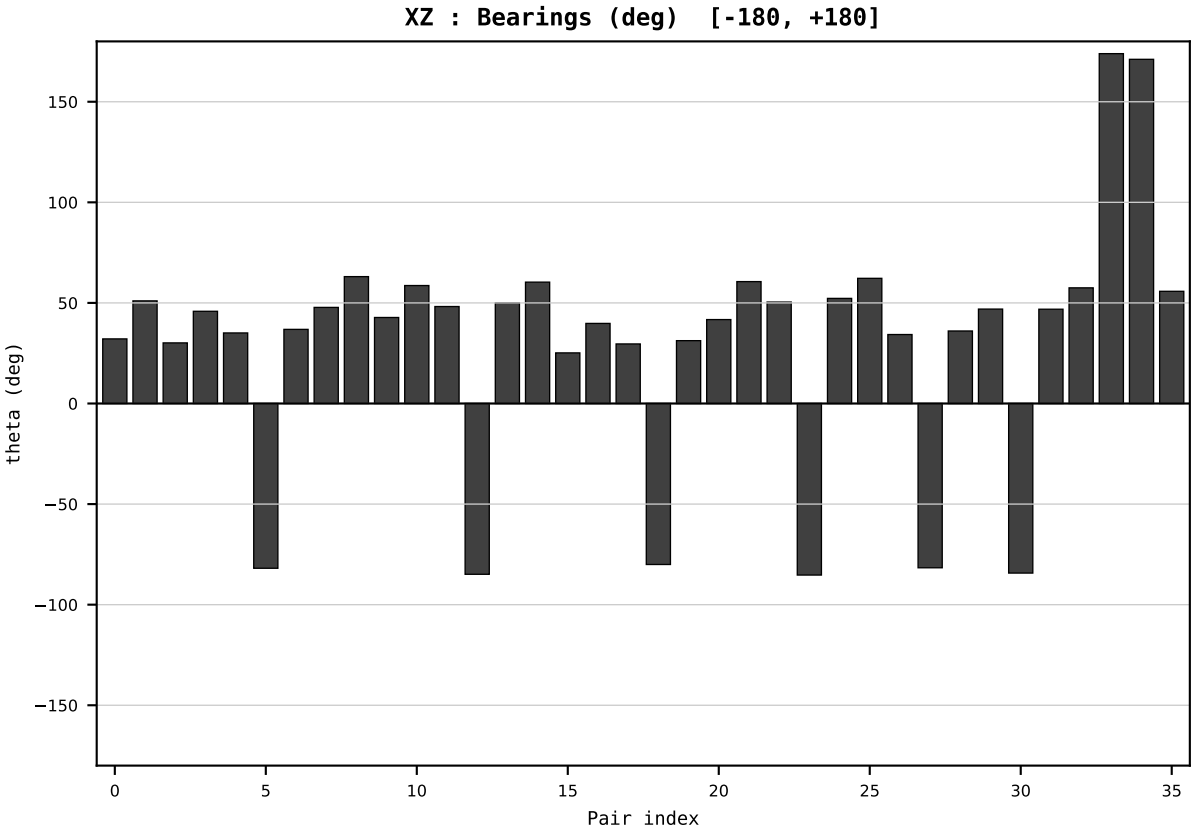
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2021
t : 21 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.255549
Ring = Hs-2
E_metric= 914.9922
kappa_HS= 12284000000000000.00
omega = 0.5927 deg
d_A = 0.368914
Helm = Hydro
Helm d = -0.225695
DR = 39.350 HLR
DR ratio= 122839999940630944.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



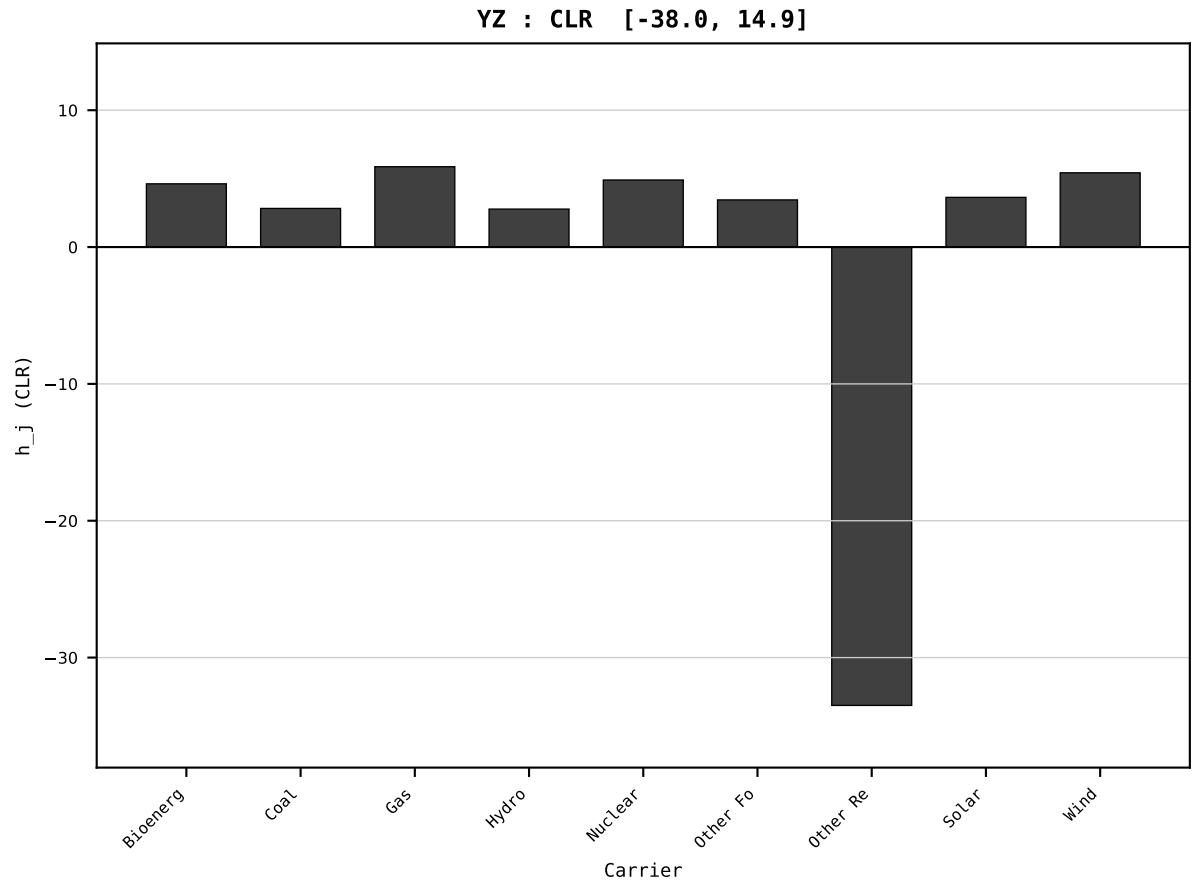
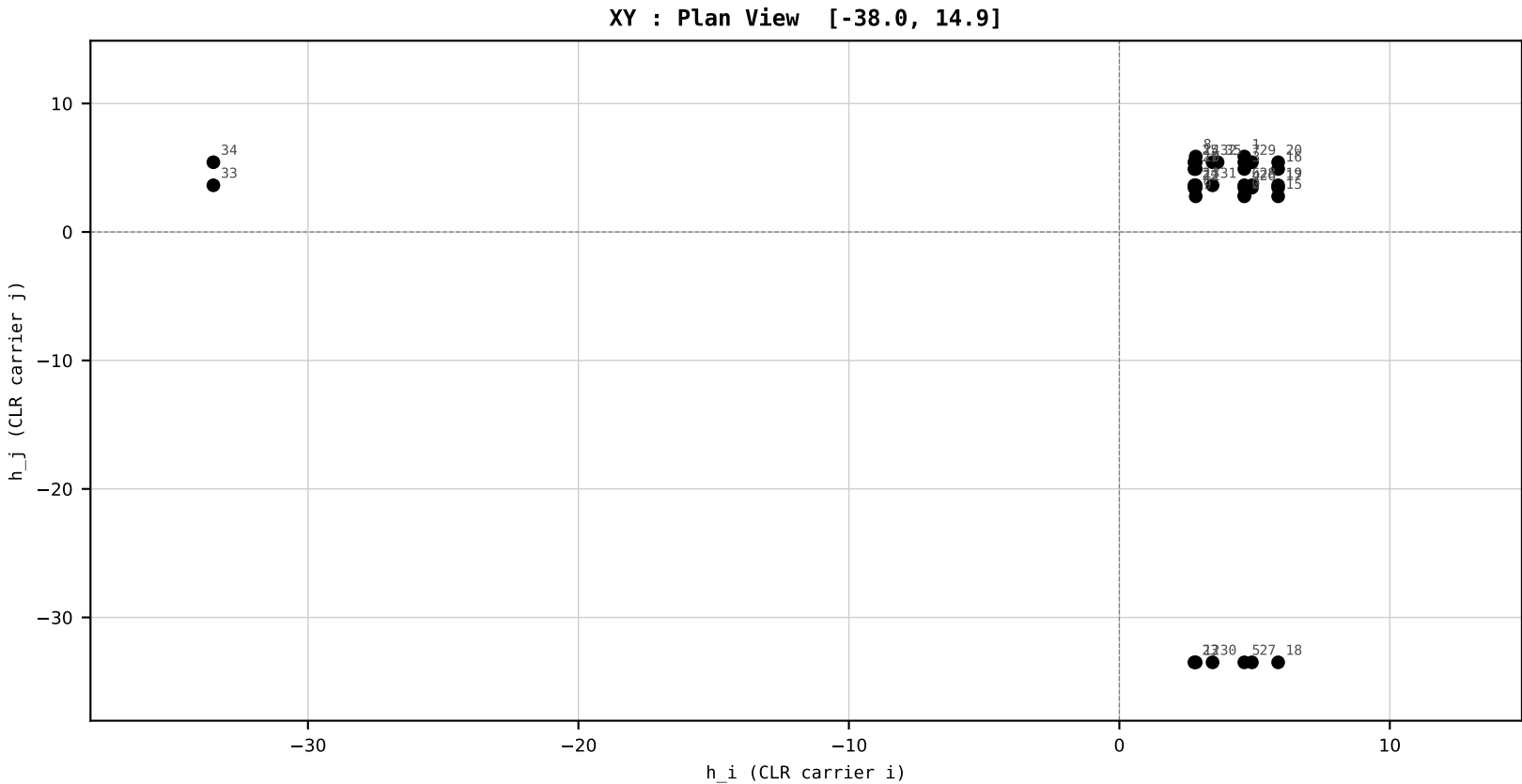
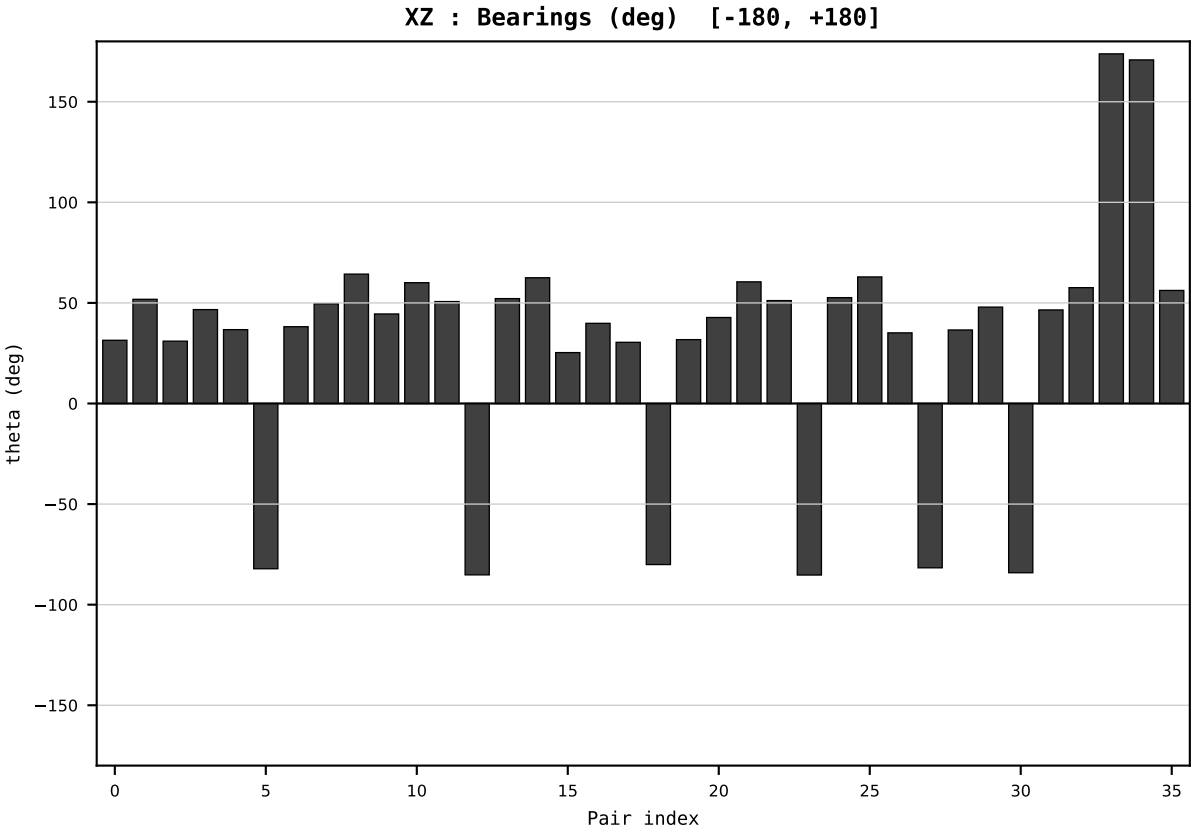
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2022
t : 22 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.259322
Ring = Hs-2
E_metric= 914.4047
kappa_HS= 125109999999999984.00
omega = 0.4928 deg
d_A = 0.309051
Helm = Wind
Helm d = +0.179503
DR = 39.368 HLR
DR ratio= 125110000595524400.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



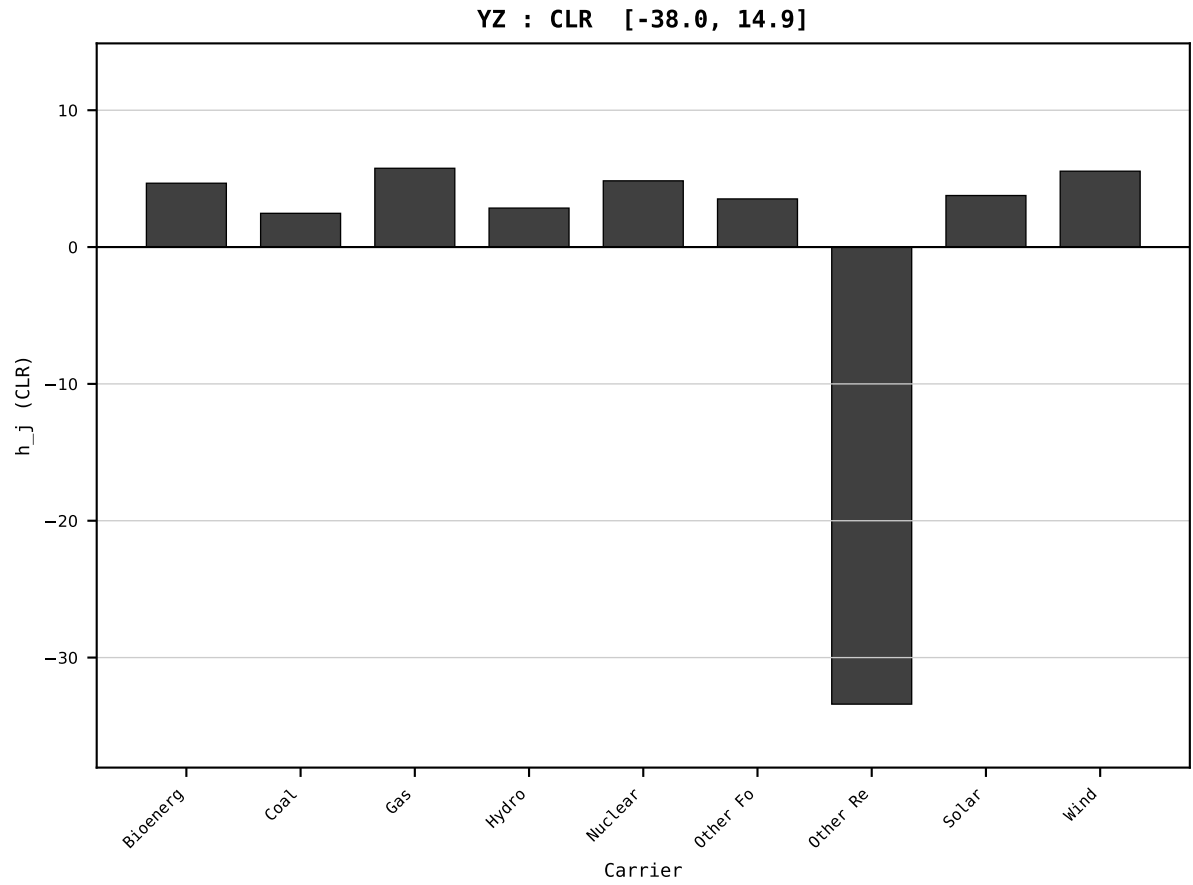
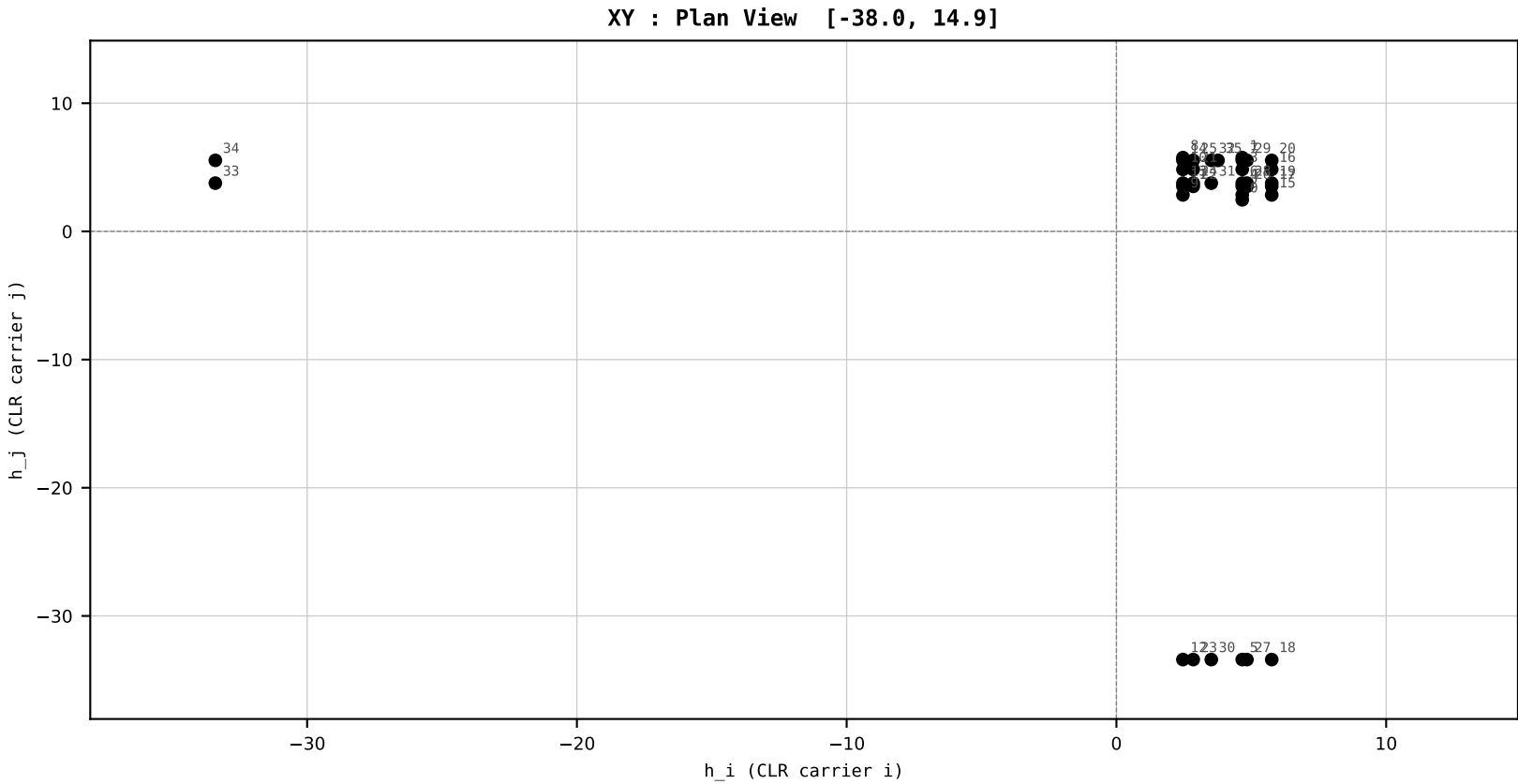
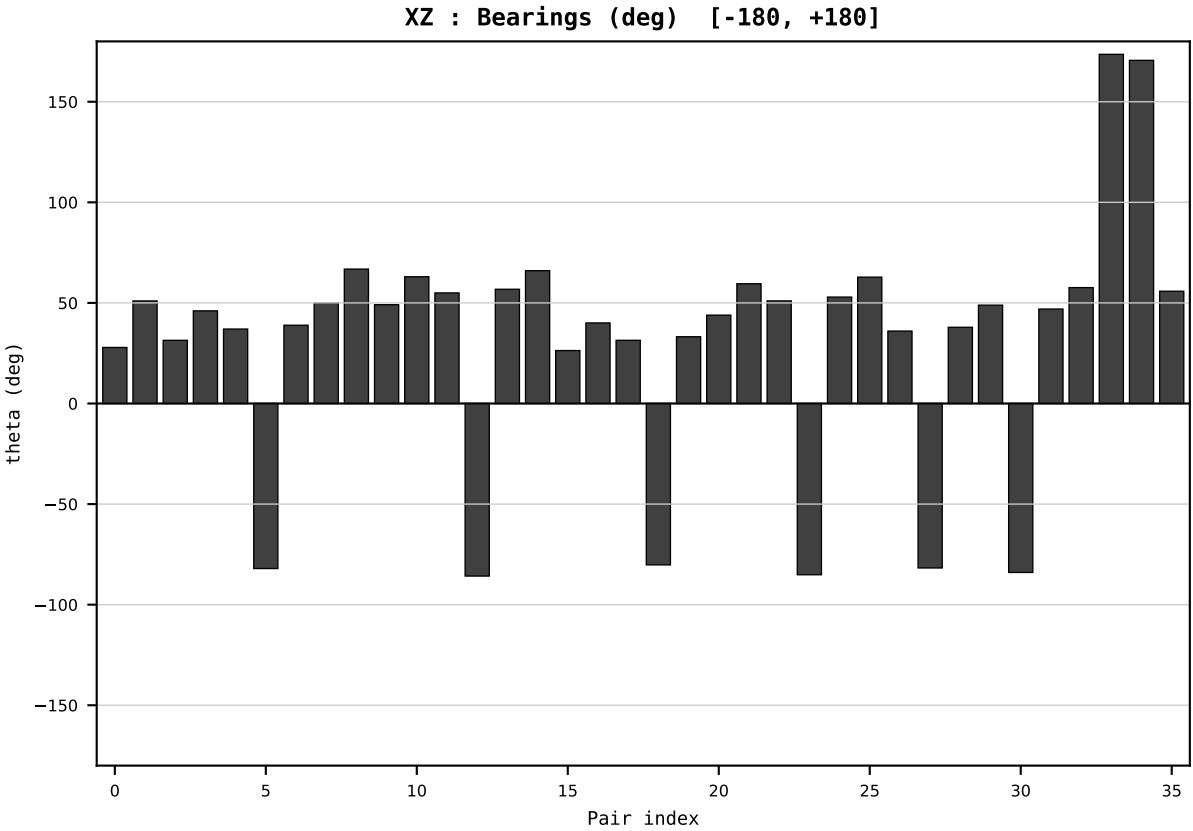
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2023
t : 23 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.250656
Ring = Hs-2
E_metric= 914.8761
kappa_HS= 10165999999999984.00
omega = 0.6998 deg
d_A = 0.444546
Helm = Coal
Helm d = -0.358081
DR = 39.160 HLR
DR ratio= 101660000312153280.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=24 | Year 2024 | D=9 N=26 pairs=36

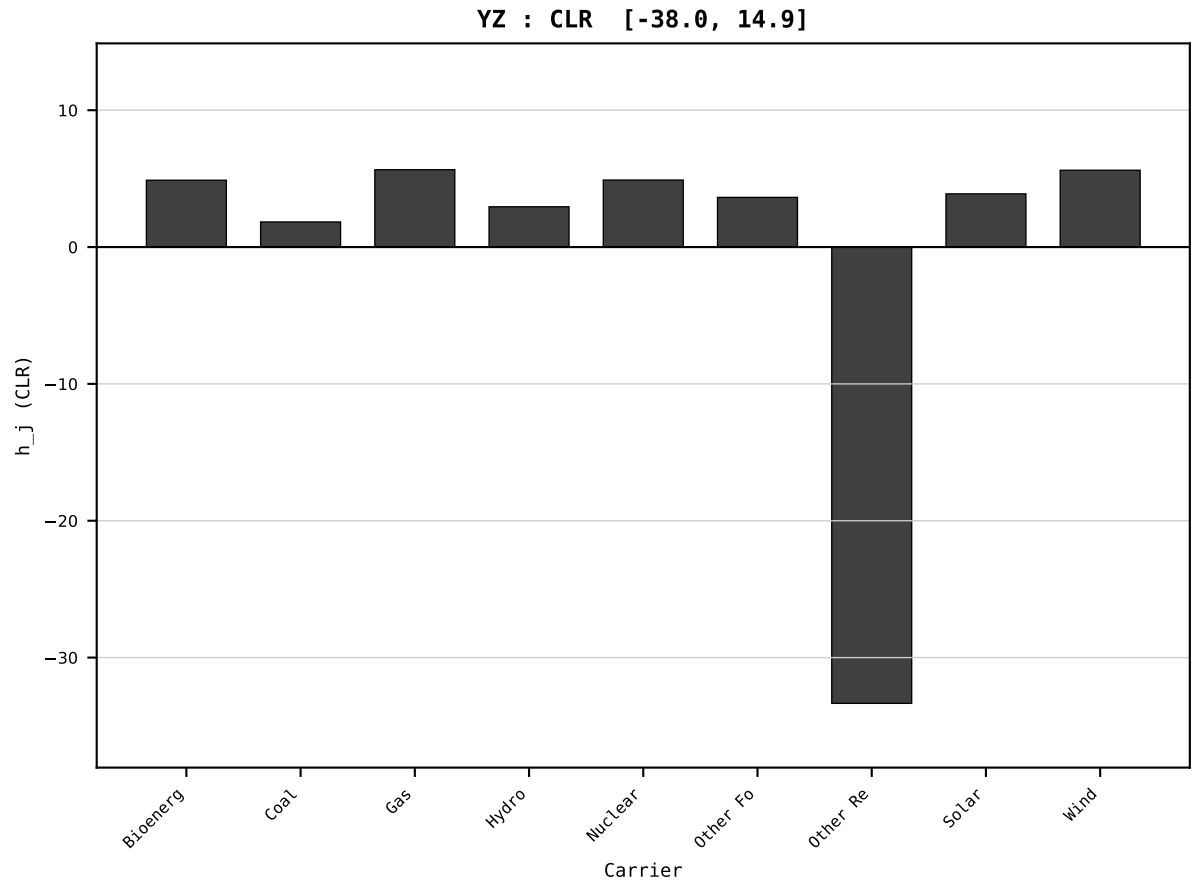
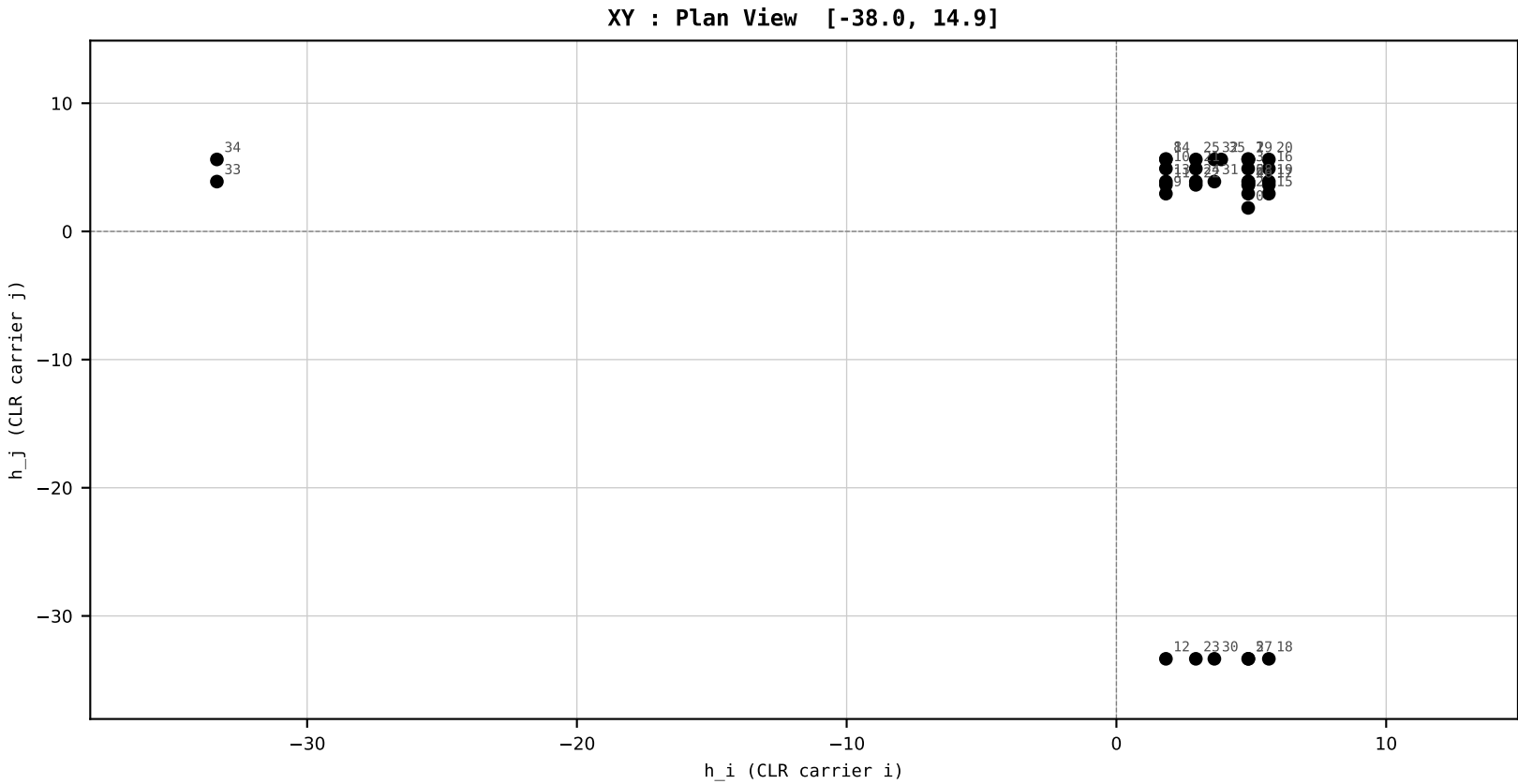
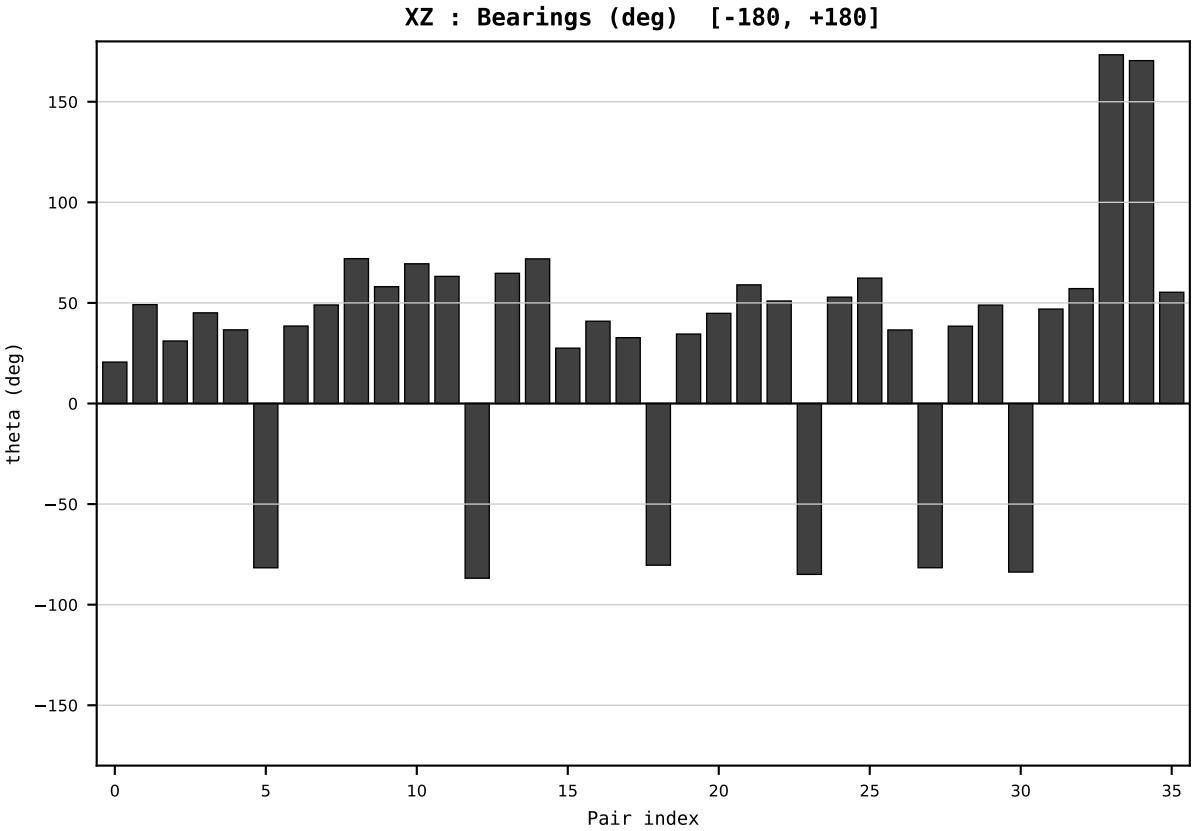
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2024
t : 24 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.239203
Ring = Hs-2
E_metric= 915.1572
kappa_HS= 8630000000000000.00
omega = 1.1445 deg
d_A = 0.710901
Helm = Coal
Helm d = -0.630382
DR = 38.997 HLR
DR ratio= 86300000604094160.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



SECTION PLATE t=25 | Year 2025 | D=9 N=26 pairs=36

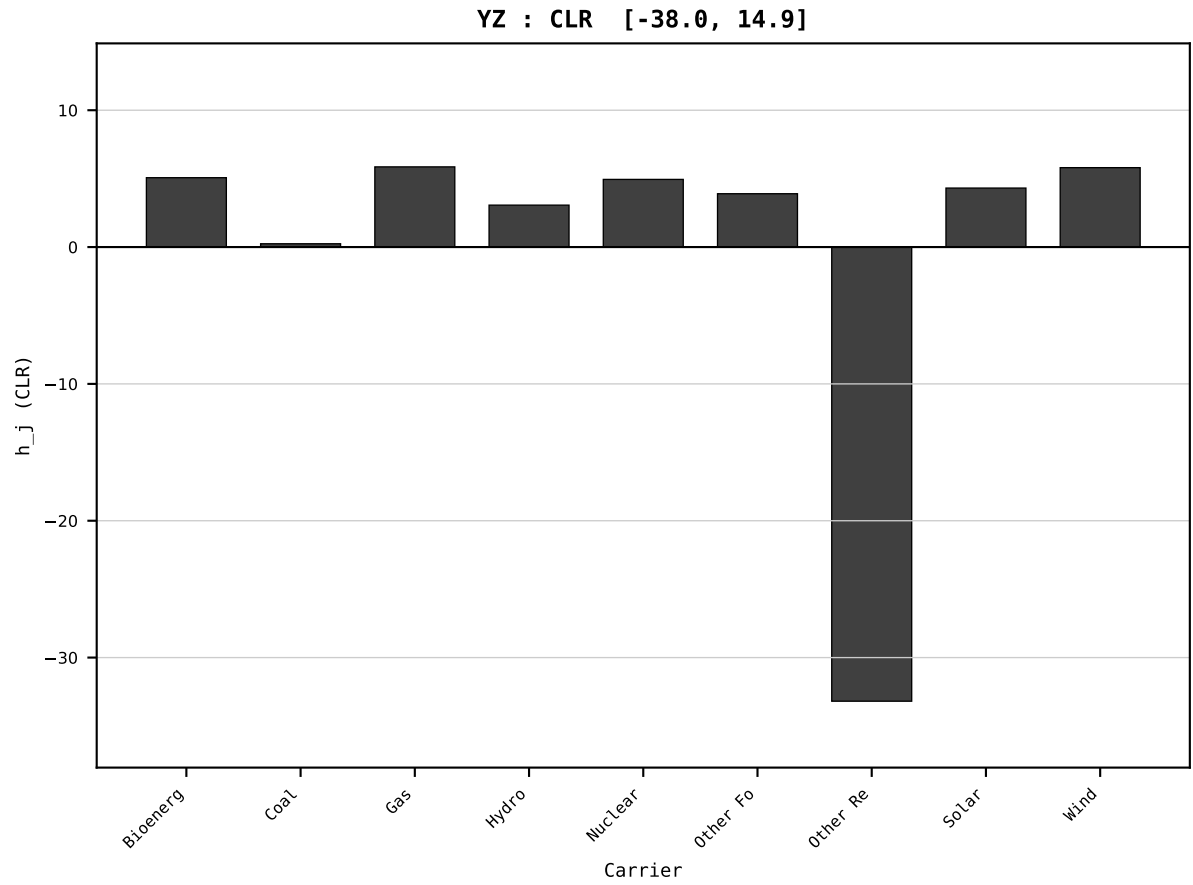
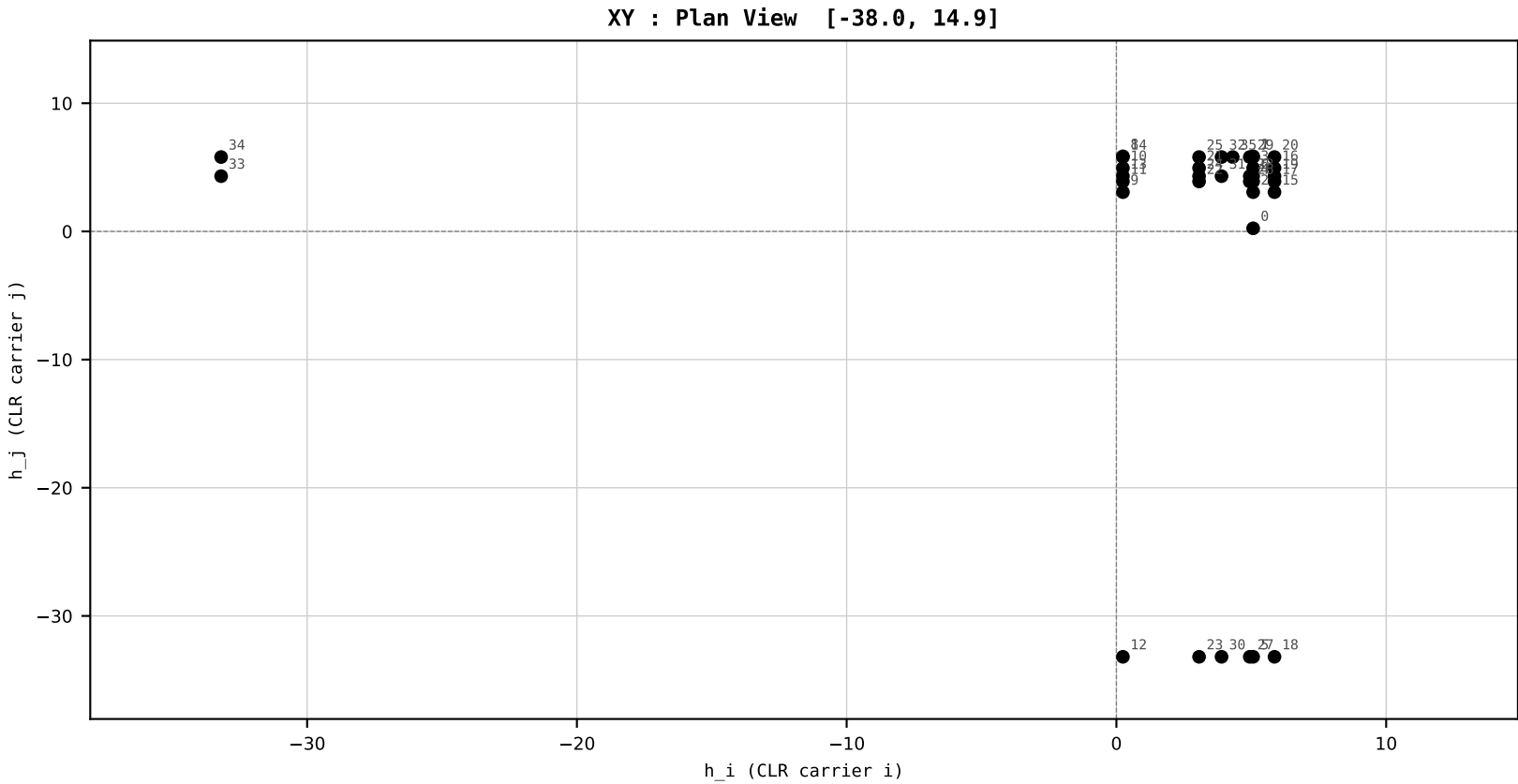
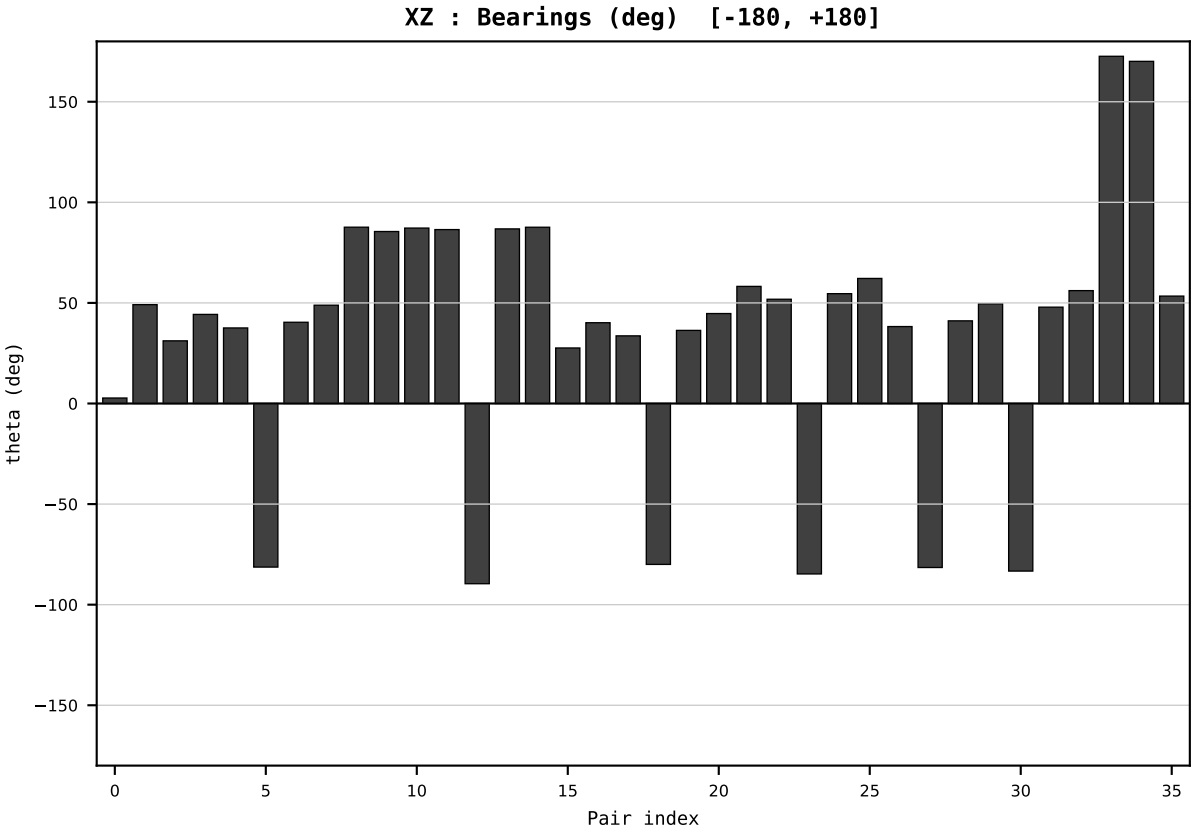
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

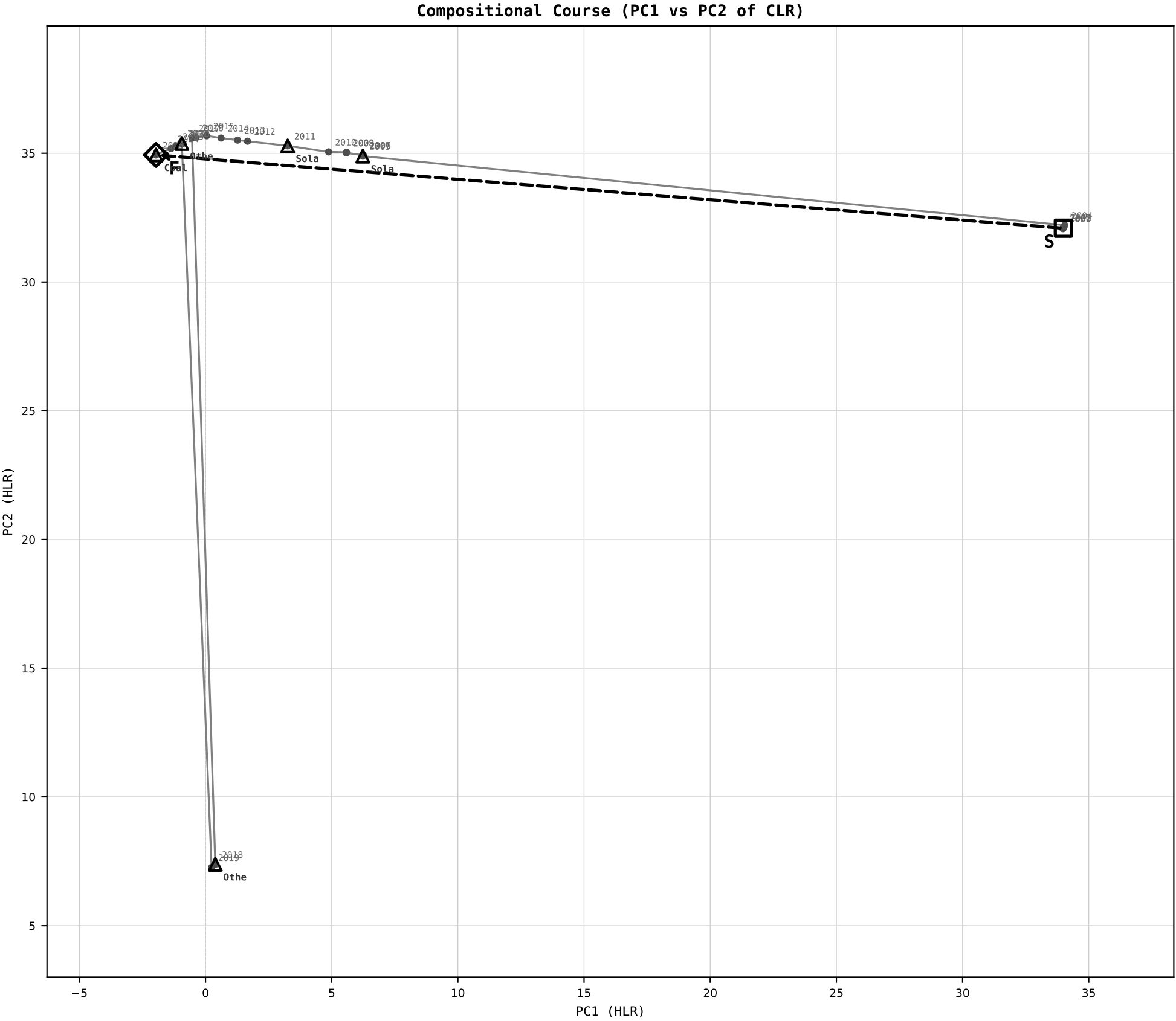
Dataset : ember_GBR_United_Kingdom_generation_TWh.csv
Year : 2025
t : 25 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.245474
Ring = Hs-2
E_metric= 914.6317
kappa_HS= 9090999999999984.00
omega = 2.7663 deg
d_A = 1.715852
Helm = Coal
Helm d = -1.593740
DR = 39.049 HLR
DR ratio= 909099999905049104.0x

XZ [-180,180]
CLR[-38.0,14.9]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind





COURSE METRICS

Scale Provenance:

$Y_j \rightarrow x_j = Y_j / T \rightarrow h_j = \text{CLR}$

$\rightarrow$ PC projection

Unit: HLR (Higgins Log-Ratio)

Net distance = 36.2591 HLR

Path length = 100.3591 HLR

Directness = 0.3613

DR start = 39.537 HLR

ratio = 148079999529708608.0x

DR final = 39.049 HLR

ratio = 90909999905049104.0x

E_metric start = 2195.0806

E_metric final = 1262.8551

PC1 loadings:

Solar -0.931

Coal +0.202

Nuclear +0.151

PC2 loadings:

Other Re -0.941

Coal +0.160

Other Fo +0.123

Top displacement events:

2018 Other Re d=28.236

2020 Other Re d=28.182

2005 Solar d=28.133

2011 Solar d=1.727

2025 Coal d=1.716

The instrument reads.

The expert decides.

The loop stays open.